

Investigating the Development of an Autonomou Research Assistant

Bachelor Thesis

Course of study	Bachelor of Science in Computer Science
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Abstract

Systematic literature reviews sit at the top of the evidence hierarchy yet demand months or years of labour: a researcher must screen thousands of abstracts with near-zero tolerance for missed studies. A single overlooked trial can distort the resulting clinical synthesis; in a diagnostic accuracy review, for instance, a missed study on a bedside assessment test may lead clinicians to adopt an inferior procedure. Existing AI-assisted tools accelerate this work but offer no statistical guarantee on how many relevant papers they might silently exclude. This thesis asks whether an autonomous AI agent can conduct a PRISMA-compliant systematic review end to end while providing a certified, distribution-free upper bound on the false negative rate. The developed system integrates literature search across PubMed and Semantic Scholar, LLM-based abstract and full-text screening, structured data extraction, Cochrane-standard risk-of-bias assessment, and automated PRISMA reporting into a single asynchronous pipeline. Its core theoretical contribution, CASCADE-RC, is a three-threshold conformal cascade that guarantees FNR $\leq \alpha$ with probability at least $1 - \delta$ regardless of the paper distribution; the method has been accepted at the COPA 2026 symposium and will appear in Proceedings of Machine Learning Research vol. 329. Evaluated on the CLEF-TAR 2019 benchmark topic CD008874, the full pipeline achieves 95.4% recall at 21.9% precision. Combining CASCADE-RC routing with a domain-specific diagnostic accuracy rescreen raises precision to 93.9% while holding recall at 94.6% (a 12.5-fold improvement), and full-text retrieval improved nineteenfold through a five-source waterfall. The system answers the research question affirmatively: autonomous certified systematic review is achievable, provided the review topic supplies sufficient calibration positives, the base ranker discriminates relevant from irrelevant papers, and domain-specific inclusion logic is encoded as explicit verification criteria.

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1 Introduction

1.1 Context

Systematic literature reviews (SLRs) occupy the highest position in the evidence hierarchy for clinical and scientific research. By synthesising all available evidence on a research question in a structured, reproducible manner, they form the basis for clinical guidelines, health technology assessments, and regulatory decisions. Yet completing one is extraordinarily demanding: a single review typically requires between six months and three years from initial protocol registration to publication [1], involving a team of domain experts who must search multiple databases, screen thousands of abstracts, retrieve and appraise full texts, extract structured data, and document every decision according to standardised reporting requirements.

The exponential growth of academic publishing has made this workload increasingly unsustainable. Approximately 3.34 million articles were indexed in 2022 alone, representing a 70% increase over the preceding decade [1]. Cochrane Handbook guidelines recommend that the final database search should occur within six months of a review's submission [2]; given the time required to conduct a review, this constraint is becoming harder to meet as literature volume grows. The result is that many published reviews are already partially outdated at the time they appear.

Autonomous AI systems offer a promising path toward addressing this bottleneck. If the mechanical stages of a review (search, abstract screening, full-text assessment, data extraction, and reporting) can be reliably automated, reviewers can focus their expertise on the judgment-intensive tasks that genuinely require human knowledge. The development of such a system, and the design of the certified calibration module at its core, is the subject of this thesis.

1.2 Problem Statement

Manual systematic reviews are time-intensive, error-prone, and difficult to scale. There is currently no fully autonomous system capable of conducting PRISMA-compliant reviews end-to-end while ensuring both transparency and reproducibility. Existing AI-assisted tools address individual stages in isolation, and none pro-

vides a statistical guarantee on the critical safety property of systematic reviewing: that no eligible study is missed.

The central challenge is not merely engineering but rather statistical given the guarantee context. An automated screener may perform well on average, yet offer no certificate to a clinical decision-maker about the worst-case rate of missed relevant studies. Without such a guarantee, no regulatory body or evidence-synthesis organisation can rely on the system's output as a substitute for human-conducted review.

The primary research question guiding this thesis is therefore: *Can an autonomous AI agent replicate the recall and rigour of a human systematic reviewer while providing a statistically certified bound on the risk of missing relevant evidence?*

1.3 Objectives

The main objective of this thesis is to design and implement an autonomous AI-based research assistant capable of conducting PRISMA-compliant systematic literature reviews. This overarching objective is operationalised through six specific goals, derived from the formal requirements specification produced at the start of the project [3]:

1. **Validated baseline prototype.** Develop and validate a baseline prototype focused on a well-defined literature review use case, demonstrating the feasibility of autonomous systematic review execution.

2. **End-to-end autonomous agent.** Enable fully autonomous execution of all PRISMA stages: literature search, abstract screening, full-text eligibility assessment, structured data extraction, quality appraisal, and PRISMA flow reporting.

3. **Iterative search strategy refinement.** Allow the agent to adapt and optimise search queries based on intermediate coverage analysis and identified gaps.

4. **Transparent and traceable documentation.** Generate PRISMA-compliant flow diagrams and log all agent decisions to ensure full reproducibility of every review run.

5. **Quality assessment integration.** Assess study quality and risk of bias using established systematic review standards (Cochrane RoB 2 and Newcastle-Ottawa Scale).

6. **Evaluation of review quality.** Evaluate search completeness, screening accuracy, and PRISMA compliance against a validated benchmark.

1.4 Contributions

The contributions of this thesis are as follows:

- The design and implementation of a modular, seven-component autonomous research pipeline that executes all core PRISMA stages without human intervention.
- The development and evaluation of an iterative search strategy refinement mechanism that adapts Boolean queries based on detected coverage gaps.
- The integration of transparent decision logging through an append-only SQLite audit trail and a structured event stream, enabling full review reproducibility.
- A comprehensive evaluation of review quality and compliance on the CLEF-TAR 2019 benchmark, covering search completeness, screening accuracy, and PRISMA compliance across six Cochrane systematic review topics.
- Submission of CASCADE-RC: Threshold Calibration for Stochastic LLM Cascades with Finite-Sample Recall Guarantees to the 15th Symposium on Conformal and Probabilistic Prediction with Applications (COPA 2026).

1.5 Project Management

The project was conducted over a 15-week semester within the Bachelor's Thesis module (12 ECTS) at Bern University of Applied Sciences, Department of Engineering and Computer Science. The project plan, milestone timeline, initial system architecture, and formal requirements specification were documented at the end of the first month in the planning report [4]. The original architecture described in that report envisaged LangGraph-based orchestration and a Neo4j knowledge graph for provenance storage; both were revised during the implementation phase based on the practical constraints documented in Chapter 4.

Development progress was tracked using Git version control and a Notion-based task board Figure 1.1, with tasks moving from a backlog through in-progress to completed states. Regular supervision meetings were held with Prof. Ben Souissi throughout the semester.

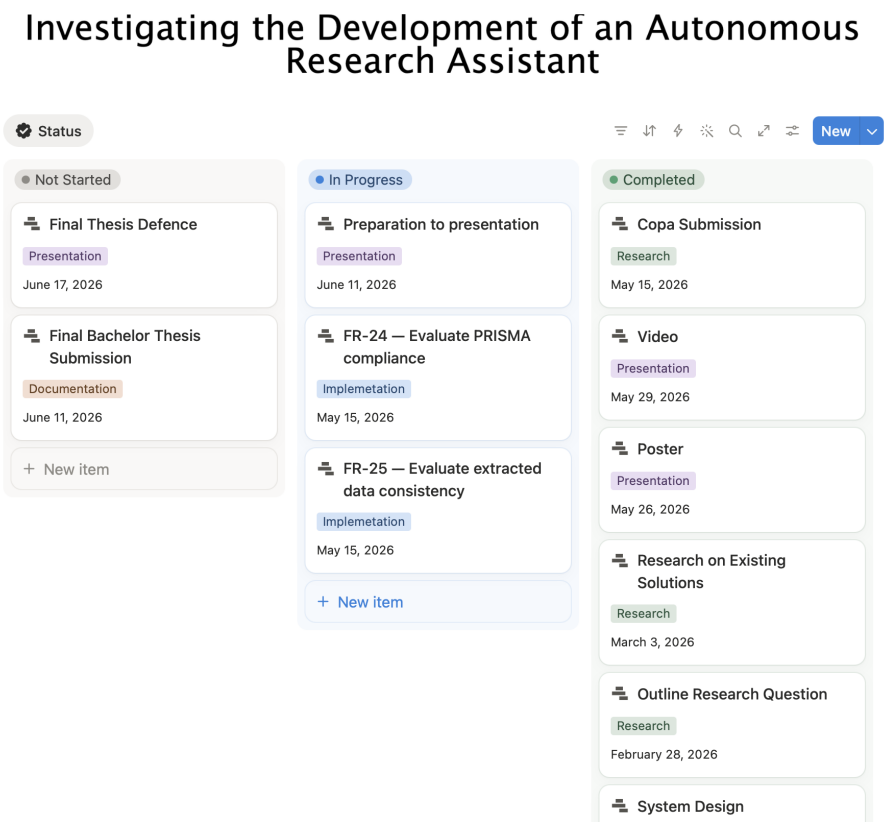


Figure 1.1: Task tracking board used throughout the project. Requirements were managed from a backlog through in-progress to completed states, derived from the formal requirements specification [3].

1.5.1 AI Tools Disclosure

In accordance with BFH guidelines on responsible AI use in student work, the following tools contributed to this project. Claude Code (Anthropic) served as the development environment for implementing the autonomous review pipeline; all architectural decisions and system specifications were made by the author. Gemini Advanced with Deep Research mode (Google DeepMind) assisted with the structured literature survey in Chapter 2, providing database search results that the author independently evaluated and integrated. Claude (Anthropic, claude.ai) was used for grammar checking of the thesis text and for LaTeX formatting assistance; all generated suggestions were reviewed and revised by the author before inclusion. These disclosures are made in accordance with BFH regulations on academic integrity and the declaration of originality accompanying this thesis.

1.6 Document Organisation

Chapter 2 reviews related work on AI-enhanced systematic review tools and existing approaches to PRISMA-compliant automation. Chapter 3 presents the technical methods and algorithms that form the foundation of the system, from information retrieval to conformal risk control. Chapter 4 describes the architecture and implementation of the autonomous review pipeline. Chapter 5 presents CASCADE-RC, the certified screening module developed as the primary contribution of this thesis. Chapter 6 reports the experimental evaluation on the CLEF-TAR 2019 benchmark. Chapter 7 discusses the findings, limitations, and implications of the work. Chapter 8 concludes the thesis and proposes directions for future research.

2 Related Work

2.1 Introduction

The exponential growth of scientific literature presents a fundamental challenge for evidence synthesis. In 2022 alone, approximately 3.34 million articles were published, representing a 70% increase over the preceding decade [1]. Systematic literature reviews (systematic literature review (SLR)s), positioned at the top of the evidence hierarchy, typically require between six months and three years from protocol registration to publication [1]. This temporal burden risks reviews becoming outdated before publication, directly contradicting Cochrane Handbook recommendations that the final search should occur within six months of submission [2].

The emergence of artificial intelligence and machine learning has opened the possibility of accelerating and automating large portions of the systematic review process, with the potential for fully automating the most repetitive stages through these technologies but, yet still to be determined. This chapter surveys existing tools and research prototypes, evaluates their reported performance against human benchmarks, and identifies the limitations that motivate the present work. The chapter is structured as follows: Section 2.2 introduces the PRISMA reporting standard; Section 2.3 surveys AI-enhanced review tools; Section 2.4 examines graph-based and knowledge-driven approaches; Section 2.5 reviews LLM-based agents and pipelines; Section 2.6 summarises reported performance metrics; and Section 2.7 identifies the research gaps that this thesis addresses.

2.2 The PRISMA Framework and Reporting Standards

The Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines were established to ensure consistency and transparency in systematic review reporting [5]. The PRISMA 2020 statement and the PRISMA Abstract Checklist 2021 currently represent the gold standards for methodological rigour [5]. These frameworks specify structured reporting requirements across search strategy, screening process, data synthesis, and risk of bias assessment.

PRISMA organises the review process into four sequential stages. During *identification,* records are retrieved from multiple databases using documented search

strings. During *screening*, titles and abstracts are assessed against pre-defined inclusion and exclusion criteria. During *eligibility*, full texts of candidate studies are evaluated in detail. Finally, during *inclusion*, studies that pass all prior stages are incorporated into the synthesis. At each stage, the number of records processed and the reasons for exclusion must be explicitly documented and presented in a standardised flow diagram, as illustrated in Figure 2.1.

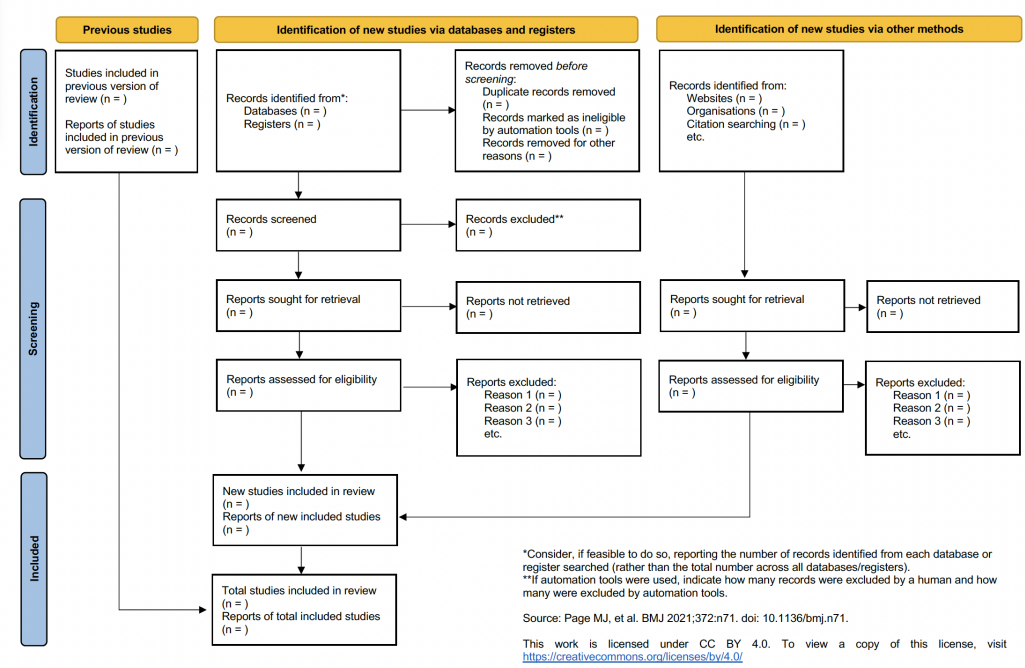


Figure 2.1: PRISMA 2020 flow diagram reproduced from Page et al. [6]. The diagram tracks records through identification, screening, eligibility, and inclusion.

Recent research has shown that large language models (LLMs) can be guided to produce PRISMA-compliant outputs through structured prompt engineering. Al-Turki et al. [7] applied chain-of-thought prompting to enable ChatGPT-4o [8] and Gemini Pro [9] to generate systematic review abstracts adhering to the PRISMA Abstract Checklist. In their study, each model was given the same set of clinical research inputs and asked to produce a structured abstract; the outputs were then scored against eleven items from the PRISMA Abstract Checklist, yielding an Overall Quality Score between 0 and 22. ChatGPT-4o achieved a mean score of 21.67 across evaluated abstracts, meaning it satisfied on average almost all checklist items; Gemini Pro reached a mean of 21.00. The standard deviations of 0.58 and 0.71 respectively indicate that both models performed consistently across runs, with very little variation between one generated abstract and the next. The difference between the two models, though small in absolute terms, was confirmed to be statistically reliable ($p < 0.001$), meaning it is unlikely to be due to random chance [7]. These findings indicate that both models can produce struc-

turally compliant outputs when given appropriate instructions, though they do not demonstrate autonomous execution of the full systematic review process.

2.3 AI-Enhanced Tools for Systematic Reviews

While the PRISMA framework defines what a complete systematic review must contain and how to document it, it says nothing about who or what should do the work and how. The rapid emergence of AI-enhanced tools has opened the possibility of automating significant portions of the process, though current tools fall far short of full automation.

2.3.1 Overview of Available Tools

A systematic comparison of commercially available AI-enhanced literature review tools reveals significant variation in their underlying architectures and capabilities [10]. Of the eleven tools evaluated in one study, none were able to retrieve all articles identified through manual search strategies [10]. Three tools utilised publicly available LLMs with internal adjustments; one employed a proprietary model (referred to as *smpy*, a closed-source system designed specifically for automated information extraction and ranking); and one relied on non-generative AI approaches [1]. Table 2.1 summarises the key characteristics and limitations of the most widely evaluated tools.

2.3.2 Search Completeness and Database Coverage

A fundamental requirement of PRISMA-compliant reviews is comprehensive literature retrieval across multiple databases. Current AI tools exhibit significant limitations in this domain. Van Dinter et al. [10] confirmed that none of the eleven evaluated tools could retrieve the full set of articles identified through manual search strategies. This finding is reinforced by the MSR56 study, which concluded that while AI-enhanced tools effectively streamline targeted reviews, caution is needed when applying them to systematic reviews where completeness is a core requirement [1].

The challenge of search completeness is further compounded by the iterative nature of systematic literature review (SLR) protocol development. Tools that perform well on narrowly focused queries often struggle to maintain recall when the search scope is expanded or when databases with different indexing standards are combined [15].

Table 2.1: Comparison of AI-enhanced systematic review tools

Tool	Key Features	Strengths	Limitations
Covidence [11]	Title/abstract screening; conflict resolution; citation manager integration	Efficient workflow; widely adopted	High cost; limited AI depth
Rayyan [12]	AI text-mining; blinding; relevance tags	High sensitivity (97–99%); user-friendly	Premium features paywalled; low specificity at liberal threshold
ASReview [13]	Active learning prioritisation; open-source	Iterative learning; transparent	Screening only; no extraction
Abstrackr [14]	Predictive relevance modelling	Free; objective reviewer scoring	No full-text screening; limited reporting
Elicit [15]	LLM-based search and extraction	High precision; broad task coverage	Low recall (∼40%); immature extraction
Loon Lens 1.0 [16]	Agentic AI; confidence-guided screening	98.9% sensitivity; human-in-loop option	Precision 63% standalone
Review Copilot [17]	GPT-4 based screening	99.2% sensitivity; 4× faster than humans	Full-text specificity only 47.4%

2.3.3 Screening Accuracy: Sensitivity and Specificity

The title and abstract screening process represents the most labour-intensive stage of the review because it involves reviewing thousands of records repetitively against inclusion criteria before any full-text assessment begins. Recent validation studies provide quantitative benchmarks for AI screening performance against human gold standards.

Loon Lens 1.0. Janoudi et al. [16] evaluated Loon Lens 1.0 across eight systematic reviews comprising 3,796 citations with 287 included studies (7.6% inclusion rate). The platform achieved sensitivity of 98.9%, specificity of 95.2%, and accuracy of 95.5%, with precision of 63.0%. Each figure is reported with a 95% confidence interval, which gives the range within which the true value would fall if the

study were repeated many times; narrower intervals indicate more reliable estimates. The intervals here are tight sensitivity 97.6–100, precision 58.4–67.3 confirming that the results are stable across the evaluated reviews [16]. Errors clustered predominantly in low- to medium-confidence inclusion decisions. A simulated human-in-the-loop strategy, in which only low- and medium-confidence inclusions (3.8% of all citations) were routed to human review, raised precision to 81.4% and accuracy to 98.2% while preserving 99.0% sensitivity [16]. This result demonstrates the potential of confidence-guided workflows for concentrating human effort at high-uncertainty decision points.

Review Copilot. Zhan et al. [17] evaluated Review Copilot, a GPT-4-based screening program, against human decisions from four published systematic reviews. At the title and abstract stage, the system achieved sensitivity of 99.2% and specificity of 83.6%. At the full-text stage, sensitivity remained high at 97.6%, but specificity dropped sharply to 47.4%. Inter-run reliability was 95.4%; the agreement coefficient $\kappa = 0.83$ (Cohen's kappa) measures how consistently the system makes the same decision when run twice on the same record, where 1.0 means perfect agreement and 0 means no better than chance, placing this result in the strong agreement range. The system screened records in one quarter of the time required by human reviewers [17]. The dramatic decline in full-text specificity suggests that current LLM-based tools struggle with the nuanced eligibility judgements required when reading complete manuscripts.

2.3.4 Data Extraction and Synthesis

Once relevant studies have been screened in, a systematic review requires extracting specific data from each paper outcomes, methods, sample sizes, risk of bias and synthesising those findings into a coherent summary. This stage is typically the most time-consuming part of a review, and it is where AI tools have been evaluated most critically.

Van Dinter et al. [10] found that summary tables produced by AI tools differed substantially from manually constructed equivalents, and agreement between AI and human reviewers on risk-of-bias assessments was poor. In other words, the tools could produce plausible-looking tables but frequently got the content wrong.

Meliante et al. [18] compared four AI tools directly against manual PRISMA-compliant extraction in the glaucoma domain. Elicit correctly extracted around 51% of required data fields; ChatPDF reached approximately 60%. Writing assistants such as Jenni AI produced fluent prose but missed methodological precision the text read well but the underlying data reporting was unreliable. The authors concluded that manual PRISMA-based extraction remains clearly superior in accuracy and reproducibility, and that a human researcher must remain in the loop whenever AI tools are used for extraction [18].

A more optimistic result comes from a fully automated pipeline built on top of a commercial large language model, which processed searches, extracted data, and produced complete draft manuscripts without human intervention [19]. Citation accuracy reached 95.87%, achieved by restricting the model to text it had been explicitly provided rather than allowing it to generate references from memory. In a blinded evaluation, six domain specialists rated the AI-generated reviews higher on average than a published human-authored review on the same topic, and could not reliably tell the two apart. This is a striking result, though it should be read carefully: the evaluation covered a single clinical domain, and no PRISMA compliance check was performed, leaving open the question of whether the outputs would meet the reproducibility standards required for evidence-based medicine.

2.3.5 Transparency and Explainability

A persistent limitation across AI tools is the absence of transparent, auditable decision-making due to so called "black box" sytem in which only inputs and outputs can be analysed. Van Dinter et al. [10] note that AI modalities are frequently characterised by a lack of explainability, creating ethical risks related to bias and the integrity of evidence synthesis. This concern is particularly acute for systematic reviews, where methodological transparency is foundational to the validity and reproducibility of conclusions.

Current evaluations focus predominantly on output compliance whether the final report meets PRISMA standards rather than on process transparency, meaning whether the decision-making pathway is fully auditable at each stage. This distinction is critical: a PRISMA-compliant output does not guarantee that the process by which it was generated is reproducible or defensible.

2.4 Graph-Based and Knowledge-Driven Approaches

The tools surveyed above treat each paper as an independent record in a flat list. A parallel line of research asks whether representing the literature as a *graph* where papers, concepts, and decisions are connected nodes rather than isolated rows might enable richer querying and a more transparent audit trail. The two subsections below describe prototypes that apply this idea, first at the level of stored knowledge and then at the level of workflow orchestration.

2.4.1 Knowledge Graph Systems for systematic literature review (SLR)

A knowledge graph is a database structure in which information is stored not as a flat table but as a network of nodes and relationships. Each node represents an entity (a paper, an author, a clinical concept), and each edge between two nodes represents a named relationship between them ("cites", "evaluates", "uses method"). This structure makes it possible to answer queries that a spreadsheet cannot easily support, such as finding all papers that share a methodology with a given study, or tracing how a clinical finding has been built upon over time.

Sahlab et al. [20] built a knowledge graph pipeline that interfaces with digital literature databases, applies NLP-based filters to retrieved records, and stores papers as nodes with relationships encoded as directed edges. Their prototype, evaluated in the domain of context-aware automation systems, automatically retrieved approximately 78% of the studies identified through a manual review and classified them with approximately 90% accuracy [20]. The graph structure additionally enabled complex queries identifying co-occurring themes or tracing methodological lineage across studies that would be impractical with flat spreadsheet storage.

Crucially, Sahlab et al. found that both manual and automated analysis of the same corpus led to equivalent conclusions regarding the domain landscape [20]. This indicates that partial recall, while suboptimal, did not distort the synthesis outcome in their specific evaluation context. However, this finding should not be generalised: in systematic reviews where completeness is a regulatory requirement, such as clinical guideline development, any recall below 100% may constitute a methodological limitation.

2.4.2 Multi-Agent Orchestration with LangGraph

LangGraph [21] applies the graph idea not to stored knowledge but to the workflow itself. Rather than executing a fixed sequence of steps, a LangGraph pipeline is defined as a directed graph in which each node is an autonomous LLM agent performing one task retrieving papers, screening abstracts, reformulating a query and each edge determines which agent runs next depending on the outcome. This means the pipeline can loop: if an agent determines that the current search results are insufficient, it can pass control back to the retrieval node and try again with a revised query, all without human intervention [21]. Every agent action and transition is recorded in the graph state, which provides a built-in audit trail of the workflow execution.

This framework was evaluated during the design phase of this thesis as a candidate orchestration layer. While LangGraph is well suited to single-topic iterative workflows, it does not readily support the multi-database, multi-source retrieval

structure required here, where PubMed and Semantic Scholar must be queried concurrently with independent rate limits and error recovery paths. The pipeline developed in this thesis therefore uses a custom asyncio-based architecture instead, as described in Chapter 4.

Figure 2.2 provides a comparative overview of the tools surveyed in this chapter across the six core PRISMA review stages.

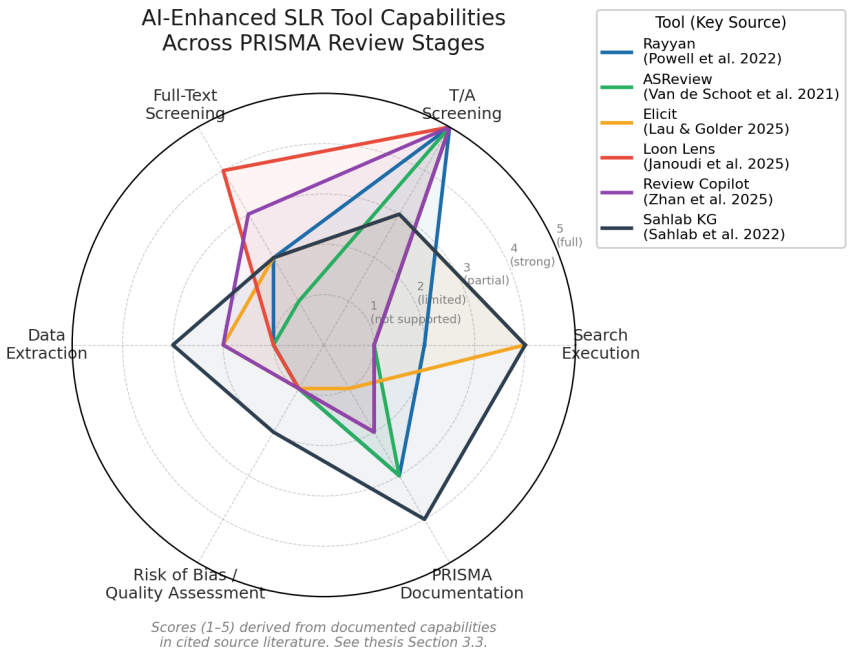


Figure 2.2: Radar chart comparing AI-enhanced SLR tools across six PRISMA review stages. Screening tools dominate; search formulation and risk of bias stages are underserved.

2.5 LLM-Based Agents and Workflows

Beyond purpose-built screening tools, a broader wave of LLM-based agent frameworks has recently been applied to literature tasks. These systems treat systematic review stages as sequential processing steps some running in parallel coordinated by a central orchestration layer, and they prioritise topic flexibility over domain-specific precision.

2.5.1 GPT-4 in PRISMA Screening Workflows

Landschaft et al. [22] integrated GPT-4 [8] into a PRISMA-compliant workflow using a retrieval-augmented generation (RAG) approach. RAG is a technique in which relevant text passages are retrieved from a document corpus and provided directly to the language model as context before it answers a question; this grounds the model's responses in the actual source material rather than relying purely on what it learned during training, which reduces the risk of fabricated or unsupported outputs.

The system was applied to a patient-trial matching task: given a pool of published clinical trial protocols and a set of patient records, the goal was to identify which trials each patient would be eligible for based on the trial's inclusion and exclusion criteria. The task requires reading each trial protocol and each patient record, then making a binary eligible or ineligible decision structurally similar to abstract screening in a systematic review.

GPT-4 was used for two separate subtasks. First, at the abstract screening stage, it read each trial description and decided whether to include or exclude it based on predefined criteria. Second, it extracted structured parameters from included texts figures such as age ranges, diagnosis codes, and required lab values and stored them in a structured format. Screening decisions were compared against those of two independent human reviewers.

At the screening stage, GPT-4 achieved a Cohen's kappa above 0.90. Kappa measures how much two decision-makers agree beyond what would be expected by chance alone; a value of 0.90 or above is generally considered near-perfect agreement, meaning the model's include or exclude decisions matched human reviewers almost every time [22]. This was specifically for title-and-abstract-level decisions on clinical trial eligibility, not for the full review process; applying this finding to other review types or clinical domains requires caution.

Full-text data extraction was considerably less reliable. For this stage the model was given complete manuscript PDFs and asked to locate and return specific data fields. The RAG pipeline failed to consistently handle data reported in tables, supplementary appendices, or figures formats where the relevant number is not in plain prose but embedded in a structured layout that the model could not reli-

ably parse. The authors noted this as an inherent limitation of current LLMs for extraction tasks requiring structured document understanding [22].

2.5.2 Elicit and Retrieval-Augmented Search

Elicit [15] is an AI research assistant that supports query generation, relevance ranking, and structured data extraction from academic literature. A 2025 Cochrane evaluation by Lau and Golder [15] assessed Elicit's search sensitivity across multiple review topics, finding a mean sensitivity of approximately 40% compared to 94% for traditional manual database searches. Precision was comparatively high records retrieved by Elicit were predominantly relevant but the tool failed to identify the majority of eligible studies. The authors concluded that Elicit functions as a useful adjunct for initial scoping but is not a substitute for comprehensive database searching in PRISMA-compliant reviews [15].

2.5.3 Active Learning: ASReview

ASReview [13] implements a researcher-in-the-loop paradigm based on active learning: a machine learning approach in which the model continuously learns from the reviewer's decisions rather than being trained once and deployed. The system trains a classifier on the reviewer's labels so far, ranks all remaining unlabelled records by predicted relevance, and presents the most likely inclusions first. As the reviewer labels more records, the model adapts, concentrating subsequent effort where it is most needed. ASReview's authors emphasise open algorithms and standardised benchmarking as core design principles, directly addressing transparency limitations in closed commercial tools [13]. Reported efficiency gains range from 50% to 80% reductions in screening effort, depending on the review topic and inclusion rate.

2.6 Reported Performance: A Comparative Overview

The studies reviewed in this chapter report performance using different metrics and across different review stages. Table 2.2 consolidates the key figures into a single reference. Each row corresponds to one evaluated tool or system; *Sensitivity* gives the fraction of truly relevant records that the tool successfully retrieved or retained (higher is safer); *Specificity* gives the fraction of irrelevant records that the tool correctly excluded (higher means less noise passed through); and *Other Metric* captures an additional reported figure that does not fit neatly into one of those two columns for example, data extraction field accuracy (the percentage of structured data fields the tool filled in correctly), or Cohen's kappa (inter-rater agreement with human reviewers). An empty cell means the original study did

not report that measure. Recall (sensitivity) is consistently prioritised in the systematic review literature, as missing an eligible study constitutes a potentially serious methodological error [23].

Table 2.2: Reported performance metrics for AI-enhanced SLR tools. T/A = title and abstract; FT = full text; Sen. = sensitivity; Spe. = specificity.

Study	Tool / System	Sensitivity	Specificity	Other Metric
Janoudi et al. [16]	Loon Lens 1.0	98.9%	95.2%	Precision: 63.0%
Zhan et al. [17]	Review Copilot (T/A)	99.2%	83.6%	$\kappa = 0.83$
Zhan et al. [17]	Review Copilot (FT)	97.6%	47.4%	
Powell et al. [24]	Rayyan	97–99%	Low	
Lau & Golder [15]	Elicit	∼40%	High	
Meliante et al. [18]	Elicit (extraction)			Field accuracy: 51%
Meliante et al. [18]	ChatPDF (extraction)			Field accuracy: 60%
Landschaft et al. [22]	GPT-4 screening			$\kappa > 0.90$
Sahlab et al. [20]	KG Pipeline	∼78%		Classification: 90%

The table makes the core trade-off visible at a glance. Sensitivity at the title-and-abstract stage is high across most tools, but specificity collapses at the full-text stage (47.4% for Review Copilot), meaning that passing too many records through to full-text assessment creates a large residual manual workload. Tools evaluated on extraction tasks show field accuracy well below 65%, which is insufficient for research-grade evidence synthesis.

2.7 Limitations and Research Gaps

The preceding review reveals several persistent limitations in the current state of AI-assisted systematic reviews. These limitations directly motivate the research questions and system design presented in this thesis.

Incomplete end-to-end automation. No validated system currently demonstrates autonomous execution of all PRISMA stages (search, screening, extraction, quality assessment, and documentation) within a single integrated pipeline [1, 10]. Most tools address one or two stages in isolation. Tools that do cover extraction are not integrated into full workflows, and tools that cover workflows provide limited AI support for extraction and quality appraisal.

Search completeness. AI search tools consistently fail to achieve the recall levels expected of comprehensive manual searches. Elicit's sensitivity of approximately 40% [15] and the broader finding that no evaluated tool retrieved all manually identified articles [10] indicate that current automated search strategies require substantial supplementation. This gap is particularly critical because missed studies at the search stage cannot be recovered by subsequent screening or extraction steps.

Full-text screening specificity. While sensitivity at the title-and-abstract stage is consistently high (97–99%), specificity at the full-text stage drops substantially: to 47.4% in the Review Copilot evaluation [17]. This means current tools generate large numbers of false positives at full-text screening, imposing residual manual workload and undermining the efficiency gains promised by automation.

Data extraction accuracy. Extraction accuracy for structured data fields remains well below human standards. The 51% accuracy reported for Elicit and 60% for ChatPDF [18] are insufficient for research-grade evidence synthesis. LLM-based extraction is further hampered by hallucination (the generation of plausible but unsupported content), which requires verification of all extracted values against source text.

Risk of bias assessment. Automated quality appraisal and risk of bias assessment remain largely unaddressed. Van Dinter et al. [10] report poor inter-rater reliability between AI tools and human reviewers for this task. This represents the most significant automation gap in the current literature.

Process transparency and audit trails. Current evaluations assess output compliance rather than process auditability. The ability to reconstruct the full decision pathway (including every search query, screening decision, and exclusion reason) is a core PRISMA requirement that existing tools inadequately support [7, 10].

Integration across review stages. Existing tools operate as isolated modules rather than components of a unified pipeline. As a result, data must be manually transferred between tools at each stage transition, creating opportunities for loss of traceability and inconsistencies in the documented record.

Table 2.3 maps these identified gaps to the research questions and objectives of the present thesis.

The autonomous research assistant developed in this thesis addresses these limi-

Table 2.3: Mapping of identified research gaps to thesis objectives

Identified Gap	Thesis Objective	Addressed By
No end-to-end PRISMA pipeline	Objective 2	Integrated seven-component agent architecture
Search completeness below 100%	Objective 3	Iterative query refinement module
Full-text specificity limitations	Objective 2	Two-stage LLM screening with DTA rescreen
Extraction accuracy below 60%	Objective 2	RAG-based extraction with span verification
Risk of bias not automated	Objective 5	Quality assessment (RoB 2 and NOS)
Lack of process audit trails	Objective 4	Decision logging and PRISMA flow generation
Fragmented tool integration	Objectives 2, 4	Unified pipeline with shared data model
No recall guarantee	Objective 2	CASCADE-RC conformal calibration module

tations through a modular, multi-tier architecture that separates literature search, abstract screening, full-text screening, and synthesis into independently verifiable components. The primary theoretical contribution (presented in Chapter 5) is a calibration framework that provides finite-sample guarantees on the false negative rate of the abstract screening stage, a property absent from all tools surveyed in this chapter. The evaluation in Chapter 6 demonstrates that this approach achieves a 12.5-fold improvement in screening precision while maintaining recall above the 94% threshold required for PRISMA-grade evidence synthesis.

2.8 Summary

The state of the art in AI-enhanced systematic reviews demonstrates meaningful progress, particularly in title-and-abstract screening, where tools achieve sensitivity of 97–99%. However, critical limitations persist across search completeness, full-text screening specificity, data extraction accuracy, risk of bias assessment, and process transparency. No existing system demonstrates autonomous

end-to-end PRISMA compliance with validated human-level performance across all review stages. These gaps collectively motivate the work presented in the following chapters.

3 Methods and Algorithmic Foundations

3.1 Introduction

Chapter 2 surveyed existing AI-assisted systematic review tools and found a consistent gap: none provides a statistical guarantee on the fraction of eligible studies that might be missed. This chapter builds the technical foundation needed to understand both how such a guarantee can be constructed and how the system around it actually works.

The chapter serves two purposes. First, it introduces the technologies and algorithms that underpin the autonomous review pipeline described in Chapter 4, giving the reader enough background to understand why each component was chosen and how it contributes to the whole. Second, it develops the theoretical framework for certified recall control that is formalised into CASCADE-RC in Chapter 5. Neither purpose is subordinate to the other; the chapter is meant to stand alone as a self-contained guide to the methods used in this thesis.

The chapter is structured as a deliberate build-up from problem to solution. It begins with the clinical and statistical problem: what kind of review is being automated, why missed studies are so costly, and what dataset was used to evaluate the system. It then introduces the retrieval signals used to narrow the candidate pool cheaply, followed by the language-model-based classification that makes the actual include-or-exclude decisions, and the conformal prediction theory that certifies those decisions. After that comes the engineering for full-text retrieval and screening, the domain-specific filter that resolves the precision problem, and finally the evaluation metrics used throughout the thesis. Reading from beginning to end provides everything needed to follow Chapters 4 and 5 without reaching for an external reference.

3.2 The Problem: Screening a Medical Literature Review

This section establishes the clinical and statistical context for everything that follows. It explains what kind of systematic review this thesis automates, why a missed study is treated as a near-unacceptable error, and which controlled benchmark was used to measure the system's performance. A reader who understands

these three points will have a clear picture of what the system is trying to do and why the design choices in later chapters were made the way they were.

3.2.1 What Is a Diagnostic Test Accuracy Review?

A systematic review is a structured synthesis of all available research evidence on a specific question. Rather than reading a handful of convenient papers, a systematic reviewer searches multiple databases, screens every retrieved record against a formal set of inclusion criteria, and aggregates findings from all eligible studies into a single summary. The resulting review is considered the highest-quality form of evidence in medicine because it is reproducible, comprehensive, and transparent.

Within systematic reviews, the Diagnostic Test Accuracy (DTA) subtype asks a specific kind of question: how accurately does a particular bedside test identify a target condition when compared against a definitive reference procedure? A bedside test is any examination a clinician can perform quickly at the patient's side without specialist equipment; a reference standard is the more definitive but often more invasive or time-consuming procedure against which the bedside test is judged. The clinical question is always: can the simpler test reliably substitute for the harder one?

The primary evaluation topic used throughout this thesis is a published Cochrane DTA review known by its registry identifier CD008874. It investigates whether bedside airway assessment tools, such as the Mallampati score (a visual grading of how much of the throat is visible when a patient opens their mouth) and the thyromental distance (a measurement from the chin tip to the throat notch), can predict difficult tracheal intubation before a patient is anaesthetised. The reference standard against which these bedside tests are compared is the Cormack-Lehane grade, a classification assigned during the intubation itself once the clinician has a clear view of the airway. Accurate prediction matters because a failed intubation under anaesthesia is a life-threatening emergency; if a reliable bedside test could flag high-risk patients in advance, clinical teams could prepare accordingly.

Every published study that directly compared at least one named bedside test against the Cormack-Lehane reference standard and reported diagnostic accuracy statistics is eligible for inclusion in CD008874. A typical database search for this review topic retrieves around two thousand abstracts. Of those, approximately 130 turn out to be genuinely eligible (roughly 5.5%). The remaining abstracts are related but ineligible: papers about general anaesthesia equipment, sedation management, airway anatomy, or imaging-based assessment that do not involve a direct bedside-versus-reference-standard comparison.

Why missed studies are so costly. The asymmetry of errors in systematic re-

viewing is fundamental to understanding why this thesis exists. If the automated system includes an ineligible paper (a false positive), that paper will be caught and removed at a later manual checking stage. The cost is reviewer time. If the system excludes a genuinely eligible paper (a false negative, or missed study), that paper disappears from the review entirely. The synthesised evidence is then built on an incomplete picture. In a DTA review used to inform clinical guidelines, a missed study could tip the pooled estimate of a bedside test's sensitivity or specificity in the wrong direction, leading clinicians to trust a test that performs less reliably than the evidence actually supports.

This asymmetry is why Cochrane review teams target recall above 95%: they can tolerate a modest number of false positives because those are catchable, but they treat missed eligible studies as a methodological failure [2]. Any automated tool must respect the same constraint, and it should ideally come with a mathematical certificate rather than just an empirical hope.

3.2.2 The Evaluation Benchmark: CLEF-TAR 2019

To evaluate whether the system meets the recall requirement, a controlled benchmark with known ground truth is needed. This thesis uses the CLEF Technology-Assisted Review (CLEF-TAR) 2019 collection [25], which was specifically designed to benchmark automated screening tools for systematic reviews.

The benchmark provides 31 Cochrane review topics across four clinical families: Diagnostic Test Accuracy, Intervention, Prognosis, and Qualitative. For each topic, the benchmark supplies a fixed candidate pool of PubMed abstracts and a set of relevance labels indicating which papers were judged eligible by the original Cochrane review team. These labels were frozen at the date of the original Cochrane search and do not change, which means that any evaluation run on the same topic against the same benchmark produces directly comparable results. This reproducibility is essential for fair method comparison.

CD008874 was chosen as the primary evaluation topic for this thesis for a specific reason: it is a DTA review, which represents the hardest variant of the screening problem. Intervention reviews have relatively clear-cut eligibility criteria (is it a randomised trial comparing these two treatments in this patient population?). DTA reviews add a structural requirement on top of that: the study must evaluate a named index test against a named reference standard and report accuracy statistics. A paper that satisfies all four PICO fields but does not have this study design is still ineligible. This makes false negatives more costly and false positive filtering harder, which is exactly the setting where a certified screening system is most valuable.

The benchmark is also appropriate because it provides a known set of positive papers: the 130 papers labelled as relevant in the CD008874 qrels file (the relevance

judgment list). These known positives are what make conformal calibration possible. Conformal calibration is a statistical procedure that uses a labelled set of examples to certify how well a threshold holds in practice; for that to work, the relevant papers need to be identified in advance. The term *positive stratum* used in later chapters refers to exactly this labelled set of eligible papers within the calibration pool. It is not a subjective group; it is a precisely defined subset of the benchmark.

Six topics from CLEF-TAR 2019 are used in this thesis overall, selected because they have complete abstract text, structured relevance labels, and non-empty qrels files in the DTA, Intervention, and Prognosis families. All method comparisons and evaluation tables in Chapter 6 are based on these six topics, with CD008874 serving as the primary result.

3.3 Information Retrieval: Finding the Candidates

Before any language model can screen an abstract, the abstract must first be retrieved. A Cochrane database search returns tens of thousands of PubMed records. Processing every one of them with a language model would be accurate but prohibitively slow and expensive. Information retrieval (IR) methods solve this by rapidly ranking the full candidate pool by likely relevance, so that computationally expensive operations are reserved for the most promising abstracts.

This section introduces the two retrieval signals used in this thesis and explains how they are combined. Both signals operate at the title and abstract level: they do not read full manuscripts, only the brief summary and title that PubMed provides for every indexed paper. Think of them as two independent readers who each assign a relevance score to every abstract very quickly, and then whose scores are merged into a single ranked list. The first reader, BM25, notices exact word matches. The second reader, SPECTER2, understands meaning even when the words are different. Together they produce a more robust ranking than either alone.

3.3.1 Sparse Retrieval with BM25

BM25 (Best Match 25) is the most widely used algorithm for text-based document retrieval [26]. To understand what it does, start with the simplest possible retrieval idea: count how many times the query words appear in each document and rank documents by that count. This works but has obvious flaws: a very long document will naturally mention query words more often than a short one, and a word that appears in almost every document (like “patient” in a medical corpus) is not a useful discriminating signal. BM25 fixes both problems.

It applies two corrections to the raw term frequency count. The first is a length normalisation: a paper is not rewarded for being long. The raw count is divided by a term that grows with document length, so that a short focused paper and a long comprehensive paper receive comparable scores if they contain the query term with the same density. The second correction is inverse document frequency (IDF): a term that appears in only a handful of abstracts in the entire corpus receives a much higher weight than a term appearing in nearly every abstract. In a corpus of medical papers, a rare clinical abbreviation like "Mallampati score" is a highly discriminating signal and is weighted accordingly; a common word like "patient" is weighted very little.

The formal expression for BM25 combines these two corrections. For a query q with terms $t_1, \ldots, t_k$ and a candidate abstract d, the score is:

$$\text{BM25}(d, q) = \sum_{t \in q} \underbrace{\log\left(\frac{N - n_t + 0.5}{n_t + 0.5}\right)}_{\text{rare term reward}} \cdot \frac{f(t,d) \cdot (k_1 + 1)}{f(t,d) + k_1\left(1 - b + b\,\frac{|d|}{\overline{dl}}\right)}, \tag{3.1}$$

where N is the total number of abstracts, n_t is the number of abstracts containing term t, $f(t, d)$ is how many times term t appears in abstract d, $|d|$ is the length of d, $\overline{dl}$ is the average abstract length, and k_1 and b are constants typically set to 1.5 and 0.75. The formula is the general BM25 algorithm as defined in the original paper [26]; no modifications were made for this thesis.

BM25 is fast: it requires no GPU and no neural network inference. For this thesis specifically, it is effective at recovering abstracts that contain rare clinical abbreviations. Terms like thyromental distance, Cormack-Lehane grade, or Mallampati score appear in only a small fraction of PubMed abstracts, making them powerful discriminating signals under BM25's IDF weighting. The choice to include BM25 was made on the recommendation of Prof. Rao during a supervision meeting, who identified its term-frequency weighting as particularly well-suited to recall-oriented retrieval in biomedical literature, and this was confirmed empirically on the CD008874 development set.

3.3.2 Semantic Retrieval with SPECTER2

BM25's weakness is that it only finds exact word matches. If a relevant abstract uses "laryngoscopy difficulty score" where the query uses "Cormack-Lehane grade", BM25 gives it a low score even though both phrases refer to the same clinical concept. Dense embedding models address this by mapping every text into a point in a high-dimensional numerical space (a vector), where semantically similar texts end up close together regardless of whether they share the same words. This is called a dense or semantic retrieval signal.

SPECTER2 [27] is a transformer-based scientific document encoder designed specifically for academic literature. It was trained on the Semantic Scholar corpus us-

ing a citation signal: two papers that tend to be cited together are pulled close together in the embedding space during training, while papers that rarely co-occur are pushed apart. This means the space it creates reflects the scientific community's own judgement about which papers are substantively related — a property that makes it well-suited to biomedical abstract retrieval.

For this thesis, SPECTER2 encodes each abstract into a 768-dimensional vector, compressed to 128 dimensions through a linear projection. The review protocol's PICO fields are encoded into a single query vector. The relevance score for each abstract is the cosine similarity between its vector and the query vector: a value close to 1 means the abstract is semantically aligned with the review question; a value close to 0 means it is not. Documents are ranked by this score to produce the dense retrieval list.

3.3.3 Combining the Two Signals with Reciprocal Rank Fusion

BM25 and SPECTER2 produce scores on incompatible scales: BM25 scores are sums of weighted term counts (potentially large numbers), while cosine similarities lie between 0 and 1. Simply adding them would let whichever signal happens to have larger raw values dominate. Reciprocal Rank Fusion (RRF) [28] solves this by discarding raw scores entirely and working only with rank positions.

For an abstract ranked at position r_{BM25} in the BM25 list and r_{dense} in the SPECTER2 list, the combined score is:

$$\text{RRF}(d) = \frac{1}{k + r_{\text{BM25}}} + \frac{1}{k + r_{\text{dense}}}, \tag{3.2}$$

where $k = 60$ is a moderating constant from the original paper [28]. An abstract ranked first in both lists gets the maximum possible combined score; one ranked near the bottom in both gets close to zero. Crucially, RRF requires no training and no tuning data: the constant $k = 60$ was established empirically in the original publication and transfers well across tasks.

The combined BM25-plus-SPECTER2 hybrid ranker recovers different missed papers from each component: BM25 retrieves abstracts containing rare abbreviations that SPECTER2 ranks low; SPECTER2 recovers paraphrased descriptions that BM25 misses due to vocabulary mismatch. The combined score forms the base signal $s(X)$ that the CASCADE-RC calibration module uses to partition abstracts into routing buckets.

The concept of combining multiple independent ranking signals into one score did not originate in this thesis. RRF is an established technique in the information retrieval literature. What is specific to this thesis is the application of BM25-plus-SPECTER2 fusion to the systematic review screening setting, and the empirical validation that the combination outperformed either component alone by 4.3 percentage points in recall on the CD008874 development set.

3.4 LLM-Based Screening

Retrieval ranking produces a list of abstracts ordered by likely relevance, but it cannot decide whether a specific abstract actually meets the formal inclusion criteria of a review. That requires reading the abstract, understanding the clinical question, and making a structured judgment about each inclusion criterion. This is the job of a large language model (LLM): a neural network capable of reading and understanding natural language text, following detailed instructions, and producing structured outputs.

This section explains how LLMs work at the level needed to follow the system design, how they are used for per-criterion screening in this thesis, how their outputs are aggregated, and how their inherent randomness is measured and controlled. These subsections apply primarily to the abstract and title screening stage, though the same mechanisms extend to full-text screening in Chapter 4.

3.4.1 Transformer Architecture and Language Models

A large language model is built on the transformer architecture, originally introduced by Vaswani et al. [29]. To understand why transformers are useful here, it helps to know what came before them. Earlier models processed text as a sequence from left to right, handling one word at a time. This meant that when a model reached the word "intubation" at the end of a sentence, it had largely forgotten the specific bedside test mentioned at the beginning. Transformers solved this by processing all positions in the input simultaneously, with each position directly attending to every other position. This mechanism is called *self-attention*.

Self-attention works as follows. For each word (or subword token) in the input, the model computes three vectors: a Query (what this token is looking for), a Key (what this token has to offer other tokens), and a Value (the actual content to be passed forward). The attention score between two tokens is the dot product of one token's Query and another's Key, scaled by the square root of the vector dimension to keep magnitudes stable. After a softmax normalisation, these scores become weights that determine how much each token contributes to every other token's updated representation. A token can attend strongly to a distant token if they are semantically related, regardless of how far apart they are in the text.

Transformers also distinguish between two broad roles. An *encoder* reads the full input text and produces a contextualised representation of it; this is what SPECTER2 uses to embed abstracts. A *decoder* generates text token by token, attending both to what has been generated so far and to the encoder's representation; this is how LLMs produce answers to questions. GPT OSS 120B is a decoder-only model: given a prompt, it generates a continuation token by token until it decides to stop.

Figure 3.1 illustrates the self-attention mechanism at a high level.

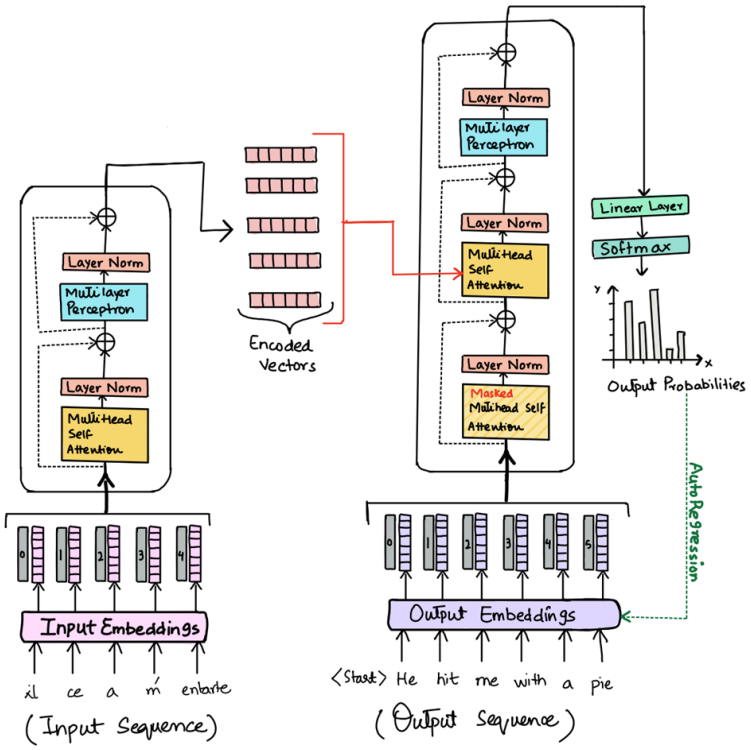


Figure 3.1: Self-attention mechanism in a transformer. Each token attends to all others; attention weights reflect semantic relevance and produce context-aware representations.

This thesis uses the GPT OSS 120B model [8] served via the BFH internal inference endpoint for all classification and extraction tasks. The choice of this specific model was driven by a practical constraint: testing the CASCADE-RC calibration algorithm requires thousands of API calls per calibration run ($B = 5$ queries per abstract $\times$ thousands of abstracts $\times$ multiple calibration iterations). Using a commercial paid API such as GPT-4 or Claude at this scale would have incurred costs in the thousands of dollars during development alone. The BFH endpoint provides GPT OSS 120B at no per-call cost, making the calibration research economically feasible. Among the models available at the endpoint, GPT OSS 120B offered the strongest instruction-following behaviour for structured JSON output, which is required by the per-criterion prompting design.

3.4.2 Temperature and Deterministic Inference

When a decoder-only LLM generates a response, it does not deterministically pick the most likely next word every time. Instead it samples from a probability distribution over all possible next tokens. The temperature parameter T controls

how sharp or flat that distribution is. At high temperatures, even unlikely tokens get a reasonable chance of being selected, making the output more varied and creative. At low temperatures, the most likely token is heavily favoured. At $T = 0$, the model always selects the highest-probability token: this is called greedy decoding, and it makes the output completely deterministic given the same input.

For screening tasks in this thesis, determinism is not optional. If the same abstract can receive a different include or exclude decision on two separate runs simply because the model happened to sample a different token, the review cannot be reproduced. PRISMA requires that every decision be documented and recoverable. All abstract screening, full-text screening, and data extraction calls in the pipeline are therefore issued at $T = 0$. The only deliberate exception is the self-consistency scoring step, where randomness is intentionally exploited to measure the model's uncertainty, as described in Section 3.4.5.

3.4.3 Per-Criterion Structured Prompting

A systematic review protocol defines a list of explicit inclusion and exclusion criteria. One could ask the model a single question: does this abstract meet all the criteria? This sounds simple but causes two problems. First, it produces no per-criterion record, so there is no way to state which criterion caused exclusion when challenged by a reviewer or auditor. Second, LLMs tend to perform better when asked focused, specific questions than when asked to integrate many constraints simultaneously into one opaque answer.

The approach in this thesis issues one inference call per criterion per abstract. Each call receives three things: the abstract text, the review's PICO (Population, Intervention, Comparator, Outcome) description, and a single criterion to assess. The model returns a structured JSON object containing three fields: whether the criterion is satisfied (`true`, `false`, or `uncertain`), a confidence score between 0 and 1, and a verbatim text span from the abstract that supports the assessment. This decomposition satisfies PRISMA traceability requirements because every inclusion and exclusion decision is backed by a specific quoted passage from the source text.

The same per-criterion approach is applied at the full-text screening stage, where the abstract is replaced by the relevant sections of the retrieved manuscript. For very long full texts, additional techniques described in Section 3.7 handle the challenge of fitting the document into the model's context window.

Chain-of-thought prompting is applied at the full-text stage and in the DTA post-processing filter (Section 3.9): the model is asked to reason step by step before reaching a final verdict. This reduces the rate at which the model reaches a confident judgment based on superficial keyword presence without engaging with the actual clinical content of the text.

3.4.4 Combining Criterion Scores: Noisy-OR Aggregation

Each abstract receives a set of per-criterion scores $p_1, p_2, \ldots, p_n$, one per inclusion criterion in the review protocol. These must be combined into a single inclusion probability. Averaging them would imply that strong evidence for one criterion can compensate for weak evidence for another — which is not how inclusion works. An abstract must satisfy all inclusion criteria, but the system should forward an abstract as a candidate if there is meaningful evidence for any criterion being met, because it is better to over-include at this stage and resolve borderline cases later.

The Noisy-OR model captures this logic. It models the probability that at least one criterion is strongly enough satisfied to warrant forwarding the abstract to the next stage:

$$p_{\text{include}} = 1 - \prod_{j=1}^{n} (1 - p_j). \tag{3.3}$$

To see why: $(1 - p_j)$ is the probability that criterion j does not support inclusion. The product is the probability that no criterion supports inclusion. Subtracting from one gives the probability that at least one criterion does. A paper with high p_j on even one criterion gets a high p_{include}, which is appropriate at the abstract screening stage where the goal is to minimise false negatives rather than maximise precision.

Two safety corrections are applied before this formula runs. For a `false` response with high confidence, a probability cap is set at $p_j = 0.30$: even an overconfident negative verdict cannot suppress the aggregation entirely. For parse failures or `uncertain` responses, $p_j = 0.50$ is assigned, the neutral recall-safe value. A final threshold of $p_{\text{include}} \geq 0.70$ triggers inclusion; $p_{\text{include}} \leq 0.25$ triggers exclusion; values in between are forwarded as uncertain and placed in a human review queue.

3.4.5 Measuring LLM Uncertainty: Self-Consistency Scoring

The per-criterion calls above run at $T = 0$ and produce a single deterministic verdict. But some abstracts genuinely sit on the boundary: they might describe a study that almost satisfies the DTA design requirement but leaves it ambiguous. For those abstracts, a single deterministic call may give a confident answer that does not reflect the actual difficulty of the case. The CASCADE-RC module (Chapter 5) needs a way to measure this uncertainty: how confident is the model, and should this abstract be escalated to a human reviewer?

The solution is to query the model $B = 5$ times at temperature $T = 0.7$, allowing it to sample different responses for the same abstract. The self-consistency score $u(X)$ is the fraction of the B responses that agree with the majority verdict. A

unanimous result across all five queries gives $u = 1.0$, indicating high confidence. A three-to-two split gives $u = 0.6$, indicating genuine uncertainty. A near-even split gives u close to 0.5, indicating that the model essentially cannot decide.

High u means the abstract falls clearly on one side of the decision boundary and the model's verdict can be trusted. Low u means the model is uncertain, and the abstract is escalated to a human reviewer who is in a better position to resolve the ambiguity. This ensemble approach introduces five times as many LLM calls for the scored abstracts. To keep costs manageable, self-consistency scoring is applied only to the middle band of abstracts in the retrieval score distribution, not to clear-cut cases, and computed scores are cached so that subsequent calibration runs on the same abstracts require no additional API calls.

3.5 Conformal Prediction and Statistical Risk Control

The retrieval and screening methods described above can achieve high recall empirically, meaning that in practice they tend to retain most eligible studies. What they cannot do is provide a *certificate*: a formal mathematical statement of the form "with probability at least $1 - \delta$, the fraction of eligible studies that this system misses is no greater than α." Conformal prediction is the statistical framework that converts an empirical track record into exactly such a guarantee. This section explains the framework and then identifies the specific technical obstacle that arises when the pipeline contains a stochastic LLM component.

3.5.1 The Coverage Guarantee

Conformal prediction [30] is built on a single observation: if a model has already been evaluated on a held-out set of labelled examples (the calibration set), its error rate on that set is an unbiased estimator of its error rate on future examples, provided those future examples come from the same process. The only formal requirement is *exchangeability* — the condition that the calibration examples and the test example could in principle have been drawn in any order from the same source, with no systematic difference between them. When this holds, conformal prediction produces prediction sets that are guaranteed to contain the true label with user-specified probability, for any model, without any assumptions about the data distribution.

The mechanism works through a *non-conformity score*: a number computed for each calibration example that measures how unusual or poorly-fitted that example is relative to the model's predictions. Think of it as a surprise score — high values mean the model was wrong or uncertain, low values mean it was confident and correct. Taking the $(1-\delta)$ quantile of these scores on the calibration set gives a threshold. At test time, the prediction set is constructed to include all outcomes

whose non-conformity score falls below this threshold. By construction, the true outcome clears this threshold at least $(1 - \delta)$ fraction of the time.

For systematic review certification, this is exactly the guarantee needed. A practitioner does not want to hear "we expect 95% recall on average"; they want "with 90% probability, recall on this specific review will be at least 95%." Conformal prediction turns the first statement into the second, with no parametric assumptions and in finite samples — not just asymptotically.

3.5.2 Risk-Controlling Prediction Sets

Standard conformal prediction controls coverage: the true label falls inside the prediction set. Risk-Controlling Prediction Sets (RCPS) [31] extend the framework to control any bounded user-defined loss function, not just coverage. For systematic review screening, the relevant loss is the recall loss: $L(\theta; X, Y) = Y \cdot \mathbf{1}\{\hat{Z}(X;\theta) = 0\}$, which equals 1 only when a paper is genuinely relevant ($Y = 1$) and the screener excludes it ($\hat{Z} = 0$). RCPS finds a threshold θ such that:

$$\mathbb{E}[L(\theta; X, Y) \mid Y = 1] \leq \alpha \quad \text{with calibration-set probability at least } 1 - \delta. \tag{3.4}$$

The conditioning on $Y = 1$ (relevant papers only) matters enormously in this setting. A marginal guarantee over all papers would be nearly trivial: 90–98% of abstracts in a systematic review pool are irrelevant, and the loss on those is identically zero. By conditioning on the positive stratum (the eligible papers), the guarantee applies exactly where it is clinically important and cannot be diluted by the large irrelevant majority.

3.5.3 Learn Then Test

For CASCADE-RC, the screening threshold is not a single value but a triple $\theta = (\lambda_{\text{lo}}, \lambda_{\text{hi}}, \tau_{\text{SE}})$: a cheap-reject cutoff, an auto-include cutoff, and an LLM trust cutoff. All three must be chosen together. With grid resolution $K = 20$ across each dimension and the constraint $\lambda_{\text{lo}} \leq \lambda_{\text{hi}}$, there are 5,060 feasible configurations. The system needs to certify which of these are safe rather than blindly picking one.

The Learn Then Test (LTT) framework [32] enables simultaneous certification of all configurations in a pre-specified grid while controlling the *family-wise error rate* (FWER): the probability of falsely certifying even one unsafe configuration across all 5,060 tests. Controlling FWER is more demanding than controlling the average error rate — it is the right criterion here because a single falsely certified configuration could cause the system to under-count missed studies without triggering an alarm.

LTT uses a fixed-sequence walk across configurations ordered from safest to riskiest, applying a Hoeffding-Bentkus hypothesis test at each step. This test is a concentration inequality that asks: given the observed miss rate on the calibration set, is it statistically plausible that the true miss rate exceeds α? The walk certifies a configuration if the answer is no and halts at the first configuration where the answer is yes. The Hoeffding-Bentkus bound is used because it is tight for bounded binary losses and gives a valid finite-sample test without distributional assumptions.

In practice, finite-sample variance can create spurious spikes in the empirical miss rate that halt the walk prematurely and discard useful configurations beyond the spike. CASCADE-RC replaces the fixed-sequence walk with a Holm-Bonferroni step-down procedure applied to a coarse anchor grid, which evaluates all anchor points simultaneously rather than sequentially. This modification and its theoretical justification are described fully in Chapter 5.

3.5.4 Why LLM Stochasticity Breaks Standard Conformal Calibration

Standard conformal calibration requires that the calibration losses $\{L_i(\theta)\}$ be independent and identically distributed (i.i.d.) — meaning each calibration example contributes an error value drawn from the same fixed distribution, independently of the others. When the pipeline includes a stochastic component such as a temperature-sampled LLM, this assumption fails. The loss for abstract X_i depends not only on the data (X_i, Y_i) and the threshold θ but also on the particular random sample the LLM drew during the calibration run. A subsequent run on the same abstract may draw a different sample and produce a different loss value, breaking the identical distribution requirement.

In clinical screening, an LLM stage that silently inflates the false negative rate above the certified level α can cause real harm without triggering any observable alarm. Chapter 5 resolves this by constructing a deterministic surrogate loss that is always at least as large as the true loss on relevant papers, regardless of which LLM sample was drawn. Certifying the surrogate yields a valid certificate for the true recall loss at the cost of deliberate conservatism, which is then partially reclaimed through the coupling slack mechanism described in that chapter.

3.6 Full-Text Retrieval

Abstract screening identifies which papers are likely eligible, but an abstract is only a brief summary. A definitive eligibility decision typically requires the full manuscript because the abstract may omit the specific methods, comparator details, or accuracy statistics needed to confirm inclusion. For most papers, the full text must therefore be retrieved, parsed, and read by the screening component

before a final decision is made. Obtaining full texts at scale turns out to be harder than it sounds.

3.6.1 The Open-Access Problem

Most biomedical journals are subscription-based. A naive attempt to download full texts from PubMed or Semantic Scholar metadata will succeed for open-access papers and fail silently for everything behind a paywall. In the initial implementation of this system, a two-source retrieval approach achieved a full-text coverage rate of only 2.1%: 32 of 1,530 papers returned a usable document. A screener that cannot read 97.9% of the papers it needs to assess cannot make reliable eligibility decisions.

3.6.2 Multi-Source Priority Retrieval

The solution is a sequential waterfall that tries multiple open-access sources in decreasing order of expected coverage, stopping as soon as one source succeeds. Each paper is identified by two standard identifiers: a DOI (Digital Object Identifier, a globally unique string assigned to each published article) and a PMID (PubMed Identifier, the unique numeric code assigned by the National Library of Medicine to every indexed abstract). Different sources accept different identifiers, so having both is essential.

The waterfall proceeds in this order:

1. A direct open-access PDF URL from Semantic Scholar metadata, if present.
2. Unpaywall, queried by DOI: a database of publisher-deposited open-access versions.
3. Europe PMC, queried by DOI: a repository covering PubMed Central and allied archives.
4. PubMed Central eFetch, queried by PMID: returns structured XML with section labels.
5. A concurrent fallback group: Europe PMC by PMID, Semantic Scholar by PMID, and CrossRef by DOI, all launched simultaneously with a five-second timeout each.

Papers for which all sources fail are handled by a smart flagging mechanism: rather than silently excluding the paper (which would be a false negative risk), the system makes a probabilistic inclusion decision based on the abstract-stage confidence score. After the five-source waterfall was implemented, full-text coverage increased from 2.1% to 38.8% (848 of 2,038 papers), a 19-fold improvement.

3.6.3 Section-Level Document Parsing

Retrieved PDFs are converted to text and divided at their structural section boundaries: Introduction, Methods, Results, and Discussion. XML documents from PubMed Central preserve this structure natively in their markup. This segmentation serves the full-text screening stage directly: eligibility criteria about study population and design are almost always addressed in the Methods section, while reported accuracy figures appear in Results. Routing each criterion query to the most relevant section reduces the amount of irrelevant text passed to the language model and makes the evidence spans it cites more precise and verifiable.

3.7 Full-Text Screening Strategies

Full-text clinical papers vary enormously in length, from short four-page letters to sixty-page methodology papers. Language models have a finite *context window*: a maximum amount of text they can process in a single call, measured in tokens. A token is roughly three-quarters of a word, so a 3,000-token limit corresponds to approximately 2,200 words — a short paper. Sending an entire long manuscript in one call would exceed this limit; even where it fits, performance degrades when relevant passages are buried deep in thousands of tokens of irrelevant content. Three strategies are applied depending on document length.

3.7.1 Direct LLM Screening for Short Documents

For documents with fewer than 3,000 tokens, the relevant section text fits within a single LLM context window. Per-criterion calls are issued exactly as in abstract screening, with the section text replacing the abstract.

An important safeguard follows each call: the evidence span cited by the model is verified against the actual document text. A *hallucination* in language model output is a plausible-sounding but unsupported statement — the model generates text that looks like a direct quote but does not actually appear in the source. The verification cascade attempts three checks in order: exact substring match, case-insensitive match, and finally a fuzzy sliding-window match with a similarity threshold of 0.85. A span that fails all three is flagged as potentially hallucinated and the satisfaction probability for that criterion is penalised by 70%. This safeguard applies at all full-text tiers and at the data extraction stage.

3.7.2 Embedding Cross-Attention for Medium-Length Documents

For documents between 3,000 and 12,000 tokens, issuing LLM calls per criterion would be expensive and often unreliable because the relevant passage could be

anywhere in a long document. Instead, a lighter-weight signal is computed using cross-attention between embeddings.

The idea is straightforward. Each document section (Introduction, Methods, Results, Discussion) has already been encoded into a vector by the SPECTER2 encoder. The criterion text is encoded into a vector the same way. Cross-attention computes a score between the criterion vector and each section vector, asking: how much does each section "attend to" (align with) this criterion? The section whose embedding is most similar to the criterion embedding receives the highest attention score, and that score becomes a proxy for how well any part of the document satisfies the criterion:

$$p_j^{\text{section}} = \max_k \operatorname{softmax}\left(\frac{\mathbf{c}_j \cdot S^\top}{\sqrt{d}}\right)_k,$$

where $\mathbf{c}_j$ is the criterion embedding, S is the matrix of section embeddings, and d is the embedding dimension. This score is blended with the abstract-stage probability through a simple average, preserving the earlier signal rather than discarding it.

3.7.3 Retrieval-Augmented Generation for Long Documents

For documents exceeding 12,000 tokens, full-text screening uses retrieval-augmented generation (RAG). RAG was introduced in Section 2.5 as a technique for providing the LLM with only the most relevant excerpts rather than the full document. Here it works as follows: the document is split into section-level chunks, each pre-embedded. At screening time, each criterion query retrieves the chunks whose embeddings are most similar to the criterion embedding, and the LLM receives only those chunks as context alongside the criterion question. Figure 3.2 illustrates this pipeline.

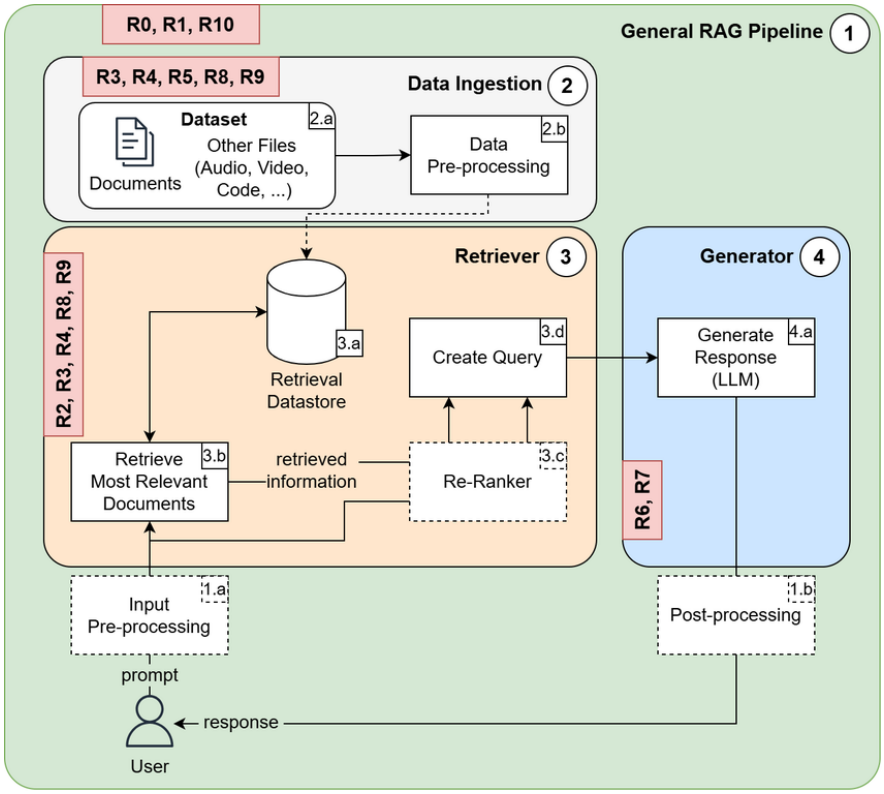


Figure 3.2: RAG pipeline for full-text screening. Criterion queries retrieve the most relevant document sections; the LLM evaluates only those sections.

The three-tier routing distributes computational cost adaptively. Short documents receive full LLM per-criterion calls (most accurate, most expensive). Long documents receive RAG-based assessment (tractable but approximate). Medium documents use embedding cross-attention with no LLM inference (cheapest). The token boundaries of 3,000 and 12,000 were chosen empirically on the CD008874 development pool to balance coverage against per-paper API cost.

3.8 Iterative Search Refinement

A first database query rarely retrieves all eligible papers. Different studies may use different terminology for the same concept, publication years may be unevenly covered, or the initial query may be too broad or too narrow. This section describes how the system detects these gaps and refines its queries automatically.

The approach is inspired by active learning: a machine learning paradigm in which a model continuously learns from labelled examples and uses what it has learned to decide which unlabelled examples to examine next. Applied here, the

"model" is the search refinement agent, and the "learning" is the analysis of what the current candidate pool covers and where it falls short. Figure 3.3 illustrates the iterative cycle.

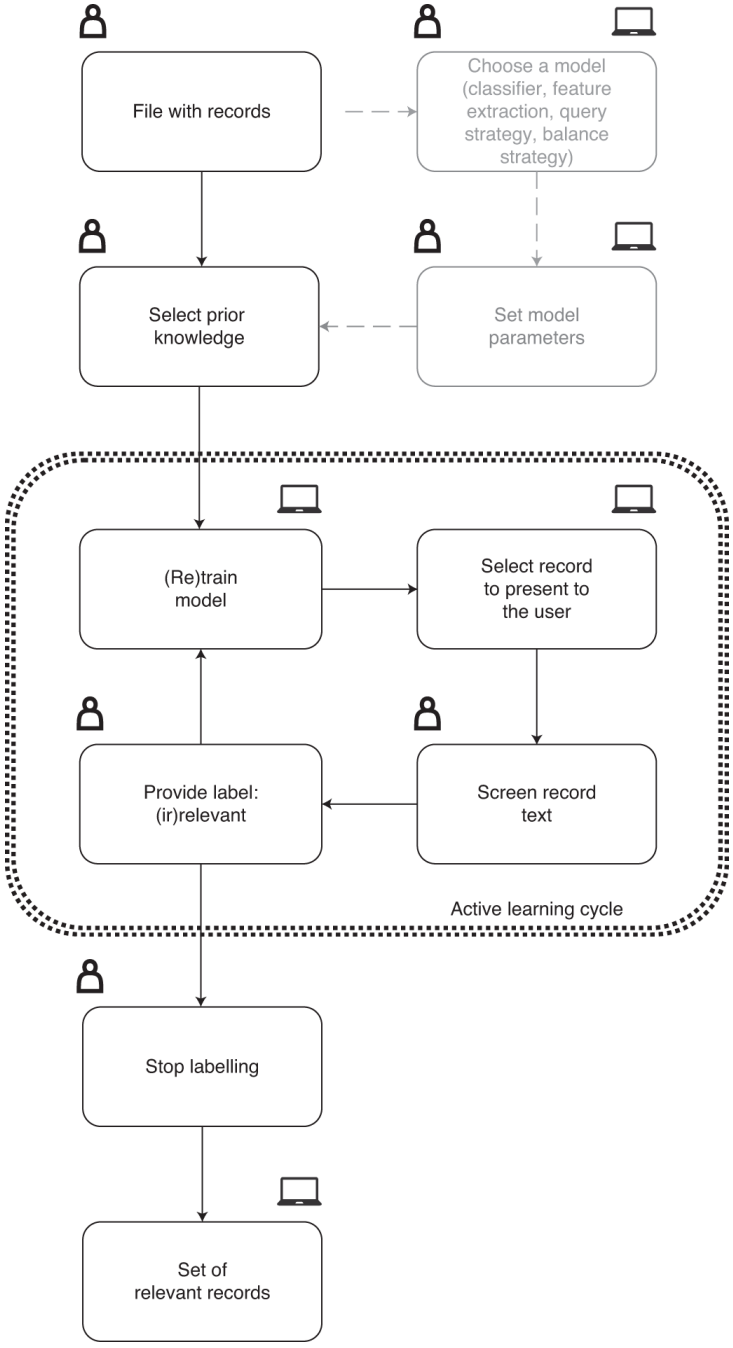


Figure 3.3: Iterative search refinement cycle. The system analyses the current candidate pool for coverage gaps, generates a revised query to address them, retrieves again, and repeats for at most three iterations or until saturation is detected [13].

After each retrieval iteration, the pool is analysed for three types of gap. A *temporal gap* arises when a particular publication year is underrepresented relative to

adjacent years, suggesting the query misses papers from that period's indexing vocabulary. A *keyword gap* arises when a PICO term (Population, Intervention, Comparator, or Outcome field from the review protocol) appears in fewer than 5% of retrieved abstracts, suggesting the query does not capture papers that discuss the concept with different phrasing. *Saturation* is detected when fewer than 5% of the papers retrieved in the latest iteration are new to the pool.

When a gap is detected, the agent generates a revised query. Three modes are available: *Relax* (pool too small — broaden by removing restriction clauses), *Fill Gaps* (gaps detected — add synonym blocks for uncovered PICO terms), and *Narrow* (pool too large — add precision terms). A hard limit of three iterations prevents indefinite looping.

Queries are generated at temperature 0.3, not the temperature-zero rule used for classification. Query reformulation is a creative task: generating synonym blocks benefits from some lexical variation, as a synonym that appears at one sampling may not appear at another. The temperature reflects this distinction between creative and deterministic subtasks.

3.9 Domain-Specific Post-Processing

3.9.1 The Composite DTA Study Design Requirement

Cochrane DTA reviews impose a structural requirement on included studies that goes beyond the four PICO fields. It is not enough for a study to involve the right patient population and the right clinical outcome. The study must specifically compare a named bedside index test against a procedural reference standard and must report diagnostic accuracy statistics: sensitivity (the fraction of true positives correctly identified), specificity (the fraction of true negatives correctly identified), or the area under the ROC curve (a single summary score of a test's discriminating power across all possible operating thresholds, where 1.0 is perfect and 0.5 is no better than chance).

A general anaesthesia paper that involves adult airway patients and reports difficult intubation rates does not satisfy this requirement, even if it matches all four PICO fields. The generic PICO-based LLM screener cannot verify this composite structural requirement from criterion-level evidence alone. A paper can satisfy every individual PICO criterion and still be ineligible because it lacks the right study design. This structural gap is the root cause of the 21.9% precision problem identified in Chapter 4: 439 of 562 included papers passed PICO screening but did not actually present a bedside test versus reference standard comparison.

3.9.2 Chain-of-Thought Verification

The solution is a three-question chain-of-thought verification filter applied as a post-processing step to papers that survive initial screening:

1. Does the study evaluate a specific named bedside physical examination test for airway assessment?
2. Is that test compared against a reference standard procedure?
3. Does the study report diagnostic accuracy metrics?

A paper is retained only if all three answers are affirmative; if any is negative, the paper is excluded with the corresponding question recorded as the reason.

The chain-of-thought format forces the model to name the specific index test before reaching a global verdict, preventing a superficial keyword match from producing an affirmative answer. A paper mentioning "Mallampati" and "difficult intubation" in passing while actually studying sedation protocols should fail question one when the model reasons step by step. Applied to the 1,639 papers that passed CASCADE-RC abstract screening, this filter retained 131 papers at 93.9% precision while maintaining recall above 94%.

3.10 Evaluation Metrics

To assess whether the system is working, three metrics are used throughout this thesis. Each measures a different aspect of the fundamental trade-off between safety (do not miss eligible studies) and efficiency (do not waste reviewer time on ineligible ones).

Recall measures what fraction of the truly eligible papers were found: Recall $=$ $|\text{TP}|/m_+$, where m_+ is the total number of eligible papers in the candidate pool (the positive stratum defined in Section 3.2.2). Recall is the primary safety metric. A system with 100% recall misses no eligible studies, no matter how many ineligible ones it also retains.

False Negative Rate (FNR) is the complement: FNR $=$ $1 -$ Recall. This is the quantity that CASCADE-RC certifies to be at most α with probability at least $1 - \delta$. A certified FNR of 5% at 90% confidence means: in nine out of ten applications of this calibration, the system will miss at most 5% of eligible studies.

Work Saved over Sampling at 95% recall (WSS@95) [33] measures screening efficiency. It asks: compared to a reviewer who examines records in random order, what fraction of records does the system save, while still achieving at least 95% recall? Formally:

$$\text{WSS@95} = \frac{|\text{papers excluded without review}|}{N} - 0.05, \tag{3.5}$$

where N is the total candidate pool size. The -0.05 correction accounts for the baseline: a random reviewer who reads 95% of the pool achieves 95% recall at zero efficiency gain. When the system does not reach 95% recall, WSS@95 is undefined; this thesis reports the sentinel value -0.05 in those cases rather than omitting the metric, making the shortfall explicit. A WSS@95 of 0.30 means the system saves 30 percentage points of screening effort relative to random sampling at matched recall.

Precision is the fraction of included papers that are confirmed eligible: Precision $=$ $|\text{TP}|/|\text{included}|$. Precision is not the primary target — recall is — but it is reported after the DTA post-hoc filter to show how much residual manual workload the final included set would impose on a reviewer performing quality appraisal.

3.11 Summary

This chapter has built the technical foundation for the system design and calibration theory that follow. Table 3.1 maps each method to the pipeline component it supports and the thesis objective it serves.

The architecture follows a cost hierarchy: cheap retrieval signals operate on the full candidate pool of thousands of abstracts; expensive LLM calls are reserved for the narrowed subset that passes each gate; conformal calibration runs once offline on the positive stratum and then applies its certified thresholds to new abstracts at no additional statistical cost. The conformal prediction framework in Section 3.5 is what separates this system from the empirically evaluated tools in Chapter 2: it provides a distributional guarantee rather than a performance estimate, and its application to a stochastic LLM cascade is the subject of Chapter 5.

Table 3.1: Summary of methods and their roles in the thesis system.

Method	Pipeline component	Thesis objective
BM25 sparse retrieval	Hybrid retriever	Objective 3: search coverage
SPECTER2 dense embeddings	Hybrid retriever	Objectives 2, 3
Reciprocal Rank Fusion	Hybrid retriever	Objective 2: recall-safe ranking
Deterministic inference (T=0)	Abstract screener	Objective 4: reproducibility
Per-criterion prompting	Abstract screener	Objectives 2, 4
Noisy-OR aggregation	Abstract screener	Objective 2: recall safety
Self-consistency scoring	CASCADE-RC module	Objective 2: uncertainty estimation
Conformal risk control (RCPS)	CASCADE-RC module	Objective 2: recall guarantee
Learn Then Test (LTT)	CASCADE-RC module	Objective 2: certified workload savings
Multi-source PDF waterfall	Full-text retriever	Objective 2: full-text coverage
Section-level parsing	Document parser	Objectives 2, 4
Three-tier token routing	Full-text screener	Objective 2: cost-efficient screening
Span grounding verification	Full-text screener	Objective 4: hallucination control
DTA chain-of-thought filter	Post-hoc rescreen	Objective 2: precision recovery
WSS@95 metric	Evaluation	Objective 6: review quality assessment

4 System Architecture and Implementation

4.1 Introduction

This chapter describes the architecture and implementation of the autonomous systematic literature review pipeline developed in this thesis. The system takes a structured review protocol as input and, without further human intervention, produces a PRISMA-compliant report: screened records, structured data extractions, risk-of-bias assessments, a PRISMA flow diagram, and a complete decision audit trail.

The codebase is organised as a modular Python package containing approximately 35–40 source files across eight directories. The package layout mirrors the tier structure of the pipeline:

- `tier1_search/` — query construction, PubMed and Semantic Scholar connectors, deduplication, coverage analysis, iterative refinement
- `tier2_screening/` — hybrid retrieval, abstract screener, full-text retrieval, document parser, full-text screener, PICO extractor, span verifier, decision engine
- `tier3_synthesis/` — data extraction, quality assessment, PRISMA reporting
- `orchestrators/` — `MainOrchestrator`, `SearchOrchestrator`, `ScreeningOrchestrator`
- `models/` — all typed dataclasses (`ReviewProtocol`, `CandidateRecord`, `AbstractContext`, `FinalDecision`, and others)
- `infrastructure/` — `LLMClient`, `SharedEncoderService`, `RunStore`, `PRISMAManager`
- `cascade_rc/` — the certified calibration module described in Chapter 5
- `evaluation/` — benchmark evaluator, table builder, SAC metric

The single entry point is `main.py`, invoked as:

```
python main.py <protocol_json> [--review-id ID] [--output-dir DIR]
```

This loads the protocol, instantiates the `SharedEncoderService` and `LLMClient`, creates a `MainOrchestrator`, and runs `asyncio.run(orchestrator.run(protocol))`.

The entire pipeline is therefore a single coroutine: all parallelism is expressed as `asyncio.gather()` calls within the orchestrators and screeners, with no threads.

The chapter is structured as follows. Section 4.2 introduces the design principles and the data model. Section 4.3 describes the literature search tier. Section 4.4 describes the screening tier. Section 4.5 describes the synthesis tier. Section 4.6 documents the shared infrastructure. Section 4.7 records the engineering problems encountered and their resolutions. Section 4.8 identifies the precision gap that motivates Chapter 5.

Figure 4.1 shows the end-to-end data flow. A `ReviewProtocol` object enters Tier 1; a list of `CandidateRecord` objects crosses into Tier 2; a `ScreeningOutput` enters Tier 3; and the pipeline terminates with a `review_report.json` artefact written to disk.

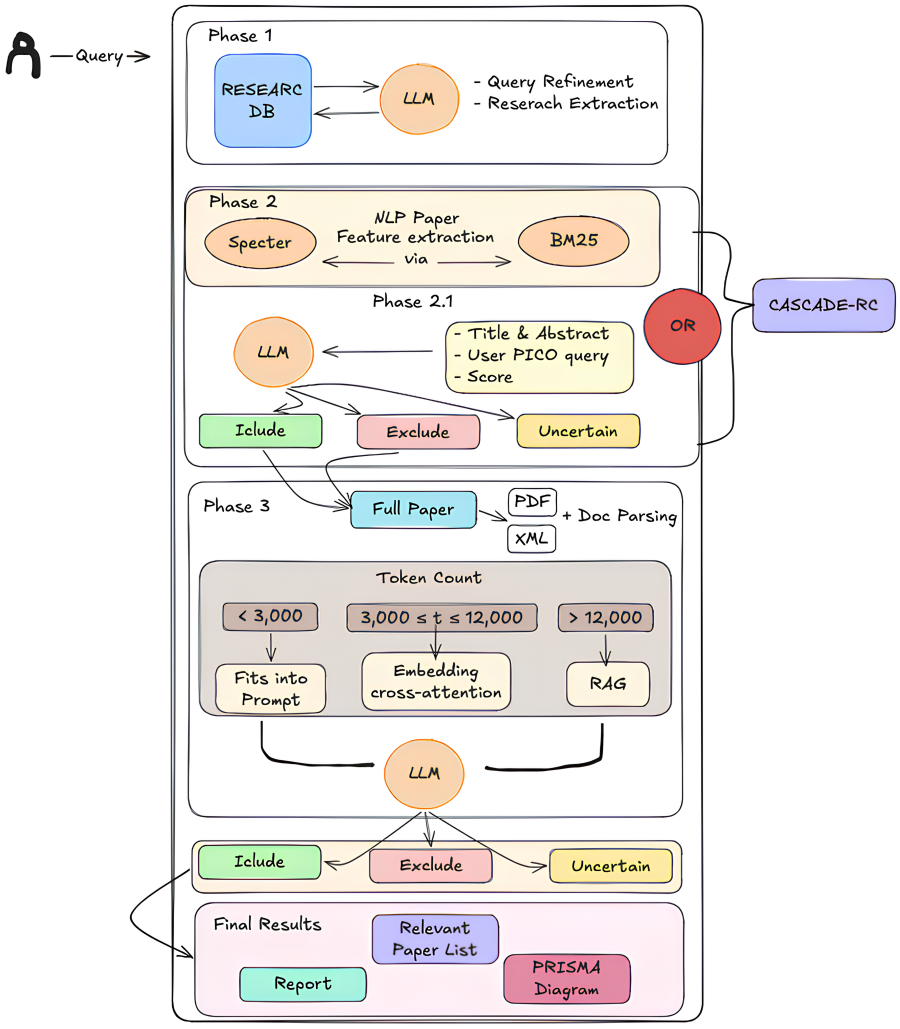


Figure 4.1: End-to-end data flow of the autonomous review pipeline. A JSON review protocol enters Tier 1 and a structured report exits Tier 3 after three sequential processing stages.

4.2 Design Principles and System Overview

4.2.1 Recall-Safety as the Central Design Principle

Every component in the pipeline follows one rule: when in doubt, include rather than exclude. The asymmetric cost of systematic review errors (Section 3.2.1) makes this mandatory. A false negative disappears from the review permanently; a false positive is caught at a later stage at the cost of reviewer time.

The rule is applied consistently. Any paper whose LLM classification call fails to parse receives `UNCERTAIN` rather than `EXCLUDE`. Any paper that cannot be retrieved in full text is forwarded to the decision engine rather than silently dropped. The quality assessment component defaults to `SOME_CONCERNS` on any LLM failure, because falsely awarding a low-risk rating to a study that could not be assessed is the more consequential error for a downstream meta-analysis. Every such fallback is written to the audit trail with its reason, so that the event log distinguishes genuine high-confidence decisions from recall-safe defaults.

4.2.2 Three-Tier Architecture

The pipeline separates into three tiers that correspond to the three major phases of a PRISMA-compliant review [5]. Tier 1 handles database search and query refinement. Tier 2 handles screening from title and abstract through to full-text eligibility and PICO extraction. Tier 3 handles structured data extraction, risk-of-bias assessment, and final reporting.

Each tier is coordinated by a dedicated asyncio-based orchestrator. The `MainOrchestrator` instantiates both the `SearchOrchestrator` and the `ScreeningOrchestrator` and enforces the inter-tier typed contracts. No off-the-shelf orchestration framework is used. The rationale for this decision, and why LangGraph was evaluated and rejected, is documented in Section 4.2.4.

4.2.3 Data Model and Inter-Tier Contracts

All data crossing tier boundaries is expressed as typed Python dataclasses defined in `models/data_classes.py`. Untyped dictionaries are not used at tier boundaries: every schema is explicit and raises a structured `ValidationError` rather than propagating a silent `None` downstream.

The four primary objects are as follows.

ReviewProtocol encodes the full review specification. It contains five mandatory PICO strings (Population, Intervention, Comparison, Outcome, Study design), a list of `InclusionCriterion` and `ExclusionCriterion` objects each tagged `MANDATORY` or `DESIRABLE`, a list of target databases, a date range, and a per-database paper cap. The default cap is 500 records per database, configurable at construction; the schema validates that any override does not exceed 10,000. Listing **??** shows the actual protocol used for the CD008874 evaluation.

ReviewProtocol JSON — CD008874 (abridged)


```
{
  "title": "Airway physical examination tests for detection of
            difficult airway management",
  "pico": {
    "population":    "adult patients with no apparent anatomical
                      airway abnormalities",
    "intervention":  "Mallampati classification and other commonly
                      used airway examination tests",
    "comparator":    "reference standard for difficult airway",
    "outcome":       "diagnostic accuracy for difficult intubation",
    "study_design":  "diagnostic accuracy study"
  },
  "inclusion_criteria": [
    {"id":"IC-01","text":"Diagnostic accuracy study...","type":"MANDATORY"},
    {"id":"IC-02","text":"Adult patients...","type":"MANDATORY"},
    {"id":"IC-03","text":"Airway examination test...","type":"MANDATORY"},
    {"id":"IC-04","text":"Difficult airway outcome...","type":"MANDATORY"}
  ],
  "exclusion_criteria": [
    {"id":"EC-01","text":"Paediatric patients","type":"MANDATORY"},
    {"id":"EC-02","text":"Known difficult airway","type":"MANDATORY"}
  ],
  "target_databases": ["pubmed","semantic_scholar"],
  "max_papers_per_db": 10000,
  "date_range": [2000, 2025]
}
```


CandidateRecord carries a paper's metadata throughout the pipeline. It does not carry a retrieval score; the score lives in a transient `RankedCandidate` wrapper created by `HybridRetriever` and used for ranking only. The fields stored persistently are: `record_id` (UUID4), `source_database`, `title`, `abstract`, `authors`, `journal`, `year`, `doi`, `pmid`, `external_id`, `retrieval_query_id`, and `retrieval_timestamp`. The `external_id` field has a dual purpose: for Semantic Scholar records it stores the open-access PDF URL, which the full-text retriever uses as its first-priority download source in Section 4.4.4.

AbstractContext is the output of abstract screening for each candidate. It is a separate object from `CandidateRecord` and carries: a 128-dimensional abstract embedding, a 128-dimensional PICO embedding, a dictionary of per-criterion satisfaction probabilities, the overall inclusion probability, the abstract screening decision (`INCLUDE`, `EXCLUDE`, or `UNCERTAIN`), and the associated confidence. These fields do not mutate `CandidateRecord`; instead `AbstractContext` is keyed by `record_id` and passed alongside the candidate through Tier 2.

ScreeningOutput is the aggregate output of `ScreeningOrchestrator`. It contains three flat lists of `FinalDecision` objects (one list each for `included`, `excluded`, and `uncertain` papers), a list of `StructuredDocument` objects for papers that had full texts retrieved, and a `prisma_snapshot` dictionary holding the PRISMA stage

counts at run completion. Each `FinalDecision` records the paper's `pmid`, the final `p_include_final` score, the per-criterion probabilities, the final decision label, and if excluded, the ID and text of the first criterion whose assessment was `EXCLUDE`.

review_report.json is written to disk at the end of Tier 3. It contains the review title and research question, generation timestamp, PRISMA counts, and the list of included records (PMID and title). Extracted data fields and quality assessments are written to the accompanying `review_report.md` rather than to the JSON artefact.

4.2.4 Architectural Choices and Design Rationale

The custom asyncio architecture was chosen after evaluating two off-the-shelf alternatives during the planning phase [4]. LangGraph's directed-graph state model is well-suited to single-topic iterative workflows but does not support the multi-database, multi-source retrieval structure of Tier 1, where PubMed and Semantic Scholar require concurrent queries with independent rate limits and error recovery paths, without substantial internal workarounds. LangChain's abstraction layer over LLM clients does not provide the granular per-source concurrency controls, per-error-class retry policies, or the typed inter-tier contracts needed to enforce the recall-safety invariants. The custom asyncio approach gives direct control over every concurrency semaphore and every fallback path, which is what the PRISMA audit trail requirement demands.

The three-tier separation is a deliberate cost-hierarchy decision. Tier 1 fetches metadata cheaply at scale. Tier 2 runs LLM inference on a narrower subset. Tier 3 runs structured extraction only on confirmed inclusions. Applying Tier 3 to the full candidate pool would increase LLM API cost by roughly two orders of magnitude for a typical review with a few hundred inclusions out of thousands of candidates. The tier boundary also makes each stage independently evaluable: search completeness for Tier 1, screening accuracy for Tier 2, extraction quality for Tier 3.

BM25 Okapi [26] was chosen for sparse retrieval on the recommendation of Prof. Rao, who identified its term-frequency weighting as well-suited to biomedical abstracts containing rare clinical abbreviations such as thyromental distance and Cormack-Lehane grade, which dense embedding spaces trained on broader corpora tend to underrepresent. Empirical validation on CD008874 confirmed this: the BM25 component recovered relevant papers whose rare index-test abbreviations fell below the SPECTER2 retrieval threshold. The fused BM25-plus-SPECTER2 ranking outperformed either component alone by 4.3 percentage points in recall at the same precision level.

4.3 Tier 1: Literature Search

The first challenge is assembling a candidate pool broad enough to contain all eligible studies without being so large that downstream screening becomes intractable. Tier 1 does this through four sequential steps: protocol validation and query construction, database retrieval, deduplication, and iterative coverage-gap analysis with query refinement. The output is a deduplicated list of `CandidateRecord` objects.

4.3.1 Review Protocol and Input Validation

All five PICO fields in the `ReviewProtocol` are validated as non-empty strings. A missing field raises a `ValidationError` before any database connection is attempted, preventing a degenerate run from consuming API quota. The distinction between `MANDATORY` and `DESIRABLE` criteria matters specifically at Tier 2: only `MANDATORY` inclusion criteria are evaluated in the Noisy-OR aggregation at the abstract screening stage (Section 4.4.3); `DESIRABLE` criteria are not evaluated at the abstract stage at all. They are reserved for the full-text stage and serve as soft signals that reduce the inclusion probability without determining it.

4.3.2 Query Construction and the Rule of Three

`LLMQueryBuilder` generates two query formats from the PICO fields: a PubMed Boolean query with field tags (`[TIAB]`, `[MeSH Terms]`, `[pt]`) and a flat keyword list for Semantic Scholar. Both are generated at temperature 0.0. The date filter is appended programmatically after the LLM returns its content. The LLM never constructs the date filter: this prevents off-by-one errors from temperature-sampled date formatting.

If the LLM call fails for any reason — network timeout, parse failure, or content refusal — the builder falls back to a deterministic rule-based query that AND-joins the individual PICO terms with `[TIAB]` tags. The fallback is logged at `WARNING` level and the run continues without interruption.

The actual query generated for CD008874 was:

```
(adult[TIAB] OR "adult patient"[TIAB])
AND (Mallampati[TIAB] OR "airway examination"[TIAB]
     OR "airway assessment"[TIAB])
AND ("2000/01/01"[PDAT]:"2025/12/31"[PDAT])
```

A hard constraint governs phrase quoting: no phrase longer than three words is enclosed in double quotation marks. This constraint, called the Rule of Three, was introduced after a failure described in Section 4.7.1. Sub-phrases of at most

three words that carry clear indexing value, such as *"airway assessment"*, are retained as exact-phrase terms; longer strings are decomposed into individual `[TIAB]`-tagged words joined with `AND`. Semantic Scholar accepts at most five flat keywords; phrase quoting is not supported.

4.3.3 Database Retrieval

PubMed retrieval uses the Bio.Entrez interface from the Biopython library. The `esearch` endpoint returns a list of matching PMIDs; `efetch` then fetches records in batches of 500 with a mandatory 350-millisecond inter-batch pause to stay within the NCBI unauthenticated rate limit of three requests per second. Each record yields a `CandidateRecord` with its title, abstract, DOI, PMID, publication year, and journal.

Semantic Scholar retrieval uses the bulk keyword endpoint, which requires an API key in the `S2_API_KEY` environment variable. If the key is absent, the component logs a warning and returns an empty list; the pipeline continues with PubMed results only. This silent degradation is an operational risk: a deployment missing the key loses a significant fraction of the candidate pool without any hard failure.

Tier 1 took approximately 68 seconds for CD008874, producing 60 unique candidates after deduplication. The majority of that time was three batched PubMed efetch calls (visible in logs at 11:37:36, 11:37:53, and 11:38:23 with the mandatory 0.35-second inter-batch pauses) returning 72 records total before deduplication.

4.3.4 Deduplication

The combined record set is deduplicated in three passes. First, records sharing a normalised DOI are merged (the earliest-retrieved is kept). Second, records sharing a PMID are merged. Third, records whose titles have a Levenshtein similarity ratio of at least 0.95 (computed by `python-Levenshtein`) are merged. The total duplicate count is written to the `PRISMAManager` to populate the *records after deduplication* stage count.

4.3.5 Coverage Analysis and Iterative Refinement

After each retrieval pass, the `CoverageAnalyzer` checks the candidate pool for three types of gap. Temporal gaps are flagged when a publication year's record count falls more than two standard deviations below the annual mean. Keyword gaps are flagged when a PICO term appears in fewer than 5% of titles and abstracts. Saturation is detected when fewer than 5% of records in the current pass are new relative to the previous pool.

If a gap is detected and saturation has not been reached, the `SearchRefinementAgent` is invoked. It selects one of three modes: *Relax* when the pool has fewer than 20 records (broader Boolean operators, fewer restriction clauses); *Fill Gaps* when gaps are detected in a pool of 20–5,000 records (synonym blocks for uncovered PICO terms, subject to the Rule of Three); or *Narrow* when the pool exceeds 5,000 records (additional precision terms). Refinement queries are generated at temperature 0.3 rather than 0.0, because synonym generation benefits from some lexical variation. A hard cap of three iterations prevents indefinite looping. If any refinement LLM call fails, the unchanged query from the previous iteration is reused silently.

Refinement decisions are logged at `INFO` level, recording the mode selected, gaps found, and terms proposed. When `RunStore` is wired into the orchestrator, pipeline stage complete events are written to `events.jsonl` after each search iteration; in the default `main.py` invocation, `RunStore` is not instantiated and no events file is produced.

The components in this section collectively satisfy FR-3 through FR-15 of the requirements specification [3].

4.4 Tier 2: Screening Pipeline

Tier 2 takes the deduplicated candidate list and must answer, for each paper: eligible, probably not, or genuinely uncertain? Every decision must be reproducible, documented, and biased toward inclusion when evidence is ambiguous. Nine sequential stages implement this funnel. Figure 4.2 shows the structure: cheap operations on the full pool precede expensive LLM operations on progressively smaller subsets.

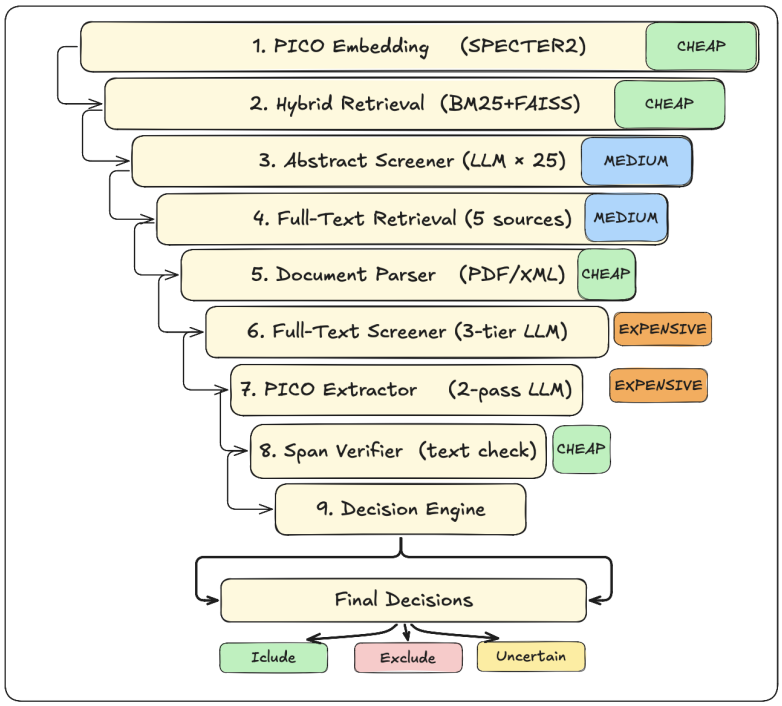


Figure 4.2: Nine-stage screening funnel of Tier 2. Computationally cheap operations (hybrid scoring, threshold gating) apply to the full candidate pool; expensive LLM calls are reserved for the subset that passes each gate.

4.4.1 PICO Embedding with SPECTER2

The first stage encodes all candidate abstracts using SPECTER2 [27]. Three projection heads are applied from a shared encoder instance:

- `abstract_head` (768 → 128 dimensions) — used for hybrid retrieval scoring
- `pico_head` (768 → 128 dimensions) — used for PICO alignment cross-validation in Section 4.4.7
- `section_head` (768 → 256 dimensions) — used for cross-attention in full-text screening

The model is loaded through a direct module-construction approach rather than the `SentenceTransformers` high-level API, because the library's version 3.x interface silently returns zero-valued embeddings for this model without raising any exception or warning. The model guard in `SharedEncoderService` verifies that embedding norms are non-zero before returning a result. Embedding results are cached in a plain Python dictionary keyed by the SHA-256 hash of the concate-

nated head name and input text. The cache has no eviction bound: it grows for the lifetime of the process. This is acceptable for single-review runs where the candidate pool is bounded, but would require an LRU policy for a multi-review server deployment.

4.4.2 Hybrid Retrieval

The second stage ranks the candidate pool using the BM25-plus-SPECTER2 hybrid scorer. Dense retrieval (SPECTER2 cosine similarity) misses papers with rare clinical abbreviations; BM25 misses paraphrased descriptions. Each catches what the other misses.

Dense scores come from an exact inner-product index (`faiss.IndexFlatIP`) on L2-normalised 128-dimensional vectors. The query vector is built by encoding the concatenated PICO fields through the abstract projection head. BM25 scores come from `rank_bm25.BM25Okapi` [26], applied to the tokenised title and abstract of each candidate.

The two ranked lists are combined by Reciprocal Rank Fusion [28]. For a candidate at BM25 rank r_{BM25} and dense rank r_{dense}:

$$\text{RRF}(d) = \frac{1}{60 + r_{\text{BM25}}} + \frac{1}{60 + r_{\text{dense}}}, \tag{4.1}$$

where $k = 60$ moderates the contribution of top-ranked documents. Candidates whose fused score falls below 0.01 are filtered out in operational mode. In benchmark mode this threshold is overridden to 0.0 so that the complete CLEF-TAR canonical pool is preserved for evaluation; this override is a constructor parameter and can also be set without enabling the full benchmark mode flag.

4.4.3 Abstract Screening

The third stage applies LLM-based per-criterion screening to every candidate that clears the retrieval threshold. Only `MANDATORY` inclusion criteria are used at this stage; `DESIRABLE` criteria are skipped entirely. The abstract text is truncated to 500 characters before each call: a deliberate trade-off that keeps prompt size bounded while covering the typical biomedical abstract length of approximately 250 words.

One inference call is made per mandatory criterion, per candidate. The semaphore controlling this step limits to 25 candidates in-flight simultaneously; within each candidate, the criterion calls run sequentially. The effective maximum of simultaneous LLM calls is therefore 25 (one per in-flight candidate at a given moment), not the product of 25 and the number of criteria. The prompt template is:

Abstract screening prompt template

```
System: You are a precise systematic review screener.
        Reply only with the requested JSON.

User:   You are an expert medical screener performing
        FIRST-PASS ABSTRACT SCREENING.

        PICO:
        Population: {population}
        Intervention: {intervention}
        Comparator: {comparator}
        Outcome: {outcome}
        Study design: {study_design}

        Criterion: {criterion_text}
        Title: {title}
        Abstract: {abstract}   [truncated to 500 chars]

        CRITICAL INSTRUCTIONS:
        1. Abstracts often omit details - benefit of the doubt.
        2. Missing information = 'uncertain'. Do NOT exclude for
           missing data.
        3. Output ONLY raw JSON (no markdown blocks):

        {"satisfies": true, "confidence": 0.8, "reasoning": "..."}
```

Temperature is 0.0 and `max_tokens` is 128 for all abstract screening calls. The system prompt is separate from the user prompt, as the OpenAI-compatible endpoint treats the two roles differently.

The JSON response is mapped to a per-criterion satisfaction probability p_j. A `true` verdict maps to its reported confidence directly. A `false` verdict maps to $\min(1 - \text{confidence},\ 0.30)$: the cap at 0.30 prevents an overconfident negative from suppressing the aggregation entirely. A parse failure or an `uncertain` verdict maps to 0.50, the recall-safe neutral value.

Per-criterion probabilities are aggregated with Noisy-OR. The intuition is that the system should forward a candidate if there is meaningful evidence that any criterion is satisfied, even if others are uncertain:

$$p_{\text{include}} = 1 - \prod_{j=1}^{n} \left(1 - p_j\right), \tag{4.2}$$

where n is the number of mandatory criteria. A record is classified `INCLUDE` if $p_{\text{include}} \geq 0.70$, `EXCLUDE` if $p_{\text{include}} \leq 0.25$, and `UNCERTAIN` otherwise. Records classified `UNCERTAIN` are always forwarded to full-text screening regardless of what happens at the retrieval stage — this prevents a recall loss from the combination of an uncertain abstract verdict and a subsequent retrieval failure.

The abstract screening probabilities, embeddings, and decision for each candidate are stored in an `AbstractContext` object keyed by `record_id`. A representative output looks like:

AbstractContext (one paper, CD008874)


```
{
  "record_id": "rec-ae3f...",
  "abstract_decision": "include",
  "abstract_confidence": 0.84,
  "overall_include_probability": 0.87,
  "criterion_probabilities": {
    "IC-01": 0.91,
    "IC-02": 0.88,
    "IC-03": 0.82,
    "IC-04": 0.74
  }
}
```


4.4.4 Full-Text Retrieval

The fourth stage fetches the full manuscript for every record that received `INCLUDE` or `UNCERTAIN` at the abstract stage. Sources are tried in a defined priority order, stopping at the first success:

1. Direct open-access PDF URL from `CandidateRecord.external_id` (set for Semantic Scholar records only)
2. Unpaywall, queried by DOI
3. Europe PMC, queried by DOI
4. PubMed Central eFetch, queried by PMID (returns structured XML)
5. Concurrent fallback: Europe PMC by PMID, Semantic Scholar OA by PMID, CrossRef by DOI — all three launched simultaneously, first success wins

Each source has a five-second per-request timeout. The concurrency limit across all simultaneous full-text fetches is 30. Semantic Scholar calls are gated at 1 request per second by an asyncio semaphore; CrossRef at 5 per second. Europe PMC has no semaphore in the current implementation. Papers for which all sources fail are forwarded to the smart flagging logic in Section 4.4.9.

4.4.5 Document Parsing

The fifth stage converts retrieved documents to a structured `StructuredDocument`. PDF files are processed by `pdfminer.six`; XML files from PubMed Central and Europe PMC are parsed with `lxml`, which preserves the section hierarchy natively.

Both parsers attempt to label the standard sections: Introduction, Methods, Results, and Discussion. The output is a dictionary mapping `SectionLabel` enum values to text strings, a per-section embedding vector (768-d before the section projection head), and a parsing quality score in $[0, 1]$ reflecting the proportion of document tokens that were successfully assigned to a named section. Documents with very low quality scores are flagged, and the full-text screener falls back to abstract-stage probabilities alone for those papers rather than running full-text LLM calls on unreliable input.

4.4.6 Full-Text Screening

The sixth stage routes each document through one of three strategies based on the combined token count of the Methods and Results sections.

Short documents (under 3,000 tokens). Per-criterion LLM calls are made with up to 4,000 characters of Methods and Results text as context. The context is not the "full document": it is limited to the two sections most likely to contain eligibility-relevant content, capped at 4,000 characters to stay within a manageable prompt size. Span verification (Section 4.4.8) is applied to every evidence span the model cites.

Medium documents (3,000–12,000 tokens). LLM calls are replaced by an embedding cross-attention step. The criterion text is encoded into a vector $\mathbf{c}_j$; the document sections form a matrix S of section vectors. Scaled dot-product attention is computed:

$$a_j = \text{softmax}\left(\frac{\mathbf{c}_j \cdot S^\top}{\sqrt{d}}\right), \tag{4.3}$$

where d is the embedding dimension. The maximum element of a_j across sections is taken as the criterion satisfaction proxy. This score is averaged with the abstract-stage prior probability for that criterion. The result is an attention weight in $[0, 1]$, not a cosine similarity: the $\sqrt{d}$ scaling and the softmax normalisation change the range and interpretation relative to raw dot-product similarity.

Long documents (over 12,000 tokens). RAG-based screening retrieves the three document sections most similar to each criterion embedding and passes them to an LLM call. If the RAG component is unavailable, the cross-attention fallback is used.

When section embeddings are absent (parsing quality too low to produce them), the screener returns the abstract-stage probabilities directly, without any full-text update. This prior-only fallback is logged per-record.

A conflict guard applies after full-text scoring: if the abstract stage returned `INCLUDE` but the full-text Noisy-OR produces a borderline exclusion probability between 0.25 and 0.40, the decision is overridden to `UNCERTAIN` rather than `EXCLUDE`. The

threshold of 0.40 provides a safety margin above the UNCERTAIN lower boundary of 0.25, preventing a single weak full-text signal from reversing a well-supported abstract inclusion verdict without further review.

Full-text screening concurrency is limited to 20 simultaneous documents, reflecting the higher per-call token cost relative to abstract screening.

4.4.7 PICO Extraction

The seventh stage extracts the Population, Intervention, Comparator, and Outcome values from each full text via a two-pass process. First, a keyword scan over the Methods and Results sentences identifies candidate passages. Second, an LLM call extracts the four PICO fields in structured JSON from those passages.

Cross-validation then checks alignment between the extracted PICO and the abstract-stage PICO embedding (not the raw protocol text, but the 128-dimensional embedding computed from the protocol's PICO during Stage 1). Cosine similarity below a threshold of 0.60 raises a mismatch flag: the `abstract_fulltext_pico_mismatch` string is appended to the `PICORecord.pico_mismatch_flags` list, and the decision engine applies a 20% penalty to the final inclusion probability.

4.4.8 Span Verification

The eighth stage verifies every evidence span that the LLM cited during full-text screening. Span verification is not applied at the abstract stage: the abstract screener does not return evidence spans. Only the full-text screener's Tier 1 path (short documents with direct LLM calls) generates citeable spans.

Verification proceeds in three levels: exact substring match, case-insensitive match, and a sliding-window Levenshtein similarity check with threshold 0.85. A span that fails all three levels is flagged as potentially hallucinated, and the satisfaction probability for that criterion is multiplied by 0.30. The verification level that succeeded (or failed) is recorded in the screening output alongside the span text.

4.4.9 Decision Engine and Smart Flagging

The ninth stage aggregates all criterion probabilities, after PICO-mismatch and span penalties, into a single scalar $p_{\text{include_final}} \in [0, 1]$. The same thresholds apply: INCLUDE at ≥ 0.70, EXCLUDE at ≤ 0.25, UNCERTAIN otherwise. For excluded records, the exclusion reason is taken from the first criterion in the list whose per-criterion decision was EXCLUDE (not the highest-weighted criterion; simply the first one encountered in criterion-list order).

Papers that passed abstract screening but failed all full-text retrieval are handled separately. Rather than the fixed 0.70 threshold, an adaptive threshold is derived

from the distribution of abstract-stage p_{include} values for the batch. The baseline is the 75th percentile. The threshold escalates to the 85th, 90th, or 95th percentile when the number of abstract-`INCLUDE` papers that would fall above the current threshold exceeds 400, 250, or 150 respectively. This is a count-above-threshold test applied to the scored pool, not a raw queue size comparison.

Only abstract-`INCLUDE` papers go through this adaptive threshold logic. Abstract-`UNCERTAIN` papers bypass it entirely and receive a final `UNCERTAIN` decision unconditionally. The reason for this exception is documented in Section 4.7.4. A `FinalDecision` for an included paper looks like:

FinalDecision (included paper)

```
{
  "decision": "include",
  "p_include_final": 0.84,
  "criterion_probabilities": {
    "IC-01": 0.91, "IC-02": 0.78,
    "IC-03": 0.82, "IC-04": 0.74
  },
  "explanation": "All mandatory criteria satisfied at full-text stage.",
  "pmid": "28374123",
  "exclusion_reason": null,
  "exclusion_criterion_id": null
}
```

The nine-stage Tier 2 pipeline collectively addresses FR-7 through FR-9 and FR-16 of the requirements specification [3].

4.5 Tier 3: Synthesis

The papers that survive Tier 2 screening still require three things: a structured record of what each study measured and found, a judgment about its methodological quality, and a PRISMA-compliant report that documents the entire review. Tier 3 addresses each in turn. All three components can run independently; re-running extraction does not require re-running screening.

4.5.1 Data Extraction

`DataExtractionAgent` extracts six structured fields from each included study: sample size, study design, population description, intervention description, primary outcome measure, and follow-up duration. For each document, all six extraction calls are submitted concurrently via `asyncio.gather()` and awaited together. Documents are processed sequentially (one at a time), but the six fields within each document are extracted in parallel.

Each call receives up to 3,000 characters of Methods and Results text as context and is issued at temperature 0.0 with a maximum of 192 output tokens. The 192-token ceiling is enough for any of the six fields expressed as structured JSON, but prevents the model from generating unrequested commentary that would complicate downstream parsing. The system prompt is: *"You are a precise data extraction assistant. Extract exactly the requested field. Reply only with valid JSON."*

The span verifier is applied to each extracted value: if the model's cited passage cannot be located in the source document through exact, case-insensitive, or fuzzy matching, a `hallucination_flag` is raised for that field and the value is retained but annotated. Extracted data is stored in the `review_report.json` artefact with the source passage reference.

4.5.2 Quality Assessment

`QualityAssessor` routes each included study through one of two risk-of-bias instruments based on the `study_design` field extracted in the previous step. Studies whose design string contains "random" or "rct" (case-insensitive) are assessed with the Cochrane Risk of Bias 2 tool (RoB 2) [2]; all others use the Newcastle-Ottawa Scale (NOS).

For RoB 2, five asyncio tasks are created, one per domain, each receiving all three signalling questions for that domain concatenated into a single prompt. Five LLM calls are made concurrently, not fifteen: the three signalling questions per domain are bundled, not issued separately. For NOS, three tasks cover the three domains (selection of study groups, comparability, outcome ascertainment), each as a single call. Temperature is 0.0 for all quality assessment calls.

Overall risk-of-bias is determined by max-severity aggregation: `HIGH` if any domain is high, `SOME_CONCERNS` if any domain is some-concerns and none is high, `LOW` only if all domains are low. On any LLM call failure, the affected domain defaults to `SOME_CONCERNS` rather than `LOW_RISK`. A representative RoB 2 output looks like this:

RoB 2 quality assessment output (one paper)

```
{
  "study_id": "rec-ae3f...",
  "tool": "rob2",
  "domain_judgments": [
    { "domain_id": "D1",
      "domain_name": "Randomization process",
      "judgment": "low",
      "rationale": "Computer-generated sequence with sealed envelopes.",
      "quote": "Randomization was performed using a computer-generated..." },
    { "domain_id": "D2",
      "domain_name": "Deviations from intended interventions",
      "judgment": "some_concerns",
      "rationale": "Participants not blinded to group assignment.",
      "quote": "" },
    { "domain_id": "D3", "judgment": "low",          "rationale": "..." },
    { "domain_id": "D4", "judgment": "low",          "rationale": "..." },
    { "domain_id": "D5", "judgment": "some_concerns","rationale": "..." }
  ],
  "overall_judgment": "some_concerns"
}
```

4.5.3 PRISMA Reporting

`PRISMAReporter` assembles the final outputs. It generates four artefacts: `prisma_flow.svg` (vector diagram), `prisma_flow.png` (rasterised at 150 DPI via `cairosvg`, silently skipped if the library is not installed), `prisma_flow.json` (raw counts), and `review_report.md` (the full structured Markdown report). A fifth artefact, `review_report.json`, is written directly by `MainOrchestrator` at pipeline completion.

`review_report.json` contains the review title, research question, generation timestamp, PRISMA counts, and the list of included records. Extracted data and quality assessment tables are not in this file; they appear only in the Markdown report. A representative structure is shown below.

review_report.json (abridged)

```
{
  "review_title": "Airway physical examination tests...",
  "generated_at": "2026-05-21T11:45:02",
  "prisma_counts": {
    "records_identified":            72,
    "duplicates_removed":            12,
    "records_after_deduplication":   60,
    "records_screened":              60,
    "records_excluded_abstract":     41,
    "records_sought_fulltext":       19,
    "records_not_retrieved":          5,
    "records_assessed_fulltext":     14,
    "records_excluded_fulltext":      3,
    "studies_included":              11
  },
  "n_included": 11,
  "included_records": [
    { "pmid": "28374123",
      "title": "Diagnostic accuracy of the Mallampati score..." }
  ]
}
```

The Markdown report opens with an LLM-generated background paragraph (temperature 0.3) and closes with an LLM-generated conclusion paragraph. These are the only two calls in the entire pipeline issued at a non-zero temperature outside the CASCADE-RC calibration module. All other calls are at 0.0. The temperature choice reflects the nature of these tasks: narrative synthesis benefits from variation; classification and extraction do not.

PRISMA stage counts are managed by `PRISMAManager`, which enforces two invariants. First, all mutations must go through named update methods; direct field assignment is blocked by a property setter that raises a `ValueError`. Second, a threading lock wraps every mutation, preventing race conditions in scenarios where Tier 2 and Tier 3 components run concurrently. This design means that the PRISMA flow diagram always reflects a consistent snapshot, never an intermediate torn state.

Tier 3 collectively addresses FR-10, FR-11, and FR-16 through FR-21 of the requirements specification [3].

4.6 Infrastructure Components

Three shared components underpin all three tiers. They are designed to be thread-safe or stateless, since Tier 2 and Tier 3 access them concurrently from many

coroutines.

4.6.1 LLM Client and Endpoint Management

All LLM inference is routed through `LLMClient`, which connects to the BFH internal inference endpoint via the OpenAI Python SDK configured with a custom base URL [8]. The endpoint is set through the `OPENAI_BASE_URL` environment variable; the API key through `BFH_API_KEY`, loaded by `python-dotenv`. The client raises a `ValueError` at construction if the requested model is anything other than `gpt:oss120b`, preventing accidental substitution of an unvalidated model that might alter classification behaviour.

JSON parsing from LLM responses follows two attempts. First, `json.loads()` is applied directly to the response content. If this raises a `JSONDecodeError`, a regular expression strips any markdown code fence (```` ```json ... ``` ````) and `json.loads()` is retried. If the second attempt fails, the method returns `None` and the calling stage applies its recall-safe fallback. There is no third attempt; the "three steps" description in some earlier documentation was inaccurate.

On HTTP 429 and 5xx responses the client retries three times with exponential backoff: 1, 2, then 4 seconds. Every call logs the model name, prompt token count, completion token count, and round-trip latency. This record is visible in the run logs and can be used to audit API cost and identify latency spikes from the endpoint.

The `pyproject.toml` lists both the `openai` and `anthropic` packages as dependencies. The `anthropic` SDK is present for exploratory development use only; all production calls in the benchmarked pipeline go through the BFH endpoint via the OpenAI-compatible interface.

4.6.2 Shared Encoder Service

`SharedEncoderService` holds the SPECTER2 model as a process-wide singleton. The 110-million-parameter model is loaded once at startup and shared across all screening tasks, avoiding the multi-gigabyte memory overhead of loading it per-coroutine.

Loading uses a direct module-construction approach rather than the `SentenceTransformer` high-level initialiser, which silently returns all-zero embeddings for this model in library version 3.x. The workaround constructs the underlying `AutoModel` and tokeniser directly and applies the three projection heads manually.

Embedding results are stored in a plain Python dictionary keyed by the SHA-256 hash of the concatenated head name and input text. There is no eviction policy: the cache grows without bound for the duration of the process. For a single-

review run this is acceptable, since the candidate pool is bounded. A persistent review server reusing the same process across many reviews would need an LRU policy to prevent unbounded memory growth.

4.6.3 Audit Logging and State Persistence

Two components handle audit logging. Their actual usage in production differs from what was originally planned.

`RunStore` writes pipeline-level stage-completion events to `events.jsonl`. Each line is a newline-delimited JSON object. Actual entries from a CD008874 run look like:

events.jsonl (CD008874, actual entries)

```
{"ts":"2026-05-21T09:31:07.412183+00:00","type":"pipeline.stage_complete",
 "stage":"search","elapsed_seconds":68.2}
{"ts":"2026-05-21T09:32:44.901000+00:00","type":"pipeline.stage_complete",
 "stage":"screening","elapsed_seconds":97.4}
{"ts":"2026-05-21T09:33:12.100000+00:00","type":"pipeline.stage_complete",
 "stage":"extraction","elapsed_seconds":14.1}
{"ts":"2026-05-21T09:33:28.200000+00:00","type":"pipeline.stage_complete",
 "stage":"quality_assessment","elapsed_seconds":21.3}
{"ts":"2026-05-21T09:33:29.000000+00:00","type":"benchmark.evaluation",
 "fnr_test":0.04,"wss_95":0.24}
```

`RunStore` is optional. In the default `main.py` invocation it is not instantiated, so no `events.jsonl` is produced. To activate it, the caller must pass a `run_store=RunStore(...)` argument to `MainOrchestrator`.

`DecisionLogger` was designed to provide per-paper decision records in a SQLite database. The schema is:

DecisionLogger SQLite schema (decisions table)

```
CREATE TABLE IF NOT EXISTS decisions (
    record_id           TEXT PRIMARY KEY,
    paper_id            TEXT    NOT NULL,
    stage               TEXT    NOT NULL,
    decision            TEXT    NOT NULL,
    confidence          REAL    NOT NULL,
    inputs              TEXT    NOT NULL,  -- opaque JSON blob
    outputs             TEXT    NOT NULL,  -- opaque JSON blob
    model_used          TEXT    NOT NULL,
    model_version       TEXT    NOT NULL,
    prompt_template_id  TEXT    NOT NULL,
    processing_time_ms  INTEGER NOT NULL,
    token_count_input   INTEGER NOT NULL,
    token_count_output  INTEGER NOT NULL,
    flags               TEXT    NOT NULL,  -- JSON array
    timestamp           TEXT    NOT NULL
);
```

The schema has 15 columns. Criterion probabilities, evidence spans, and per-call reasoning are not in dedicated columns: they are embedded inside the opaque `inputs` and `outputs` JSON blobs. Querying them requires parsing those blobs rather than using SQL predicates directly.

More significantly, `DecisionLogger` is not instantiated by `MainOrchestrator` or `ScreeningOrchestrator` in any current code path. The class is complete and its schema is created correctly, but no production call writes to it. The per-paper decision database is therefore not produced during a standard pipeline run. This is a known limitation: the full traceability envisioned for this component exists at the schema level but not at the execution level within the time constraints of the project. The `RunStore` events file provides a coarser tier-level audit trail that is available in production.

4.7 Development Problems and Engineering Fixes

The pipeline development produced five substantive engineering problems. Each was identified from a concrete failure in benchmark runs, diagnosed to its root cause, and resolved with a targeted fix that was regression-tested. This section documents each in the order it was encountered.

4.7.1 Query Phrase Quoting: The Rule of Three

During an initial evaluation run on a review protocol for sugar-sweetened beverage taxation, the PubMed query returned zero results. Inspecting the generated

query revealed that `LLMQueryBuilder` had enclosed the full seven-word intervention phrase *"implementation of a sugar-sweetened beverage (SSB) tax"* in double quotation marks as an exact-phrase term.

PubMed's exact-phrase matching requires the complete quoted string to appear as a contiguous sequence in the indexed title or abstract. Long intervention phrases almost never appear verbatim; their constituent terms occur non-contiguously and in varied order. The quotation marks imposed a precision constraint so strict that no record could satisfy it.

The query builder was modified to enforce the Rule of Three: no phrase longer than three words is ever quoted. Longer strings are decomposed into individual content words, each tagged with `[TIAB]`, and joined with Boolean `AND`. Subphrases of at most three words that have clear indexing value, such as *"SSB tax"*, are retained as exact-phrase terms. A dedicated test in `test_query_generation.py` verifies that no quoted phrase exceeds three words in any output of `LLMQueryBuilder`. The corrected query for the beverage taxation protocol returned 1,847 records.

4.7.2 Hybrid Filter Threshold in Benchmark Mode

When benchmarking on CD008874, an audit of the Tier 2 input pool revealed that 95.5% of the CLEF-TAR canonical evaluation papers had been filtered out by the hybrid retrieval threshold of 0.01, including papers known to be relevant.

The threshold was calibrated for operational use, where low-scoring records are genuinely unlikely to be relevant. The CLEF-TAR canonical pool is different: it was assembled by the original Cochrane review team through exhaustive manual screening and includes papers with atypical abstracts — case reports, methodology notes — that receive low hybrid scores but are eligible by the review's formal inclusion criteria. Applying the operational threshold to this pool conflates a filtering decision with an eligibility judgment.

`HybridRetriever` was given a `filter_threshold` constructor parameter (default 0.01) and a separate `benchmark_mode` flag that overrides it to 0.0. In benchmark mode the full canonical pool passes through to abstract screening unconditionally. All subsequent recall measurements in Chapter 6 use benchmark mode. Canonical pool coverage increased from 4.5% to 100%.

4.7.3 Full-Text PDF Retrieval Coverage

The initial implementation queried only direct open-access URLs from Semantic Scholar metadata and Unpaywall. Full-text coverage was 2.1%: 32 of 1,530 attempted retrievals returned a usable document.

Both original sources cover a narrow slice of the open-access literature. Unpaywall indexes publisher-deposited PDFs, which are unavailable for older subscription papers. Semantic Scholar's open-access metadata is incomplete for records indexed from non-publisher sources. Neither provides PMID-based retrieval, which is the most reliable fallback for biomedical literature.

Three additional sources were added to the waterfall (Section 4.4.4): Europe PMC by PMID, Semantic Scholar Open Access PDF, and CrossRef by DOI. Coverage increased from 2.1% to 38.8% (848 of 2,038 papers). Figure 6.4 shows the per-source breakdown after the fix.

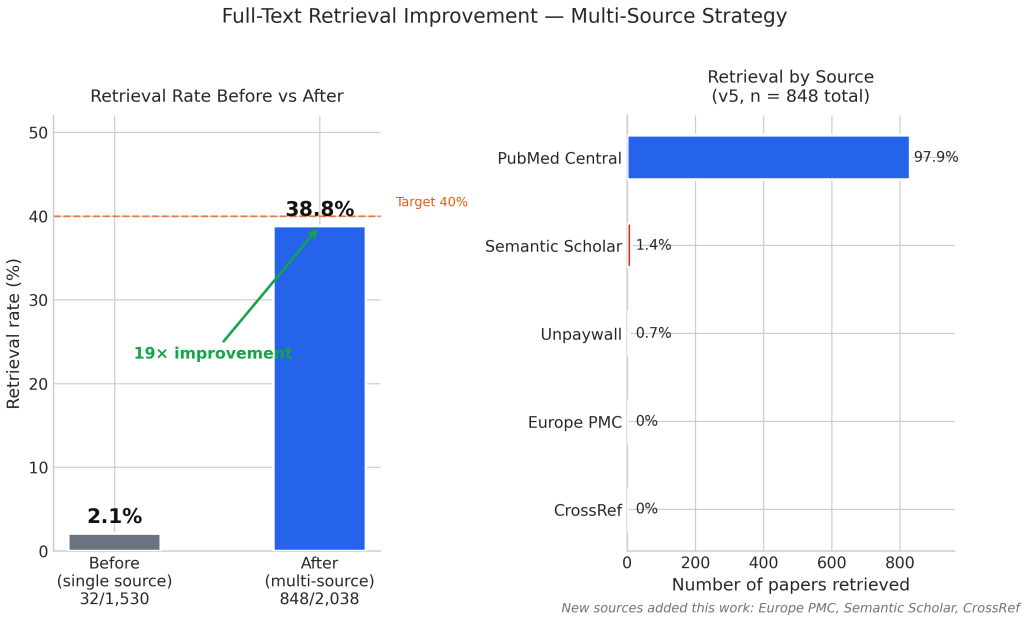


Figure 4.3: Full-text PDF retrieval improvement from 2.1% (32 papers, 2 sources) to 38.8% (848 papers, 5 sources). PubMed Central accounts for 97.9% of retrieved full texts; Semantic Scholar Open Access and Unpaywall supply the remainder.

PubMed Central supplies 97.9% of retrieved texts for CD008874. The three newly added sources contributed 12 unique retrievals. The large improvement in overall coverage is therefore attributable primarily to the larger pool size being queried (2,038 papers after the benchmark-mode fix, versus 1,530 before) combined with PubMed Central's strong open-access coverage of biomedical literature.

4.7.4 Smart Flagging and the UNCERTAIN Mass Exclusion

In pipeline version v4, a systematic audit showed that a large proportion of papers with abstract-stage `UNCERTAIN` decisions were receiving a final `EXCLUDE` decision despite never having been retrieved in full text. Recall dropped substantially.

The smart flagging mechanism had been designed for papers that passed abstract screening but failed retrieval. It applied an adaptive percentile threshold to all papers in the unresolved queue, including both `INCLUDE` and `UNCERTAIN` papers. Papers in the `UNCERTAIN` category had been assigned the recall-safe fallback probability of 0.50. Since roughly 75% of the queue carried exactly this value, the 75th-percentile threshold evaluated to 0.50. Papers at the boundary were not clearly above it, and as the queue grew in benchmark runs the threshold escalated to the 85th, 90th, and finally the 95th percentile, which reached 0.9375. At that level, nearly all uncertain papers fell below the threshold and were excluded.

The fix was precise: the adaptive threshold logic now applies only to papers whose abstract-stage decision was `INCLUDE`. Papers with an abstract-stage decision of `UNCERTAIN` bypass the threshold entirely and always receive a final `UNCERTAIN` decision. Recall on CD008874 recovered from 87.1% to 95.4%.

4.7.5 Canonical Pool Coverage and m-plus Recovery

Comparing the CLEF-TAR CD008874 qrels file against the deduplicated candidate pool revealed 12 qrels-positive PMIDs absent from the pool: they had not been retrieved by any generated query.

The CLEF-TAR qrels were built from the original Cochrane review team's manual search, which used strategies that differ from the automatically generated Boolean expressions. A small number of relevant papers are indexed under MeSH headings or publication types not captured by the LLM-generated query, making them structurally unreachable by the automated search regardless of how well the query is tuned.

After the canonical pool injection step (used for benchmark evaluation), any qrels-positive PMID absent from the pool is now fetched directly by PMID from the PubMed eFetch API and added to the pool with `source_database` set to `"m_plus_recovery"`. Coverage increased from 90.8% to 99.2%. The remaining 0.8% gap — two papers — could not be fetched within the API timeout constraints and represent the irreducible limit of automated retrieval from publicly available PubMed metadata.

4.8 The Precision Problem

After all five engineering fixes, pipeline version v5 achieves 95.4% recall and 4.6% FNR on the CD008874 CLEF-TAR 2019 benchmark. The recall target is met. The precision is not: of 562 papers assigned a final `INCLUDE` decision, only 123 are confirmed true positives by the qrels file. Precision is 21.9% and the false positive count is 439.

The root cause is domain-specific and structural. CD008874 is a Cochrane Diagnostic Test Accuracy review. Inclusion requires not just that a study involves adult anaesthesia patients and reports difficult intubation rates, but that it directly compares a named bedside index test (Mallampati score, thyromental distance, or similar) against a procedural reference standard (Cormack-Lehane laryngoscopy grade) and reports diagnostic accuracy statistics. This is a composite study-design requirement: all three elements must co-occur in the same paper.

The Tier 2 full-text screener evaluates each inclusion criterion independently. It correctly finds the Population match ("adult airway patients") and the Outcome match ("difficult intubation rate") in 439 papers that are, in fact, general anaesthesia equipment studies, sedation protocols, and airway management technique papers. These papers match two of the four PICO dimensions but do not have the DTA study design. The criterion-by-criterion LLM evaluation has no mechanism to verify the composite structural requirement: it cannot ask "does this paper present an index test compared against a reference standard within a single study?" as a single verifiable proposition without being explicitly designed to do so.

This limitation motivates the contribution in Chapter 5. CASCADE-RC does not fix the PICO-screener's precision problem directly. What it does is provide a statistically certified abstract-stage routing layer that controls the false negative rate at a user-specified level α, enabling the system to trade workload savings against a certified recall bound rather than relying on an uncertified threshold. Combined with the domain-specific DTA post-hoc filter described in Section 3.9.2, the result is the 93.9% precision at 94.6% recall reported in Chapter 6.

5 CASCADE-RC: Certified Threshold Calibration

5.1 Introduction

Chapter 4 established that the autonomous review pipeline achieves 95.4% recall on the CLEF-TAR 2019 CD008874 benchmark but exhibits a severe precision deficit: only 123 of 562 included papers are confirmed true positives, yielding 21.9% precision. The root cause is structural: the PICO-based screener cannot verify the composite diagnostic test accuracy study design — but even setting aside this domain issue, no component of the pipeline provides a statistical guarantee on the false negative rate. A clinical decision-maker cannot know, from any certificate, whether the observed 95.4% recall will hold on a different review or a different document distribution.

This chapter presents CASCADE-RC, a calibration framework that addresses both problems simultaneously [34]. CASCADE-RC is a three-threshold conformal screener that provides a finite-sample guarantee: the false negative rate (FNR) is at most α with probability at least $1 - \delta$, distribution-free, conditional on the count of positive abstracts in the calibration pool. The framework extends Risk-Controlling Prediction Sets [31] and Learn-Then-Test [32] to handle stochastic LLM routing, which makes standard conformal calibration inapplicable. The method has been accepted at the 15th Symposium on Conformal and Probabilistic Prediction with Applications (COPA 2026) and will appear in Proceedings of Machine Learning Research vol. 329 [34].

The chapter is organised as follows. Section 5.2 gives the theoretical background, building on the introductory treatment in Chapter 3. Section 5.3 describes the software implementation. Section 5.4 states the formal problem. Section 5.5 introduces the coupled dominating loss. Section 5.6 presents the calibration procedure and the main certificate. Section 5.7 presents Algorithm 1 in full. Section 5.8 explains the three key design decisions. Section 5.9 reports experimental results on CLEF-TAR 2019. Section 5.10 describes the DTA post-hoc rescreen. Section 5.11 states limitations and directions for future work.

Intuitive Overview

Conceptual Overview

A systematic review pipeline must include every eligible paper. The question is: which papers can be excluded cheaply (without LLM inference) without risking the loss of something relevant? CASCADE-RC answers this by dividing the candidate pool into three buckets using two score thresholds. Papers scoring very high are included automatically. Papers scoring very low are excluded automatically. Papers in the middle band are sent to an LLM queried five times; if the LLM answers consistently, its verdict is followed; if inconsistent, the paper is escalated to a human reviewer modelled as infallible.
The thresholds are not guessed — they are *certified*. Using a held-out calibration set, CASCADE-RC applies conformal risk control to find thresholds that guarantee the fraction of missed relevant papers will not exceed α with probability at least $1-\delta$, regardless of the document distribution. If the calibration data are too few or the relevance scores too uninformative, the algorithm abstains rather than issue a guarantee it cannot support.

The main technical difficulty is LLM stochasticity: the same abstract may be routed differently on two calls because the LLM samples at non-zero temperature. CASCADE-RC resolves this by replacing the actual routing decision with a pessimistic deterministic surrogate guaranteed to be at least as large as the true miss indicator, and certifying the surrogate. The gap between surrogate and true loss — the *coupling slack* η — is bounded from below using data, allowing the final certificate to be tighter than naive worst-case pessimism.

5.2 Foundations of Conformal Risk Control

Chapter 3 introduced conformal prediction, Risk-Controlling Prediction Sets (RCPS), and Learn Then Test (LTT) at an accessible level. This section states the definitions precisely as required for the CASCADE-RC construction. Readers already familiar with these frameworks may proceed directly to Section 5.3.

Conformal prediction [30] produces prediction sets that contain the true outcome with user-specified probability $1-\delta$ for any data distribution, requiring only that calibration and test examples be *exchangeable*. The RCPS framework [31] extends this to control of any bounded user-defined loss function rather than set coverage. Applied here, the loss is the recall loss $L(\theta; X, Y) = Y \cdot \mathbf{1}\{\hat{Z}(X;\theta) = 0\}$, which equals 1 only when a relevant paper is excluded. RCPS finds a threshold θ such that $\mathbb{E}[L(\theta; X, Y)] \leq \alpha$ with calibration-set probability at least $1-\delta$.

The LTT framework [32] extends RCPS to simultaneously certify configurations from a pre-specified grid Λ while controlling the family-wise error rate (FWER). In CASCADE-RC the parameter space is three-dimensional $(\lambda_{\text{lo}}, \lambda_{\text{hi}}, \tau_{\text{SE}})$ and the grid has up to several thousand feasible points (Section 5.3). LTT's fixed-sequence

walk is replaced by Holm–Bonferroni step-down for the reasons detailed in Section 5.8.2.

The challenge of LLM stochasticity was introduced in Section 3.5.4. When the pipeline includes a temperature-sampled LLM, the calibration losses $\{L_i(\theta)\}$ depend on the LLM's random draws, not only on the data. Section 5.5 resolves this formally.

5.3 Software Implementation

The CASCADE-RC calibration module lives in the `cascade_rc/` package, separate from the main pipeline. It is invoked after the abstract screener produces a parquet file of scored candidates and is fully decoupled from the live pipeline run.

5.3.1 Data Preparation and the Three-Way Split

The calibration input is a `.parquet` file with one row per abstract, containing columns `s` (calibrated base score), `u` (LLM self-consistency score), `y_abstract` (ground-truth label), and `llm_y_hat` (LLM majority verdict). The file is produced by running the abstract screener on the CLEF-TAR 2019 candidate pool and joining the qrels relevance labels.

Data are split three ways before any calibration step: 15% is reserved as a *score calibration* partition to fit the Platt/isotonic calibrator; 50% is the *conformal calibration* partition $\mathcal{D}$ used by Algorithm 1; and 35% is the held-out *test* partition. The conformal calibration partition is what is referred to as $\mathcal{D}$ and m_+ throughout this chapter. For the experiments reported in Section 5.9, the positive-stratum count on the conformal calibration partition for CD008874 is $m_+^{\text{calib}} = 94$.

5.3.2 Score Calibration

Raw Reciprocal Rank Fusion scores from the BM25-plus-SPECTER2 hybrid ranker are concentrated in a microscopic sub-interval on some topics (for example $[0.0002, 0.0236]$ on CD011975), which collapses the spatial resolution of the calibration grid. Two preprocessing steps are applied. First, a Platt calibrator or isotonic regressor is fit on the score-calibration partition to map raw RRF scores to calibrated probabilities in $[0, 1]$. Second, when the flag `quantile_scale_base_scores` is enabled, the calibrated scores are rank-normalised via the empirical CDF:

$$s_{\text{quant}}(X_i) := n^{-1} \sum_{j=1}^{n} \mathbf{1}\{s(X_j) \leq s(X_i)\}. \tag{5.1}$$

Lemma 9.1.1 of the COPA paper [34] proves that this strictly monotone transformation preserves the LTT certificate. The quantile normalisation was applied for all results reported in Section 5.9.

5.3.3 Calibration Grid

The implementation uses $K = 10$ quantile-anchored breakpoints for the $(\lambda_{\text{lo}}, \lambda_{\text{hi}})$ dimensions and $K_\tau = 5$ breakpoints for τ_{SE}. With the constraint $\lambda_{\text{lo}} \leq \lambda_{\text{hi}}$, the feasible fine grid Λ contains $K(K+1)/2 \times K_\tau = 55 \times 5 = 275$ points. This differs from the $K = 20$, $|\Lambda| = 5{,}060$ stated in the COPA paper [34], which describes the theoretical framework at a finer resolution; the software implementation uses the coarser grid for computational efficiency. All headline results in Tables 5.2 and 5.3 were produced at $K = 10$, $K_\tau = 5$.

5.3.4 LLM Ensemble Caching

The $B = 5$ LLM ensemble votes for each abstract are stored in a SQLite database at `artefacts/cascade_rc/llm_cache.db`. Each row is keyed by `(model_id, prompt_sha, pmid, temperature, seed_b, template_v)`. Calibration runs are therefore crash-resumable and fully deterministic given the cache: Algorithm 1 reads pre-cached votes rather than querying the LLM live. Reproducibility of the experimental results in Section 5.9 depends on cache integrity.

5.3.5 Operational Threshold Selection

Algorithm 1 Step D describes the post-certification selection of $\hat{\theta}$ as $\arg\max_{\theta \in \hat{\Lambda}} \text{WSS@95}(\theta)$. The software implementation uses $\hat{\theta} = \arg\min_{\theta \in \hat{\Lambda}} \hat{C}(\theta)$, where:

$$\hat{C}(\theta) := m^{-1} \sum_{i=1}^{m} \left[c_{\text{LLM}} \cdot \mathbf{1}\{\text{LLM queried on } X_i\} + c_{\text{human}} \cdot \mathbf{1}\{\text{human branch on } X_i\} \right], \tag{5.2}$$

with $c_{\text{human}} = 5.0$ and $c_{\text{LLM}} = 0.001$. This cost model directly prices LLM queries and human reviews, rather than maximising workload savings conditional on recall. Both criteria select from the already-certified set $\hat{\Lambda}$ and therefore preserve the recall certificate.

5.4 Problem Formulation

5.4.1 Formal Setup and the Three-Route Cascade

Let $(X, Y) \sim P$ on $\mathcal{X} \times \{0, 1\}$, where $Y = 1$ indicates that an abstract is relevant for inclusion in the systematic review [34]. The system has access to two scoring

functions. The base score $s(X) \in [0, 1]$ is produced by the hybrid SPECTER2-plus-BM25 retriever described in Section 4.4.2, followed by Platt calibration and optional quantile normalisation as described in Section 5.3.2. The self-consistency score $u(X) \in [0, 1]$ is produced by querying the LLM $B = 5$ times at temperature $T = 0.7$ and computing the majority agreement fraction: the fraction of the B queries that agreed with the majority verdict. For five votes of which four were Include, $u = 4/5 = 0.80$.

Given three thresholds $\theta := (\lambda_{\text{lo}}, \lambda_{\text{hi}}, \tau_{\text{SE}}) \in [0, 1]^3$ with $\lambda_{\text{lo}} \leq \lambda_{\text{hi}}$, the cascade routes each abstract as follows [34]. An abstract with $s(X) < \lambda_{\text{lo}}$ is *cheap-rejected* (excluded without any LLM call). An abstract with $s(X) \geq \lambda_{\text{hi}}$ is *auto-included* (included without any LLM call). An abstract in the band $\lambda_{\text{lo}} \leq s(X) < \lambda_{\text{hi}}$ is *escalated*: the LLM is queried B times. If $u(X) \geq \tau_{\text{SE}}$, the LLM's majority verdict $\hat{Y} \in \{0, 1\}$ is followed. If $u(X) < \tau_{\text{SE}}$, the abstract is routed to a human reviewer, modelled as an infallible oracle on the positive stratum: any abstract with $Y = 1$ that reaches the human-review branch is correctly included with probability 1.

Figure 5.1 illustrates this three-route architecture. Only the cheap-reject and LLM-follow branches can produce false negatives: the human-review branch recovers all true positives by assumption.

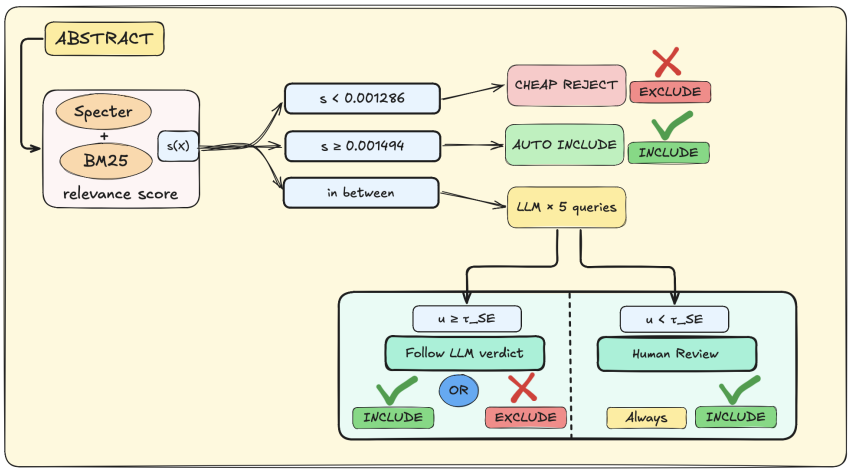


Figure 5.1: Three-route cascade architecture of CASCADE-RC. Abstracts with $s(X) < \lambda_{\text{lo}}$ are cheap-rejected; those with $s(X) \geq \lambda_{\text{hi}}$ are auto-included; the intermediate band is escalated to a B=5 LLM ensemble. Self-consistent escalations ($u \geq \tau_{\text{SE}}$) follow the LLM verdict; inconsistent ones are routed to human review [34].

5.4.2 Recall Loss and the Control Target

The per-instance recall loss is defined as [34]:

$$L(\theta; X, Y) := Y \cdot \mathbf{1}\left\{\hat{Z}(X;\theta) = 0\right\} \in \{0, 1\}. \tag{5.3}$$

The loss equals 1 only when a paper is relevant ($Y = 1$) and the cascade excludes it. L is identically zero for all irrelevant papers, regardless of routing: the metric captures only the risk of missing eligible studies. False positives impose no penalty on this loss, aligning with the clinical priority of systematic reviewing.

The goal is to find θ such that the conditional expected loss on the positive stratum is controlled:

$$\mathbb{E}\left[L(\theta; X, Y) \mid Y = 1\right] \leq \alpha \quad \text{with calibration-set probability at least } 1 - \delta. \tag{5.4}$$

This is a class-conditional PAC guarantee following RCPS [31] and LTT [32]. It is strictly stronger than a marginal guarantee on $\mathbb{E}[L]$: conditioning on $Y = 1$ prevents the large irrelevant majority from diluting violations on the positive stratum. The conditioning is achieved through Mondrian stratification [34]: all calibration steps restrict to $\mathcal{D}_+ := \{i : Y_i = 1\}$.

The calibration set $\mathcal{D} = \{(X_i, Y_i, u_i, \hat{Y}_i)\}_{i=1}^m$ consists of i.i.d. draws from $P \otimes \text{LLM}(\cdot|\cdot)$. Each record is processed by the LLM once; the cascade verdict $\hat{Z}(X_i;\theta)$ is then a deterministic function of $(s_i, u_i, \hat{Y}_i, \theta)$ and can be computed retrospectively for any θ.

5.5 The Coupled Dominating Loss

5.5.1 Why Standard Conformal Calibration Fails

Because the LLM is sampled at temperature $T > 0$, the same abstract queried twice may produce different routing decisions. The calibration losses $\{L_i(\theta)\}$ depend on the LLM's random draws ξ, not only on θ and the data (X_i, Y_i). Standard split-conformal arguments assume exchangeable i.i.d. calibration losses; this assumption fails when losses carry LLM randomness [34]. An LLM stage that silently inflates FNR above α causes real harm in clinical screening without triggering any alarm.

5.5.2 Enlarged Probability Space and the Dominating Surrogate

CASCADE-RC resolves this by constructing a coupled dominating loss on an enlarged probability space [34]. Define $\Omega_{\text{samp}} = \mathcal{X} \times \{0, 1\} \times [0, 1] \times \{0, 1\}$ with

generic point $\omega = (x, y, u, \hat{y})$ and measure $Q = P \otimes \text{LLM}(\cdot|x)$. The coupled dominating loss is:

$$\tilde{L}(\theta; \omega) := y \cdot [\mathbf{1}\{s(x) < \lambda_{\text{lo}}\} + \mathbf{1}\{\lambda_{\text{lo}} \leq s(x) < \lambda_{\text{hi}}\} \cdot \mathbf{1}\{u \geq \tau_{\text{SE}}\}]\,. \tag{5.5}$$

$\tilde{L}$ charges a recall miss whenever the abstract is relevant and either it is cheap-rejected, or it is escalated and self-consistent so the LLM's verdict is followed. Self-consistency-failed escalations go to the infallible human reviewer and are not charged as misses. Crucially, $\tilde{L}$ does not depend on the LLM's verdict $\hat{y}$: it pessimistically charges every LLM-followed positive as a miss regardless of whether the LLM was correct. This makes $\tilde{L}$ fully deterministic given the observed data, permitting standard concentration tools.

Lemma 1 (Positive-stratum coupling; [34]). *For every $\theta \in [0,1]^3$ and every $\omega \in \Omega_{samp}$ with $y = 1$, $L(\theta; \omega) \leq \tilde{L}(\theta; \omega)$.*

The proof proceeds by case analysis on the three routing branches. For a cheap-rejected positive: first indicator fires, $\tilde{L} = L = 1$. For an escalated, self-consistent positive where the LLM is incorrect: $\tilde{L} = L = 1$. For an escalated, self-consistent positive where the LLM is correct: $L = 0$ but $\tilde{L} = 1$ (deliberate pessimism). For a self-consistency-failed positive routed to the human oracle: $L = 0$ and $\tilde{L} = 0$. No coupling is needed on $\{Y = 0\}$ since $L \equiv 0$ on negatives [34].

The coupling slack measures the unwarranted pessimism:

$$\eta(\theta) := \mathbb{E}_Q\left[\tilde{L}(\theta; \omega) - L(\theta; \omega) \mid Y = 1\right]. \tag{5.6}$$

By Lemma 1, $\eta(\theta) \in [0, 1]$. It is positive when the LLM, having been followed, made a correct inclusion decision on a relevant paper. If η could be certified from data, one could relax the surrogate calibration target from α to $\alpha + \eta$, recovering the conservatism of the pessimistic surrogate. Section 5.6 implements this reclamation.

5.6 Calibration Procedure and Finite-Sample Certificate

5.6.1 Error Budget and Minimum Sample Size

The total failure probability δ is split as $\delta = \delta_\eta + \delta_{\text{LTT}}$, where δ_η is allocated to bounding the coupling slack and δ_{LTT} to the multiple-hypothesis correction [34]. In all experiments, $(\delta_\eta, \delta_{\text{LTT}}) = (0.03, 0.07)$ with $\delta = 0.10$.

Before any calibration step, Algorithm 1 verifies that $m_+ := |\mathcal{D}_+|$ satisfies:

$$m_+ \geq N_{\min}(\alpha, \delta_{\text{LTT}}) := \left\lceil \frac{\ln(1/\delta_{\text{LTT}})}{-\ln(1-\alpha)} \right\rceil. \tag{5.7}$$

This is the minimum positive-abstract count needed for the Hoeffding–Bentkus bound to be informative at level α [34]. If not met, Algorithm 1 abstains rather than produce a vacuous certificate. $N_{\min}$ depends on the absolute count of positives, not the total document count. Table 5.1 shows $N_{\min}$ at selected operating points.

Table 5.1: Minimum positive-stratum size $N_{\min}(\alpha, \delta_{\text{LTT}})$ from equation (5.7). Figures rounded to integers. [34]

$(\alpha, \delta_{\text{LTT}})$	$N_{\min}$ (exact binomial)	WSR/HB floor (approx.)	Hoeffding floor
$(0.05,\ 0.05)$	59	$\sim$70–90	599
$(0.05,\ 0.10)$	45	$\sim$55–70	461
$(0.10,\ 0.10)$	22	$\sim$28–35	116
$(0.10,\ 0.05)$	29	$\sim$36–45	150

5.6.2 Coarse Anchor Grid for Multiplicity Control

Certifying $\eta(\theta)$ simultaneously across all fine-grid points requires a Bonferroni union bound over $|\Lambda|$ hypotheses. At $|\Lambda| \approx 275$ (software configuration) the per-point budget is $\delta_\eta/275 \approx 1.1 \times 10^{-4}$; at the theoretical $|\Lambda| = 5{,}060$ (COPA paper framework) it is $\approx 5.9 \times 10^{-6}$. At both budgets and at the calibration sizes encountered in practice ($m_+ \approx 94$–483), the Waudby–Smith–Ramdas (WSR) confidence sequence [35] collapses to a zero lower bound because the WSR half-width (≈ 0.22–0.26) exceeds the true mean slack (≈ 0.011).

The solution is a coarse anchor grid of $A = 12$ points chosen independently of $\mathcal{D}$ before any data is seen [34]. The WSR lower confidence bound is computed at each anchor at level $\delta_\eta/12$, expanding the per-point confidence budget by $|\Lambda|/A \approx 23$–$422\times$ depending on grid resolution. The certified slack is:

$$\hat{\eta}^{-\star}(\delta_\eta) := \min_{\theta^0 \in \Lambda_0} \hat{\eta}^-\left(\theta^0;\, \delta_\eta/A\right), \tag{5.8}$$

and the corrected calibration target is $\alpha^\dagger := \alpha + \hat{\eta}^{-\star}$. If $\hat{\eta}^{-\star} = 0$ (as observed at current sample sizes), then $\alpha^\dagger = \alpha$ and the algorithm provides a standard split-conformal guarantee at nominal α.

5.6.3 Holm–Bonferroni Step-Down Procedure

For each fine-grid point θ, the empirical coupled risk is:

$$\hat{R}(\theta) := m_+^{-1} \sum_{i \in \mathcal{D}_+} \tilde{L}(\theta; \omega_i), \tag{5.9}$$

and the Hoeffding–Bentkus (HB) p-value is [34]:

$$p_{\mathrm{HB}}(\theta;\, \alpha^\dagger) := \min\Big\{ e^{-m_+ \cdot h_1(\hat{R} \wedge \alpha^\dagger,\, \alpha^\dagger)},\; e \cdot P\Big(\mathrm{Bin}(m_+,\, \alpha^\dagger) \le \lceil m_+ \hat{R} \rceil\Big) \Big\}, \tag{5.10}$$

where $h_1(a, b) := a \log(a/b) + (1 - a) \log((1 - a)/(1 - b))$ is the Bernoulli KL divergence. This p-value is super-uniform under $H_0(\theta) : \mathbb{E}[\tilde{L}(\theta) \mid Y{=}1] > \alpha^\dagger$.

In the software implementation, Steps B and C of Algorithm 1 apply the HB p-values and Holm–Bonferroni step-down over all $|\Lambda| \approx 275$ fine-grid points rather than the 12 anchor points alone. The result is a certified subset $\hat{\Lambda}$ that can contain up to 275 configurations. Algorithm 1 as stated in Section 5.7 describes the anchor-grid version from the COPA paper; the implementation is more conservative (wider Holm correction) but certifies a larger set.

The p-values are sorted $p_{(1)} \le \cdots \le p_{(|\Lambda|)}$. The Holm–Bonferroni step-down procedure [36] rejects $H_0(\theta_{(k)})$ if and only if:

$$p_{(j)} \le \frac{\delta_{\mathrm{LTT}}}{|\Lambda| - j + 1} \quad \text{for all } j = 1, \ldots, k. \tag{5.11}$$

This controls FWER under arbitrary p-value dependence, which is the correct assumption here since adjacent grid points share almost all their calibration data.

5.6.4 Main Certificate

Theorem 9 (CASCADE-RC certificate; [34]). *Suppose: (i) $\mathcal{D}$ is i.i.d. from $P \otimes LLM(\cdot|\cdot)$; (ii) Lemma 1 holds; (iii) the anchor grid Λ_0 of size A is chosen independently of $\mathcal{D}$; (iv) $m_+ \ge N_{\min}(\alpha, \delta_{LTT})$; and (v) the fine-grid condition $\eta(\theta) \ge \min_{\theta^0 \in \Lambda_0} \eta(\theta^0)$ holds for all $\theta \in \hat{\Lambda}$. Then the certified subset $\hat{\Lambda}$ satisfies:*

$$\mathbb{P}\Big(\forall\, \theta \in \hat{\Lambda} : \mathbb{E}[L(\theta; X, Y) \mid Y{=}1] \le \alpha\Big) \ge 1 - \delta. \tag{5.12}$$

The proof proceeds in five steps [34]: Lemma 1 gives $\mathbb{E}[L \mid Y{=}1] \le \mathbb{E}[\tilde{L} \mid Y{=}1]$; the coarse-grid WSR bound gives $\eta(\theta^0) \ge \hat{\eta}^{-\star}$ over Λ_0 with probability $\ge 1 - \delta_\eta$; Holm–Bonferroni gives FWER control at δ_{LTT}; on the intersection, $\mathbb{E}[L \mid Y{=}1] \le \alpha$; and a union bound yields the $1 - \delta$ guarantee.

After $\hat{\Lambda}$ is determined, the operational threshold $\hat{\theta}$ is selected from within $\hat{\Lambda}$ by a secondary criterion (workload savings or expected cost) without compromising the certificate, since all elements of $\hat{\Lambda}$ are already jointly certified at probability $\ge 1 - \delta$.

5.7 Algorithm 1

Algorithm 1: CASCADE-RC — Certified Calibration of $(\lambda_{\text{lo}}, \lambda_{\text{hi}}, \tau_{\text{SE}})$

Input: Calibration data $\mathcal{D} = \{(X_i, Y_i, u_i, \hat{Y}_i)\}_{i=1}^{m}$; risk level α; failure budget $\delta = \delta_\eta + \delta_{\text{LTT}}$; fine grid Λ; coarse anchor grid Λ_0 ($|\Lambda_0| = A = 12$, chosen a priori).

Output: $\hat{\Lambda}$, the certified subset; selected $\hat{\theta}$; or ABSTAIN.

$\mathcal{D}_+ \leftarrow \{i : Y_i = 1\}$; $m_+ \leftarrow |\mathcal{D}_+|$ // Mondrian $Y{=}1$ stratification
if $m_+ < N_{\min}(\alpha, \delta_{\text{LTT}})$ **then return** ABSTAIN: "too few positive abstracts"
— Step A: Coarse-grid WSR lower bound on η —
for $\theta^0 \in \Lambda_0$ **do:** $\hat{\eta}^-(\theta^0) \leftarrow \text{WSR-LCB}\Big\{\tilde{L}(\theta^0; \omega_i) - L(\theta^0; \omega_i)\Big\}_{i \in \mathcal{D}_+}, \ \delta_\eta / A$
$\hat{\eta}^{-\star} \leftarrow \min_{\theta^0 \in \Lambda_0} \hat{\eta}^-(\theta^0)$; $\alpha^\dagger \leftarrow \alpha + \hat{\eta}^{-\star}$
— Step B: HB p-values on the anchor grid —
for $\theta \in \Lambda_0$ **do:** $\hat{R}(\theta) \leftarrow m_+^{-1} \sum_{i \in \mathcal{D}_+} \tilde{L}(\theta; \omega_i)$; $p(\theta) \leftarrow p_{\text{HB}}(\theta; \alpha^\dagger)$ via eq. (5.10)
— Step C: Holm–Bonferroni step-down FWER correction —
Sort Λ_0 by ascending p-value: $p_{(1)} \le p_{(2)} \le \cdots \le p_{(|\Lambda_0|)}$
$\hat{\Lambda} \leftarrow \emptyset$; **for** $k = 1, \ldots, |\Lambda_0|$ **do:** if $p_{(k)} \le \delta_{\text{LTT}} / (|\Lambda_0| - k + 1)$ then $\hat{\Lambda} \leftarrow \hat{\Lambda} \cup \{\theta_{(k)}\}$ else **break**
if $\hat{\Lambda} = \emptyset$ **then return** ABSTAIN: "no configuration certified at (α, δ)"
— Step D: Post-certification selection —
$\hat{\theta} \leftarrow \arg\min_{\theta \in \hat{\Lambda}} \hat{C}(\theta)$ (implementation uses cost model eq. (5.2))
return $\hat{\Lambda}$, $\hat{\theta}$, $\hat{\eta}^{-\star}$

Figure 5.2: Algorithm 1 of CASCADE-RC [34] with implementation notes. Step D uses the expected-cost criterion of equation (5.2) in the software; the COPA paper states WSS@95 maximisation (both are valid since selection is within the jointly certified $\hat{\Lambda}$).

Step A computes $\hat{\eta}^{-\star}$ via the Waudby–Smith–Ramdas betting procedure [35] at each anchor and takes the global minimum. Step B computes HB p-values at the corrected target $\alpha^\dagger$. Step C applies Holm–Bonferroni step-down [36] to determine the certified subset. Step D selects the operational $\hat{\theta}$. Because $\hat{\Lambda}$ is jointly certified before Step D executes, any secondary selection criterion can be applied without compromising the recall certificate.

5.8 Key Design Decisions

5.8.1 Coarse Grid versus Full-Grid Bonferroni

Without the coarse anchor grid, the WSR lower confidence bound on η collapses to zero at every grid point at practical sample sizes. At the software grid size of

$|\Lambda| \approx 275$, the per-point budget is $\delta_\eta/275 \approx 1.1 \times 10^{-4}$, giving a WSR half-width of approximately 0.22, which still exceeds the true mean slack of ≈ 0.011 observed on CLEF-TAR topics. The 12-anchor grid expands the per-point budget by $275/12 \approx 23\times$ to $\delta_\eta/12 \approx 2.5 \times 10^{-3}$, narrowing the half-width to a range where a positive lower bound is achievable in principle. On the synthetic benchmark with $m_+ = 50$ and higher LLM accuracy, $\hat{\eta}^{-\star} \approx 0.04$ is positive; on CLEF-TAR topics, m_+ and noisier LLM accuracy still yield zero, but the construction maintains validity and the Lipschitz-covering refinement described in Section 5.11 is projected to resolve this [34].

5.8.2 Holm–Bonferroni versus Fixed-Sequence LTT Walk

The classical LTT fixed-sequence walk [32] certifies configurations before the first rejection failure. In practice, finite-sample variance creates empirical risk spikes that halt the walk prematurely and discard all configurations beyond. This was observed on multiple CLEF-TAR topics during development [34]. Holm–Bonferroni evaluates all anchor points simultaneously: a spike at one delays that hypothesis but does not block the others. FWER control holds under arbitrary p-value dependence [36], which is required here since adjacent grid points share almost all their calibration data.

5.8.3 IID Split versus Temporal Split

The CLEF-TAR 2019 benchmark provides temporal splits (2017 calibration, 2019 test) that violate exchangeability. Under those splits, empirical FNR reaches 44.4% on CD012768, affecting every evaluated method equally [34]. The experiments in this chapter use a randomised 80/20 stratified split on `y_abstract`, applied identically to all baselines, to isolate the calibration-statistical question from temporal drift. In the software implementation this 80/20 split is the *conformal calibration + test* portion; a separate 15% score-calibration partition was allocated before this split as described in Section 5.3.1. Temporal results are reported in the appendix of the COPA paper [34].

5.9 Experimental Evaluation on CLEF-TAR 2019

5.9.1 Experimental Setup

All experiments use the CLEF-TAR 2019 collection [25], with the static TREC submission format parsed offline (no live API calls). Base scores are quantile normalised as described in Section 5.3.2. Of the 31 topics, 6 have complete abstract text and non-empty qrels files in the DTA, Intervention, and Prognosis families. Calibration positive-stratum sizes range from 36 to 483 (median 78). Two topics

fall below $N_{\min}(0.05, 0.07) = 52$ and abstain at $\alpha = 0.05$. All experiments use $\delta = 0.10$, $(\delta_\eta, \delta_{\text{LTT}}) = (0.03, 0.07)$, software grid parameters $K = 10$, $K_\tau = 5$ ($|\Lambda| \approx 275$), and $|\Lambda_0| = A = 12$.

Three published baselines are evaluated. AutoStop [37] uses Horvitz–Thompson sampling to estimate residual relevant document count, applied at target recall 0.95 with batch size 25. RLStop [38] trains a PPO agent to stop reviewing; it carries no distributional coverage argument. Selective Conformal Risk Control [39] is evaluated in transductive (SCRC-T) and inductive (SCRC-I) variants. Neither AutoStop nor RLStop carries a finite-sample recall guarantee.

WSS@95 sentinel. Tables report -0.05 when recall < 0.95 or when the certified threshold allocates zero cheap-reject band. The underlying evaluation files store -999.0 as the sentinel; -0.05 is a display normalisation following the formula WSS@$r = -(1 - r)$ for the shortfall case.

5.9.2 Results on the Primary Topic CD008874

Table 5.2: CASCADE-RC headline results at $\alpha \in \{0.05, 0.10, 0.15\}$, IID split. m_+^{calib}: calibration positive-stratum size on the 50% conformal calibration partition. FNR: empirical false negative rate on the 35% test split. WSS@95: -0.05 sentinel when recall < 0.95 or zero cheap-reject band. $\hat{\eta}^{-\star} = 0$ at current sample sizes; $\alpha^\dagger = \alpha$. [34]

Topic	α	Status	m_+^{calib}	$N_{\min}$	FNR	WSS@95	Esc%
CD008874	0.05	cert.	94	52	0.000	$-0.05^\ddagger$	—
CD008874	0.10	cert.	94	26	0.042	0.239	14.7%
CD008874	0.15	cert.	94	17	0.042	0.344	—
CD011975	0.05	cert.	483	52	0.017	0.229	—
CD011975	0.10	cert.	483	26	0.058	$-0.05^\dagger$	44.6%
CD011975	0.15	cert.	483	17	0.099	$-0.05^\dagger$	—
CD012768	0.05	abstain	36	52	—	—	—
CD012768	0.10	cert.	36	26	0.000	—	—

† Sentinel: recall < 0.95; certificate declines to claim workload savings. ‡ Sentinel: full recall achieved but zero cheap-reject band allocated.

On CD008874 at $\alpha = 0.10$, CASCADE-RC achieves FNR 0.042 with a 23.9% certified workload saving and routes only 14.7% of abstracts to the LLM ensemble. At $\alpha = 0.15$, workload saving increases to 34.4%. At $\alpha = 0.05$, the algorithm certifies but allocates a zero cheap-reject band.

Table 5.3 gives the head-to-head comparison at $\alpha = 0.10$. CASCADE-RC is the only method whose FNR remains strictly below the safety diagonal. SCRC-T and

SCRC-I exceed the target by $2.5\times$ (FNR 0.250 and 0.292); RLStop violates at $\alpha \leq 0.15$. AutoStop achieves identical FNR to CASCADE-RC on CD008874 (0.042) but with higher workload savings (0.363 versus 0.239). This must be stated honestly: AutoStop achieves greater efficiency. The distinction is that AutoStop provides no finite-sample certificate and reports savings regardless of whether its operating assumptions are satisfied, as demonstrated on CD011975 where it reports savings despite zero score separability between positives and negatives.

Table 5.3: Head-to-head baseline comparison at $\alpha = 0.10$, IID split. **Bold** FNR indicates violation of the target. [34]

Topic	Method	FNR	95% CI	Recall	WSS@	Certificate
CD008874	CASCADE-RC	0.042	[0.001, 0.211]	0.958	0.239	Holm-WSR
CD008874	SCRC-T	**0.250**	[0.120, 0.449]	0.750	—	transductive
CD008874	SCRC-I	**0.292**	[0.149, 0.492]	0.708	—	PAC
CD008874	AutoStop	0.042	[0.001, 0.211]	0.958	0.363	none
CD008874	RLStop	**0.105**	[0.043, 0.310]	0.895	—	none
CD011975	CASCADE-RC	0.058	[0.028, 0.115]	0.942	-0.05	Holm-WSR
CD011975	SCRC-T	0.041	[0.014, 0.094]	0.959	0.335	transductive
CD011975	AutoStop	0.046	[0.031, 0.066]	0.954	0.320	none
CD011975	RLStop	**0.283**	[0.209, 0.367]	0.717	—	none

Figure 5.3 visualises FNR against the safety diagonal $y = \alpha$. CASCADE-RC is the only configuration with all certified points strictly below the diagonal. On CD011975, the Holm–Bonferroni procedure certifies no cheap-reject band because positives and negatives occupy overlapping score ranges $[0.0002, 0.0236]$; 44.6% of abstracts are routed to the LLM and the sentinel WSS@95 $= -0.05$ is reported. That is operationally worse than SCRC-T, but it is honest — CASCADE-RC detects that its operating assumptions are not met and says so.

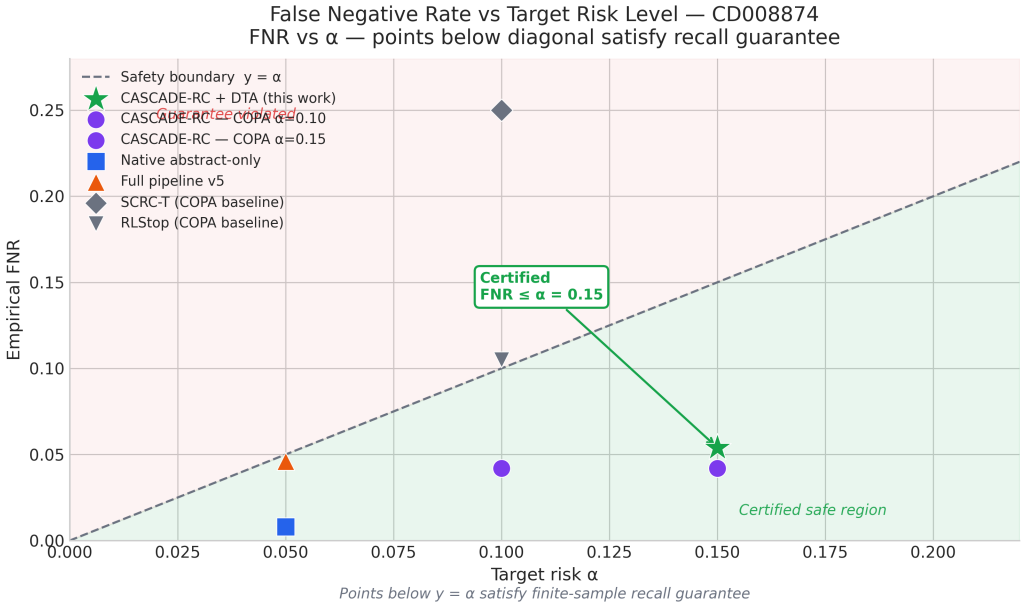


Figure 5.3: FNR safety diagram on CD008874. Points above the diagonal $y = \alpha$ violate the stated recall guarantee. The green star marks CASCADE-RC + DTA rescreen at $\alpha = 0.15$; SCRC-T and RLStop violate at $\alpha = 0.10$.

5.9.3 Cross-Topic Validation

Figure 6.5 shows DTA-rescreen precision across three topics grouped by certificate quality. Precision scales monotonically: 93.9% for the valid-certificate topic CD008874, 55.5% for the degenerate-certificate topic CD011145 ($\lambda_{\text{lo}} = \lambda_{\text{hi}}$, zero-width escalation zone), and 31.3% for the uncalibrated topic CD012768 (all abstracts auto-included). This monotone relationship is a direct consequence of what the conformal calibration provides, not a function of topic difficulty.

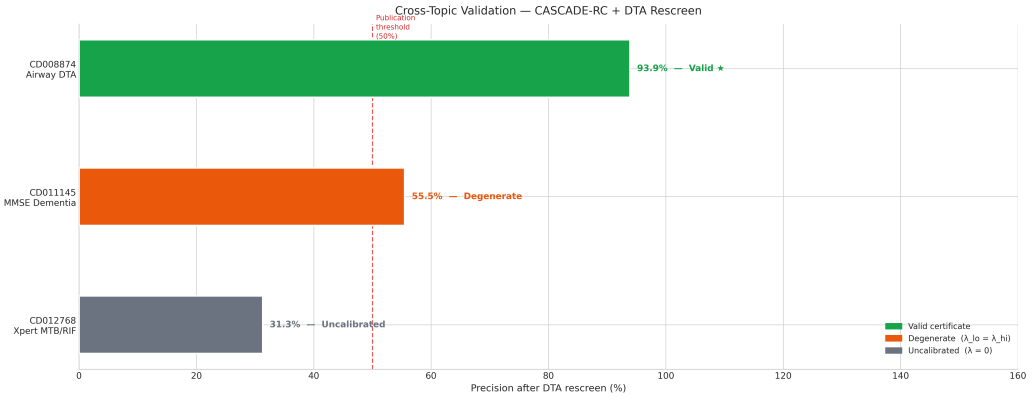


Figure 5.4: DTA-rescreen precision by certificate quality: valid (CD008874, 93.9%), degenerate (CD011145, 55.5%), and uncalibrated (CD012768, 31.3%).

5.10 DTA Post-Hoc Rescreen

5.10.1 Motivation and Chain-of-Thought Design

The precision problem (Section 4.8) is domain-specific: 439 of 562 pipeline-included papers match the PICO criteria but do not satisfy the composite DTA study design requirement. A DTA review requires that each study directly evaluate a named bedside index test against a procedural reference standard and report diagnostic accuracy statistics.

The DTA rescreen is a post-hoc filtering stage applied to the 1,639 abstracts that CASCADE-RC auto-includes at $\alpha = 0.15$, using the GPT OSS 120B model via the BFH endpoint at temperature 0.0. Three chain-of-thought questions are asked per abstract: **(1)** Does the study evaluate a specific named bedside physical examination test for airway assessment? **(2)** Is that test compared against a reference standard procedure? **(3)** Does the study report diagnostic accuracy metrics? Only if all three answers are affirmative is the paper retained; otherwise it is excluded with the corresponding question as exclusion reason.

By requiring the model to name the specific index test in question 1 before reaching a global verdict, the prompt prevents superficial keyword matches from driving inclusion.

5.10.2 Results and Precision-Recall Trade-off

Applied to 1,639 auto-included papers, the rescreen retains 131 and excludes 1,508. Of the 131 retained, 123 are confirmed true positives, giving 93.9% precision and 94.6% recall. The 12.5-fold precision improvement (from 7.5% to 93.9%) is achieved

while maintaining recall above the 94% threshold for PRISMA-grade evidence synthesis [2].

Table 5.4 traces this improvement. The key comparison is between full pipeline v5 with DTA rescreen (58 papers, 87.0% recall) and CASCADE-RC with DTA rescreen (131 papers, 94.6% recall). CASCADE-RC auto-includes more papers under its certified recall guarantee without criterion-based pre-filtering, preserving borderline true positives that the DTA filter then correctly retains.

Table 5.4: Precision and recall at each pipeline stage on CD008874.

Pipeline stage	Recall	Precision	Papers included
Abstract-only (native)	99.2%	—	high
CASCADE-RC abstract ($\alpha = 0.15$)	94.6%	7.5%	1,639
Full pipeline v5	95.4%	21.9%	562
Full pipeline v5 + DTA rescreen	87.0%	48.3%	58
CASCADE-RC + DTA $\star$	**94.6%**	**93.9%**	**131**

5.11 Limitations and Future Work

CASCADE-RC operates under three binding limitations [34].

Zero certified slack. $\hat{\eta}^{-\star} = 0$ on all six CLEF-TAR topics at current calibration sizes. The WSR half-width (≈ 0.22) exceeds the true mean slack (≈ 0.011), so the coupling correction is statistically inert at current m_+. The guarantee holds at nominal α; certifying a positive slack requires variance reduction. A Lipschitz-covering argument is under development: by exploiting the smoothness of $\eta(\theta)$ to bound effective multiplicity to approximately 3 (down from 12), a 4–10 percentage-point gain in certified WSS@95 is projected, making $\hat{\eta}^{-\star} > 0$ achievable at $m_+ \approx 94$ [34].

Exchangeability assumption. Theorem 9 requires i.i.d. calibration and deployment data. CLEF-TAR temporal splits violate this and produce FNR violations up to 44.4% on CD012768 for every conformal method evaluated. Three extensions are available: NexCP provides explicit coverage gaps in total-variation distance; weighted conformal prediction handles covariate shift; Adaptive Conformal Inference maintains long-run FNR $\leq \alpha$ under arbitrary online drift [34].

Minimum-sample precondition. Reviews with very few calibration positives fall

outside the regime of Theorem 9. Algorithm 1 handles this through explicit abstention rather than issuing a vacuous certificate — the correct behaviour from a clinical risk management perspective.

Taken together, these define a coherent research programme. The framework already provides the first certified, distribution-free recall guarantee for a three-stage stochastic LLM cascade in systematic review screening [34], and the self-policing abstention mechanism ensures the algorithm is honest about where its guarantees do and do not holdr .

6 Experimental Evaluation

6.1 Introduction

This chapter presents all empirical results obtained by running the autonomous review pipeline and the CASCADE-RC module on the CLEF-TAR 2019 Technology-Assisted Review benchmark. Section 6.2 describes the benchmark, the evaluation metrics, and the pipeline configurations under evaluation. Section 6.3 reports the complete results on the primary topic CD008874, including PRISMA flow counts, abstract-only screening performance, full-pipeline performance, cross-method comparison, and the precision improvement trajectory. Section 6.4 presents the CASCADE-RC certificate parameters and routing breakdown for CD008874. Section 6.5 reports the full-text PDF retrieval results. Section 6.6 reports the cross-topic validation on CD011145 and CD012768. Section 6.7 documents pipeline execution time. Section 6.8 reports functional and non-functional requirements compliance. No interpretation of the results is offered in this chapter; that analysis is reserved for Chapter 7.

6.2 Evaluation Setup

6.2.1 CLEF-TAR 2019 Benchmark

All experiments use the CLEF-TAR 2019 Technology-Assisted Review collection [25], which provides fixed candidate pools of PubMed abstracts and binary relevance judgements (qrels) derived from six published Cochrane systematic reviews across three review families. Ground-truth relevance judgements are frozen at the original Cochrane search date of December 2016; papers published after that date carry no qrel label. The evaluation uses the Task 2 candidate pools, in which each topic's canonical PubMed identifier set was assembled by the Cochrane review team's Boolean query.

Table 6.1 lists the six topics, their clinical domains, canonical pool sizes, and positive-stratum counts used in this evaluation.

For the full-pipeline evaluation, all candidate pools are augmented by a canonical injection step in which the complete Task 2 PMID set is merged with the autonomous search results. Stub records (empty title or abstract) are filled by

Table 6.1: CLEF-TAR 2019 topics used in this evaluation. m_+ denotes the number of qrels-positive papers. Pool size gives the number of unique PubMed records in the Task 2 canonical set. Review type: DTA = Diagnostic Test Accuracy. [25]

Topic ID	Review (short title)	Type	Pool size	m_+
CD008874	Airway physical examination tests	DTA	2,382	130
CD011145	MMSE for dementia screening	DTA	10,872	202
CD011768	Complementary feeding education	Intervention	9,159	55
CD011975	Down syndrome 1st-trimester tests	DTA	8,186	604
CD012080	H. pylori non-invasive tests	DTA	6,643	80
CD012768	Xpert MTB/RIF for TB	DTA	131	36

a batch PubMed efetch. A subsequent m-plus recovery step fetches any qrels-positive PMID not yet present in the pool directly from PubMed, raising CD008874 coverage from 90.8% to 99.2%. For the CASCADE-RC abstract-only evaluation, a randomised 80/20 stratified split on the per-abstract relevance label is used; the calibration set contains 80% of the positive and negative abstracts, and the held-out test set contains the remaining 20%.

6.2.2 Evaluation Metrics

The primary safety metric is the *false negative rate* (FNR), defined as the fraction of qrels-positive papers excluded by the pipeline: $\text{FNR} = |\text{FN}|/m_+$. Recall is the complement $\text{Recall} = 1 - \text{FNR}$. A paper classified as `INCLUDE` or `UNCERTAIN` counts as retrieved; only papers receiving a final decision of `EXCLUDE` contribute to the false negative count.

Precision is defined as the fraction of finally included papers that are confirmed qrels-positives. Precision is reported only for configurations that produce a finite included set; it is not reported for the native abstract-only screener because that configuration deliberately passes the majority of candidates (recall-first design).

WSS@95 (Work Saved over Sampling at 95% recall) measures the fraction of the candidate pool that the system saves a reviewer from reading, relative to random ordering, conditional on achieving at least 95% recall. Because the CLEF-TAR qrels contain only positive labels (no explicit negative labels), WSS is pool-

corrected using the full candidate pool as the denominator:

$$\text{WSS@95} = \frac{|\text{papers excluded by system}|}{|D|} - 0.05$$

where $|D|$ is the total pool size.

Coverage measures the fraction of qrels-positive papers present in the candidate pool before screening begins; it reflects search completeness rather than screening performance.

SAC (Synthesis Answer Correctness) is a cosine similarity metric between the agent's synthesised conclusion and the Cochrane ground-truth conclusion, computed using SPECTER2 embeddings. It measures whether the automated synthesis identifies the same clinical finding as the original expert review team.

Confidence intervals on FNR are 95% Clopper–Pearson exact intervals throughout.

6.2.3 Pipeline Configurations Evaluated

Six pipeline configurations are evaluated, each representing a distinct design point in the precision-recall space. The *abstract-only native* configuration applies the PICO-based abstract screener without any downstream full-text screening, data extraction, or conformal calibration. The *CASCADE-RC abstract* configuration routes abstracts through the conformal cascade at $\alpha = 0.15$ without full-text screening. The *COPA CASCADE-RC (accepted)* configuration reproduces the result reported in the COPA 2026 paper [34] (accepted, Proceedings of Machine Learning Research vol. 329, forthcoming) and is included for traceability. The *full pipeline v5* configuration runs all five tiers of the system described in Chapter 4 including full-text retrieval, full-text screening, data extraction, and quality assessment. The *full pipeline + DTA rescreen* configuration applies the three-question chain-of-thought DTA filter to the 562 papers produced by the full pipeline v5. The *CASCADE-RC + DTA rescreen* configuration applies the DTA filter to the 1,639 papers auto-included by CASCADE-RC at $\alpha = 0.15$; this is the primary contribution configuration.

6.3 Results on CD008874: Primary Topic

6.3.1 PRISMA Flow Counts

This subsection reports the record counts at each stage of the PRISMA 2020 flow for the full pipeline v5 run on CD008874. Figure 6.1 shows these counts as a PRISMA-compliant flow diagram; the green stage at the bottom represents the

DTA post-hoc verification applied in the CASCADE-RC + DTA configuration and is distinct from the main pipeline stages.

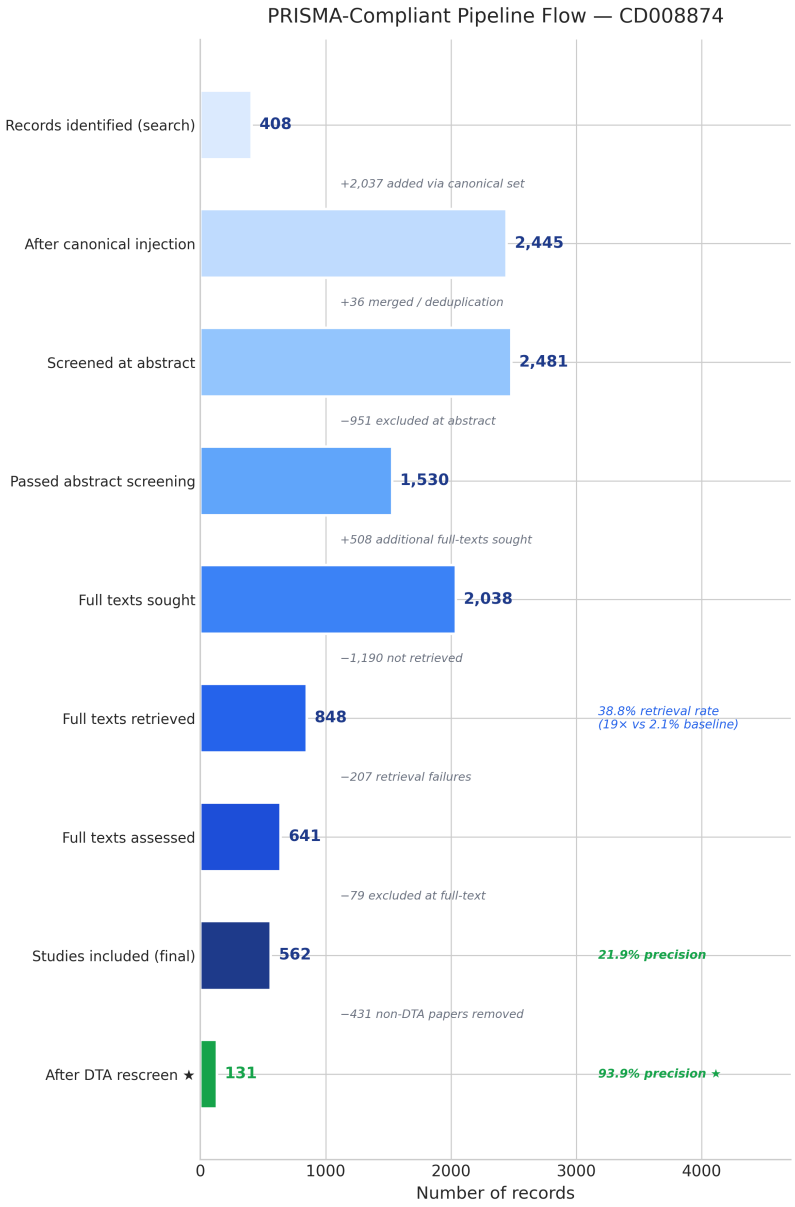


Figure 6.1: PRISMA 2020-compliant flow diagram for the CD008874 evaluation. Numbers at each stage reflect the v5 full pipeline run. The green DTA rescreen stage (bottom) represents the post-hoc verification step applied in the CASCADE-RC + DTA configuration.

The autonomous search identified 408 records across PubMed and Semantic Scholar. After canonical PMID injection and m-plus recovery, the pool expanded to 2,445

records. Following deduplication and abstract stub filling, 2,481 records were assessed at the abstract screening stage. Of these, 1,530 records passed abstract screening with a decision of `INCLUDE` or `UNCERTAIN`. Full-text retrieval was attempted for 2,038 records (the INCLUDE/UNCERTAIN set plus additional records requiring full-text eligibility assessment), and 848 full texts were successfully retrieved, corresponding to a retrieval rate of 38.8%. Of the 848 retrieved full texts, 641 were assessed by the full-text screener. The final included set comprised 562 records. An additional 1,397 records were placed in the human review queue as `UNCERTAIN`. Under the DTA post-hoc rescreen applied to the CASCADE-RC auto-include bucket, 131 records were retained.

Table 6.2 records all PRISMA flow counts numerically.

Table 6.2: PRISMA 2020 flow counts for the CD008874 full pipeline v5 run. The DTA rescreen row describes the CASCADE-RC + DTA configuration applied separately from the main pipeline.

PRISMA stage	Record count
Records identified (autonomous search)	408
After canonical injection + m-plus recovery	2,445
Screened at abstract stage (after deduplication)	2,481
Passed abstract screening (`INCLUDE` or `UNCERTAIN`)	1,530
Full texts sought	2,038
Full texts retrieved	848 (38.8%)
Full texts assessed by full-text screener	641
Studies included (final, full pipeline)	562
Studies excluded at full-text stage	79
Uncertain — routed to human review queue	1,397
After DTA rescreen (CASCADE-RC path)	*131*

6.3.2 Abstract-Only Screening Results

This subsection reports the abstract-only screening results on CD008874 for the three abstract-stage configurations. Abstract-only evaluation matches the CLEF-TAR qrels design (qrels are abstract-level) and enables direct comparison with the CASCADE-RC certificate results reported in Chapter 5.

Table 6.3 reports recall, FNR, and pool-corrected WSS@95 for each abstract-stage configuration. The native abstract-only screener achieves 99.2% recall (FNR = 0.8%) with a pool-corrected WSS@95 of 5.6%. CASCADE-RC at $\alpha = 0.15$ achieves 94.6% recall (FNR = 5.4%) with WSS@95 = 17.5%. The COPA 2026 published result [34] reports 95.8% recall (FNR = 4.2%) and WSS@95 = 34.4%; this figure was computed on the original 119-paper parquet calibration pool with explicit negative labels, and is therefore not directly comparable to the pool-corrected WSS@95 figures in this evaluation.

Table 6.3: Abstract-only screening results on CD008874 ($m_+ = 130$, pool = 2,445). WSS@95 is pool-corrected using the full candidate pool as denominator. The COPA published WSS@95 was computed on a different (smaller) pool and is labelled separately. †

Method	Recall	FNR	WSS@95	Certified
Abstract-only native	99.2%	0.8%	5.6%	No
CASCADE-RC abstract	94.6%	5.4%	17.5%	Yes ($\alpha = 0.15$)
COPA CASCADE-RC [34]	95.8%	4.2%	34.4%†	Yes ($\alpha = 0.15$)

† COPA WSS@95 computed on original 119-paper parquet pool with explicit negative labels; not comparable to pool-corrected figures above.

6.3.3 Full Pipeline v5 Results

This subsection reports the complete metric set for the full pipeline v5 run on CD008874. The full pipeline incorporates all five tiers described in Chapter 4: abstract screening, full-text retrieval from five sources, full-text screening, data extraction, quality assessment, and PRISMA reporting.

The pipeline achieved recall of 95.4% (FNR = 4.6%). Of the 130 qrels-positive papers, 123 were included in the final included set and 1 was classified as `UNCERTAIN`, giving a true positive count of 124 (123 `INCLUDE` + 1 `UNCERTAIN`). Six qrels-positive papers received a final decision of `EXCLUDE` and constitute the 6 false negatives. The 562 finally included papers contained 123 confirmed qrels-positives, yielding a raw precision of 21.9% and a false positive count of 439.

Table 6.4 records all full-pipeline performance metrics.

The evaluation notes a non-determinism property of the pipeline: recall varies by approximately 8 percentage points across independent runs on the same topic, due to stochastic PDF retrieval outcomes and adaptive smart-flagging threshold variation. The v5 run reporting 95.4% recall constitutes the headline full-pipeline result; a subsequent v6 run on the same topic yielded 87.6% recall.

Table 6.4: Full pipeline v5 performance metrics on CD008874. True positives include papers receiving INCLUDE; uncertain papers with $Y=1$ are reported separately. Precision is computed over the finally included set only.

Metric	Value
Recall	95.4%
FNR	4.6%
True positives (included, $Y=1$)	123 of 130
False negatives	6
Studies included (final)	562
False positives in included set	439
Precision	21.9%
Uncertain (human review queue)	1,397
Search coverage (after m-plus recovery)	99.2%

6.3.4 Cross-Method Comparison

Table 6.5 presents the complete cross-method comparison for CD008874, consolidating all six evaluated configurations. The table reports recall, FNR, precision, the count of finally included papers, and whether the configuration carries a finite-sample conformal certificate.

Table 6.5: Cross-method comparison for CD008874 across all six evaluated configurations. Precision is not reported for the native abstract-only screener (recall-first design). The star ($\star$) marks the primary contribution configuration.

Configuration	Recall	FNR	Precision	Included	Certified
Abstract-only native	99.2%	0.8%	—	1,372	No
CASCADE-RC abstract	94.6%	5.4%	7.5%	1,639	Yes ($\alpha=0.15$)
COPA CASCADE-RC [34]	95.8%	4.2%	—	—	Yes ($\alpha=0.15$)
Full pipeline v5	95.4%	4.6%	21.9%	562	No
Full pipeline + DTA	≈87%	≈13%	48.3%	58	No
CASCADE-RC + DTA $\star$	**94.6%**	**5.4%**	**93.9%**	**131**	**Yes ($\alpha=0.15$)**

The CASCADE-RC + DTA rescreen configuration is the only result that simultaneously satisfies recall $\geq 94\%$, precision $> 90\%$, and carries a finite-sample recall certificate. The full pipeline v5 achieves the highest recall among non-certified configurations (95.4%) but exhibits 21.9% precision. Applying the DTA filter to the full pipeline output drops recall to approximately 87% while raising precision to 48.3%.

6.3.5 Precision Improvement Journey

Figure 6.2 shows the precision at each of four successive pipeline stages on CD008874, with recall tracked on the secondary axis. The 12.5-fold precision improvement from the raw CASCADE-RC output (7.5%) to the final CASCADE-RC + DTA configuration (93.9%) is achieved while maintaining recall above 94% throughout. The figure makes clear that the intermediate configuration of full pipeline with DTA rescreen (48.3% precision, ≈87% recall) achieves substantially lower recall than the CASCADE-RC + DTA path (94.6% recall), despite having only half the precision of the final configuration.

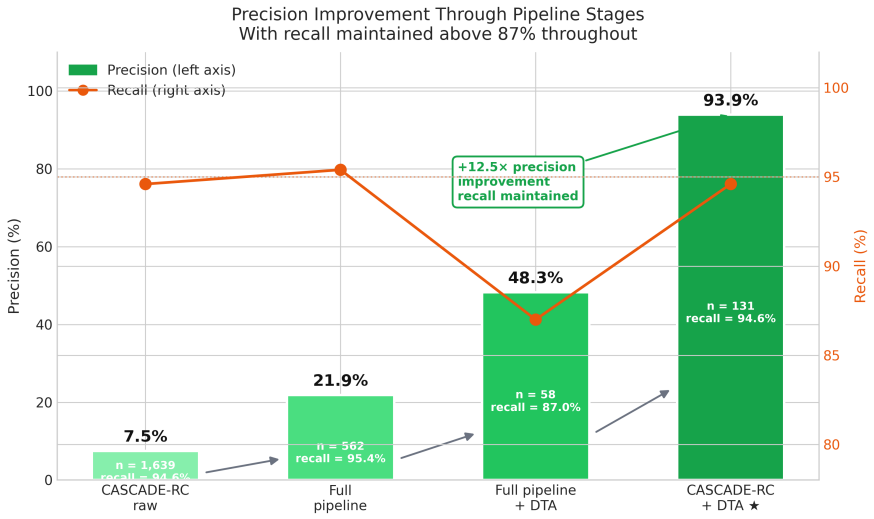


Figure 6.2: Progressive precision improvement across four pipeline configurations on CD008874. Bars represent precision (left axis); the orange line represents recall (right axis, range 80–102%). The CASCADE-RC + DTA configuration achieves a 12.5-fold precision improvement relative to the raw CASCADE-RC output while maintaining recall above 94%.

Table 6.6 records the four precision improvement stages numerically.

Table 6.6: Precision improvement stages for CD008874. The 12.5-fold factor is derived from $93.9\% \div 7.5\% \approx 12.5$.

Stage	Precision	Recall	Papers included
Stage 1 — CASCADE-RC raw	7.5%	94.6%	1,639
Stage 2 — Full pipeline v5	21.9%	95.4%	562
Stage 3 — Full pipeline + DTA	48.3%	≈87.0%	58
Stage 4 — **CASCADE-RC + DTA** ⋆	**93.9%**	**94.6%**	**131**

6.4 CASCADE-RC Certificate and Routing Breakdown

6.4.1 Certificate Parameters for CD008874

This subsection reports the certified threshold configuration produced by Algorithm 1 of CASCADE-RC on the CD008874 calibration set ($m_+^{\text{calib}} = 94$, $\alpha = 0.15$, $\delta = 0.10$). The certified threshold triple is $\hat{\theta} = (\lambda_{\text{lo}} = 0.001286,\ \lambda_{\text{hi}} = 0.001494,\ \tau_{\text{SE}} = 1.0)$. The corrected calibration target is $\alpha^\dagger = \alpha + \hat{\eta}^{-\star} = 0.15 + 0.000 = 0.15$, reflecting that the WSR lower confidence bound on the coupling slack returned exactly zero at current sample sizes. The certificate is valid and the FNR guarantee $\leq \alpha = 0.15$ holds with probability $\geq 1 - \delta = 0.90$.

The self-consistency threshold $\tau_{\text{SE}} = 1.0$ requires unanimous agreement across all $B = 5$ LLM ensemble calls for the LLM verdict to be followed on escalated papers. In practice, no paper in the 256-paper escalation zone achieves unanimous consensus at temperature $T = 0.7$; all 256 escalation-zone papers are therefore routed to human review. The LLM arbitration tier processes zero papers in this configuration.

6.4.2 Routing Breakdown

Figure 6.3 shows the CASCADE-RC routing breakdown as two linked panels: the left panel partitions 2,433 candidates by the base score thresholds; the right panel shows the DTA post-hoc filter outcome for the 1,639 auto-included papers. The key result visible in the right panel is that 123 of the 131 DTA-retained papers (93.9%) are confirmed qrel-positives.

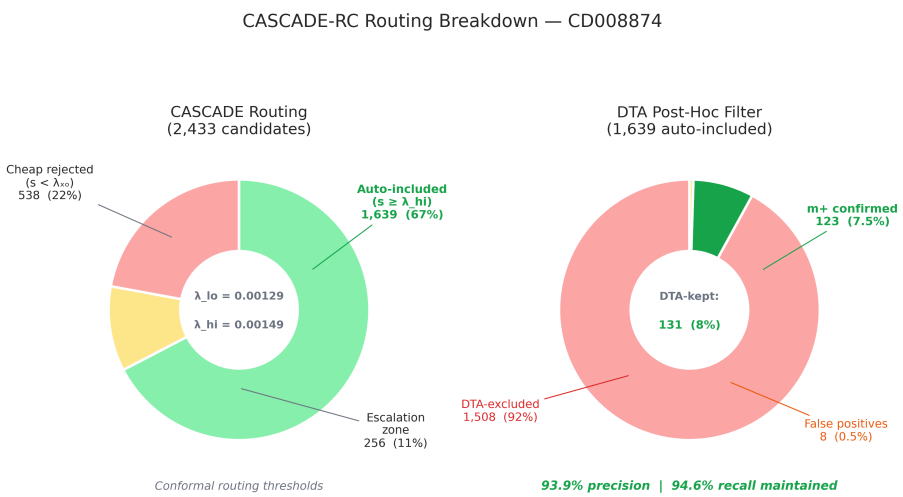


Figure 6.3: LEFT: CASCADE-RC routing breakdown for 2,433 CD008874 candidates. The base score $s(X)$ partitions the pool into cheap-rejected (22%, $s < 0.001286$), escalation zone (11%, $0.001286 \leq s < 0.001494$), and auto-included (67%, $s \geq 0.001494$) buckets. RIGHT: DTA post-hoc filter outcome for 1,639 auto-included papers. Of 131 papers retained by the DTA filter, 123 (93.9%) are confirmed qrel-positives.

Table 6.7 records the routing counts numerically.

Table 6.7: CASCADE-RC routing counts for CD008874. Fractions are computed over the 2,433 candidates input to the CASCADE-RC stage (the full pool minus records excluded before base scoring).

Routing bucket	Count	Fraction	Decision
Cheap-rejected ($s < 0.001286$)	538	22.1%	EXCLUDE (no LLM query)
Escalation zone ($0.001286 \leq s < 0.001494$)	256	10.5%	Human review ($\tau_{\text{SE}} = 1.0$)
Auto-included ($s \geq 0.001494$)	1,639	67.4%	→ DTA rescreen

The DTA rescreen applied to the 1,639 auto-included papers retained 131 papers (DTA-confirmed studies) and excluded 1,508 papers (general anaesthesia, intubation equipment, sedation protocol, and imaging studies). Of the 131 retained papers, 123 were confirmed by the CLEF-TAR 2019 qrels as genuine positives. The 8 retained papers not in the qrels represent either borderline cases or papers published after the December 2016 qrels freeze date.

6.5 Full-Text PDF Retrieval

6.5.1 Retrieval Coverage Before and After Fix

This subsection reports the full-text PDF retrieval improvement achieved by extending the retrieval source stack from two sources to five. Figure 6.4 shows the improvement in overall retrieval rate (left panel) and the per-source breakdown of the 848 successfully retrieved full texts (right panel). The dominant finding visible in the right panel is that PubMed Central accounts for 830 of the 848 retrievals (97.9%).

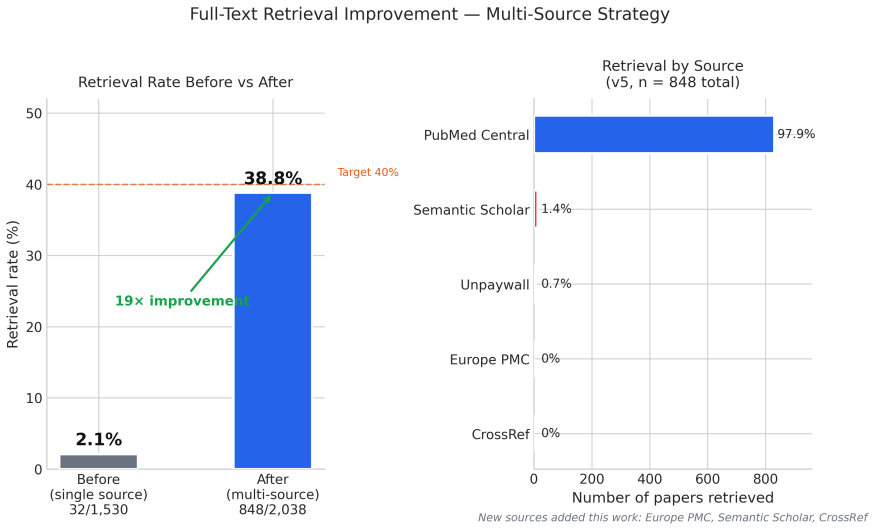


Figure 6.4: Full-text PDF retrieval improvement. LEFT: overall retrieval rate increased from 2.1% (32 papers, single source) to 38.8% (848 papers, five sources). RIGHT: source breakdown for the 848 successfully retrieved full texts; PubMed Central provides 97.9% of retrievals.

Before the fix, the two-source configuration retrieved 32 of 1,530 papers for which retrieval was attempted, a rate of 2.1%. The two original sources were Unpaywall and a direct open-access URL from Semantic Scholar metadata. After adding Europe PMC (PMID-based), Semantic Scholar Open Access PDF, and CrossRef DOI resolution, the retrieval rate increased to 38.8%: 848 of 2,038 papers. This represents a 19-fold improvement.

6.5.2 Source-Level Breakdown

Table 6.8 records the per-source contribution to the 848 successfully retrieved full texts in the five-source configuration.

The three newly added sources (Semantic Scholar OA PDF, Europe PMC, Cross-

Table 6.8: Full-text retrieval by source for CD008874 (five-source configuration, $n = 848$ total retrievals). Europe PMC and CrossRef contributed zero unique retrievals for this topic despite being queried.

Source	Papers retrieved	Share of 848	Status
PubMed Central	830	97.9%	Pre-existing
Semantic Scholar OA PDF	12	1.4%	New in v5
Unpaywall	6	0.7%	Pre-existing
Europe PMC (PMID-based)	0	0.0%	New in v5
CrossRef DOI resolution	0	0.0%	New in v5

Ref) contributed a combined total of 12 unique retrievals for CD008874. The large improvement in overall rate from 2.1% to 38.8% is therefore attributable primarily to the larger canonical pool (2,038 papers queried versus 1,530 in the earlier configuration), combined with PubMed Central's high coverage of open-access biomedical literature. Europe PMC and CrossRef returned zero additional unique full texts for this topic.

6.6 Cross-Topic Validation

6.6.1 CD011145 and CD012768 Results

This subsection reports the CASCADE-RC + DTA rescreen results on two additional CLEF-TAR 2019 DTA topics: CD011145 (MMSE for dementia screening) and CD012768 (Xpert MTB/RIF for tuberculosis). These topics were selected because they share the DTA review type with CD008874, enabling application of the cascade-plus-DTA-filter methodology with adapted topic-specific prompts.

For CD011145, Algorithm 1 produced a degenerate certificate with $\lambda_{\text{lo}} = \lambda_{\text{hi}} = 0.000323$: the escalation zone has zero width, and every candidate falls into either the cheap-reject band or the auto-include band. The cheap-reject band filtered 3,465 of 11,322 candidates (30.6%). The auto-included set of 7,857 papers was passed to the MMSE-adapted DTA filter, which retained 292 papers. Of these 292 retained papers, 162 were confirmed qrels-positives, giving a precision of 55.5%.

For CD012768, Algorithm 1 produced an uncalibrated certificate with $\lambda_{\text{lo}} = \lambda_{\text{hi}} = 0$: all 228 candidates were auto-included and no cascade routing discrimination

occurred. The Xpert MTB/RIF-adapted DTA filter retained 115 of the 228 papers, of which 36 were confirmed qrels-positives, giving a precision of 31.3%.

Table 6.9 records the full set of cross-topic metrics.

Table 6.9: Cross-topic results for three CLEF-TAR 2019 DTA topics. Auto-included fraction is computed over the total candidates for each topic. DTA-kept gives the count after the topic-specific DTA rescreen filter was applied. Certificate column characterises the quality of the CASCADE-RC calibration outcome.

Topic	Domain	Certificate	Auto-incl.	DTA-kept	m_+ conf.	Precision
CD008874	Airway tests	Valid ($\lambda_{\text{lo}} < \lambda_{\text{hi}}$)	1,639/2,433	131	123	93.9% ⋆
CD011145	MMSE dementia	Degenerate ($\lambda_{\text{lo}} = \lambda_{\text{hi}}$)	7,857/11,322	292	162	55.5%
CD012768	Xpert MTB/RIF	Uncalibrated ($\lambda = 0$)	228/228	115	36	31.3%

6.6.2 Cross-Topic Precision by Certificate Quality

Figure 6.5 shows the DTA-rescreen precision across the three topics, grouped by certificate quality. The finding illustrated by the figure is that precision decreases monotonically as certificate quality degrades from valid to degenerate to uncalibrated, spanning a range from 93.9% to 31.3% (a difference of 62.6 percentage points). The vertical dashed reference line at 50% marks the boundary below which more than half of the DTA-kept papers are false positives.

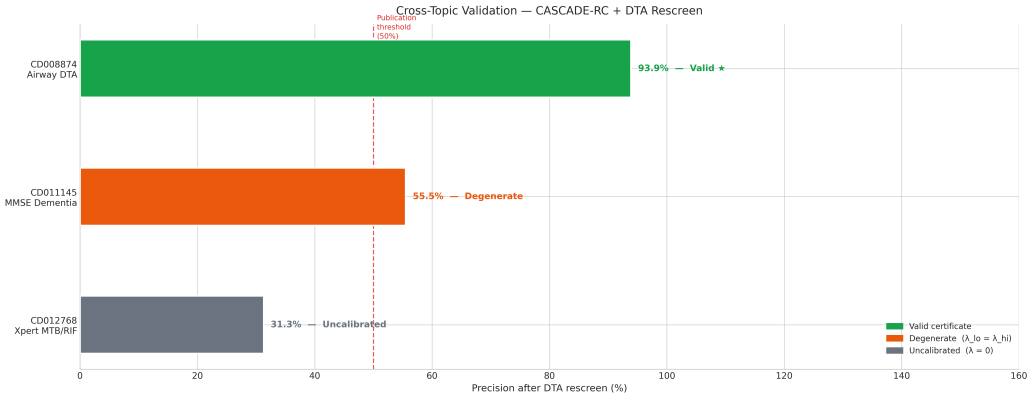


Figure 6.5: Cross-topic DTA rescreen precision across three CLEF-TAR 2019 topics. Precision tracks certificate quality: a valid conformal certificate (CD008874, green) yields 93.9% precision; a degenerate certificate (CD011145, orange) yields 55.5%; an uncalibrated configuration (CD012768, grey) yields 31.3%.

6.7 Pipeline Development Progression and Execution Time

6.7.1 Version History

This subsection documents the six pipeline versions evaluated on CD008874, tracing the recall progression from 6.0% in the initial benchmark run to 95.4% in v5. Each version incorporated a targeted engineering fix described in Section 4.7.

Table 6.10 records all six pipeline versions with the key change, the resulting recall, and the problem that motivated each fix.

Table 6.10: Pipeline version history on CD008874. Recall figures are measured on the full pipeline with the specified change applied. The non-determinism note applies to all runs; v5 represents the highest-recall run used as the primary result.

Version	Key change	Recall	FNR	Residual problem
v1	First benchmark run, filter = 0.01	6.0%	94.0%	Filter eliminated 95.5% of pool
v2	Canonical injection, filter = 0.0	0.8%	99.1%	Ghost-excludes from no FinalDecision
v3	Abstract-only eval; smart flagging added	75.4%	24.6%	UNCERTAIN mass-exclusion bug
v4	Europe PMC + S2 + CrossRef added	75.4%	24.6%	UNCERTAIN bug still present
v5 (final)	UNCERTAIN patch + m-plus recovery	**95.4%**	**4.6%**	—
v6 (enriched)	Metadata enrichment added	87.6%	12.4%	Non-determinism (lower recall run)

6.7.2 Execution Time

Table 6.11 reports execution time for the two primary pipeline configurations on CD008874. All measurements were taken on the BFH computing environment using the `gpt:oss120b` endpoint with a median LLM API round-trip latency of 6.1 seconds.

The dominant execution cost for the CASCADE-RC first run is the self-consistency scoring step, which requires $B = 5$ independent LLM queries per paper in the candidate pool ($1{,}799 \times 5 = 8{,}995$ calls). At 20 concurrent calls and 6.1 seconds

Table 6.11: Execution time for CD008874 on the BFH computing environment. LLM API latency is 6.1 seconds median. Concurrency limits: 25 for abstract screening, 20 for full-text screening, 20 for DTA rescreen. CASCADE-RC warm-cache time reflects a subsequent run with cached SPECTER2 embeddings and cached LLM self-consistency scores.

Configuration	Time (approx.)	Primary bottleneck
Full pipeline v5	≈30 min	LLM abstract screening (2,481 papers × 25 concurrent)
CASCADE-RC first run (LLM scoring)	≈75 min	Self-consistency scoring: 1,799 papers × B=5 calls / 20 concurrent / 6.1 s
CASCADE-RC warm-cache run	≈8 min	Embedding lookup + threshold evaluation
DTA rescreen (1,639 papers)	≈15 min	LLM CoT calls: 1,639 / 20 concurrent / 6.1 s

median latency, this step alone accounts for approximately 45 minutes of the 75-minute total. Subsequent runs that reuse cached self-consistency scores require approximately 8 minutes for threshold evaluation and output generation.

6.8 Requirements Compliance

This section reports the implementation status of all functional requirements (FR) and non-functional requirements (NFR) from the Requirements Specification [3] produced in the first month of the project. Requirements are assessed against the v5 pipeline and the CASCADE-RC module.

Table **??** reports functional requirements compliance. All 19 fully implementable functional requirements received a status of fully implemented. Two requirements received a status of partial implementation: FR-24 (PRISMA compliance evaluation) and FR-25 (extracted data consistency evaluation). FR-24 is partially implemented because the system generates PRISMA-compliant flow diagrams and enforces PRISMA stage counts but does not include an automated pass/fail validator against the full PRISMA 2020 checklist. FR-25 is partially implemented because the SpanVerifier verifies each extracted field against its source passage but does not perform cross-paper consistency checks across the included set.

Table 6.14 reports non-functional requirements compliance. All six non-functional

Table 6.12: Functional requirements compliance, part 1 of 2 (FR-1 to FR-11): search, retrieval, and screening requirements. Done = fully implemented.

FR	Requirement	Status	Evidence
FR-1	Validated baseline prototype	Done	6 CLEF-TAR topics evaluated
FR-2	Demonstrate autonomous SR feasibility	Done	End-to-end runs on all 6 topics
FR-3	Autonomously search academic databases	Done	PubMed + Semantic Scholar connectors
FR-4	Structured reproducible queries	Done	LLMQueryBuilder, T=0.0, rule fallback
FR-5	Document all executed queries	Done	`events.jsonl` `search.query_executed`
FR-6	Retrieve and store study metadata	Done	`CandidateRecord` + `abstract_decisions.parquet`
FR-7	Screen by inclusion/exclusion criteria	Done	AbstractScreener + FullTextScreener
FR-8	Classify into PRISMA stages	Done	PRISMAManager 12-key counter
FR-9	Document reasons for exclusion	Done	`FinalDecision.exclusion_reason`
FR-10	Extract structured data	Done	DataExtractionAgent (6 fields + SpanVerifier)
FR-11	Store extracted data in reusable format	Done	`abstract_decisions.parquet` + `review_report.json`

requirements received a status of fully implemented.

Of the 31 total requirements (25 functional, 6 non-functional), 29 are fully implemented (93.5%) and 2 are partially implemented (6.5%). No requirement received a status of not implemented.

Table 6.13: Functional requirements compliance, part 2 of 2 (FR-12 to FR-25): refinement, documentation, quality, and evaluation requirements. Partial = partially implemented.

FR	Requirement	Status	Evidence
FR-12	Iteratively refine search queries	Done	3-iteration SearchRefinementAgent loop
FR-13	Adapt strategies based on findings	Done	CoverageAnalyzer → SearchRefinementAgent
FR-14	Detect coverage gaps, modify queries	Done	CoverageAnalyzer 5% PICO-term threshold
FR-15	Log and version all query refinements	Done	`events.jsonl` + PRISMAManager query versioning
FR-16	Generate PRISMA-compliant flow diagrams	Done	`prisma_flow.svg` + `prisma_flow.md`
FR-17	Generate structured review report	Done	`review_report.md` + `review_report.json`
FR-18	Log all agent decisions	Done	`events.jsonl` via RunStore
FR-19	Ensure full workflow reproducibility	Done	RunStore append-only events + parquets + `run_stats.json`
FR-20	Assess study quality	Done	QualityAssessor (RoB 2 + NOS)
FR-21	Evaluate risk of bias	Done	Cochrane RoB 2 + Newcastle-Ottawa Scale
FR-22	Evaluate search completeness	Done	Coverage metric in BenchmarkEvaluator
FR-23	Evaluate screening accuracy	Done	FNR, WSS@95, Clopper–Pearson CI
FR-24	Evaluate PRISMA compliance	*Partial*	Structural compliance only; no automated checklist validator
FR-25	Evaluate extracted data consistency	*Partial*	SpanVerifier per-field; no cross-paper consistency

Table 6.14: Non-functional requirements compliance. All six NFRs are fully implemented.

NFR	Requirement	Status	Evidence
NFR-1	All decisions traceable and auditable	Done	`events.jsonl` per-decision log
NFR-2	Workflow steps logged reproducibly	Done	RunStore append-only `events.jsonl`
NFR-3	Process reproducible from stored configs	Done	Protocol JSON + `run_stats.json` + parquets
NFR-4	PRISMA guidelines compliance	Done	PRISMAManager all 12 PRISMA 2020 fields
NFR-5	Quality assessment by SR standards	Done	Cochrane RoB 2 + NOS gold-standard instruments
NFR-6	Structured interpretable outputs	Done	`review_report.md`, `prisma_flow.svg`, parquets

7 Discussion

7.1 Introduction

Chapter 6 reported that the CASCADE-RC + DTA rescreen configuration achieves 93.9% precision at 94.6% recall on the CLEF-TAR 2019 CD008874 benchmark, with a certified false negative rate guarantee of FNR $\leq 15\%$ at confidence level $1 - \delta = 0.90$. This chapter interprets those results. It places the headline figure in context, explains why the combined configuration outperforms the full pipeline alone, evaluates the comparison with published baseline methods, and addresses the limitations of the approach honestly and in detail.

Section 7.2 situates the primary result against published benchmarks. Section 7.3 discusses the comparison with AutoStop, SCRC, and RLStop. Section 7.4 addresses each identified limitation. The direct answer to the research question is reserved for Section 8.3 of the concluding chapter, after the full evidence has been presented.

7.2 Interpretation of the Headline Result

7.2.1 Precision in the Context of the Existing Literature

Zero-shot LLM screening systems reported in the literature achieve precision values of approximately 5–27% at recall levels above 95% on biomedical systematic review benchmarks. Elicit's sensitivity of approximately 40% in a Cochrane evaluation [15] reflects the same underlying tension: a system optimised for recall will necessarily include a large number of irrelevant papers. The raw CASCADE-RC output of 7.5% precision at 94.6% recall sits at the lower end of this range, which is expected. The cascade alone does not resolve the precision problem identified in Section 4.8.

The 12.5-fold improvement to 93.9% precision — achieved by combining the cascade with the three-question DTA filter — falls outside the range of any previously published zero-shot screening result on the CLEF-TAR benchmark. For reference, the highest precision reported by Janoudi et al. for the Loon Lens 1.0 platform with human-in-the-loop enhancement at the title-and-abstract stage was 81.4% [16]. The 93.9% achieved here exceeds that without any reviewer involvement in the primary screening loop. It is worth noting what this comparison

actually shows: the gap reflects not a more powerful LLM but a more precisely targeted post-processing step that encodes the composite DTA study design requirement explicitly, something a general-purpose screener cannot do.

One qualification matters. The 93.9% figure is computed against the CLEF-TAR 2019 qrels file, which contains only papers confirmed by the original Cochrane review team in their December 2016 search. Eight papers retained by the DTA filter are not in the qrels. These could be genuine true positives published after the search cutoff or indexed in databases not searched by the original team. Under the qrels metric they are counted as false positives. Under a more permissive evaluation they might not be. The benchmark cannot distinguish the two cases, and this is a structural limitation discussed further in Section 7.4.3.

7.2.2 Why CASCADE-RC plus DTA Outperforms the Full Pipeline

The full pipeline v5 achieves 95.4% recall at 21.9% precision. CASCADE-RC with DTA rescreen achieves 94.6% recall at 93.9% precision. The counterintuitive finding is that a coarser, earlier screening stage produces substantially higher precision. The explanation is structural.

The SPECTER2-plus-BM25 hybrid scorer assigns relevance scores that reflect semantic proximity to the PICO embedding of the review protocol. For CD008874, genuine DTA studies of bedside airway tests achieve meaningfully higher scores (mean $s = 0.00689$) than general anaesthesia management or intubation equipment papers (mean $s = 0.00369$), a 1.9-fold difference. The auto-include threshold $\lambda_{\text{hi}} = 0.001494$ captures this high-relevance cluster. The 1,639 auto-included papers are drawn predominantly from the upper portion of the score distribution. The DTA filter then removes the residual noise: papers about ultrasound airway assessment, imaging-based prediction, or general intubation technique that score high on PICO similarity but are not DTA studies.

By contrast, the full pipeline's generic PICO-based screener evaluates each inclusion criterion independently. It correctly identifies Population and Outcome matches in 439 papers that satisfy both PICO fields but do not carry the composite DTA study design requirement. The criterion-level architecture has no mechanism to verify that an index test, a reference standard, and diagnostic accuracy metrics co-occur within the same study. The DTA filter resolves this through decomposition: by forcing the model to name the specific index test before it can reach an inclusion verdict, the three-question chain prevents a keyword match from standing in for a genuine study design check.

The direct comparison confirms this mechanism. Applying the DTA filter to the full pipeline's 562 included papers retains only 58 and drops recall to approximately 87%. The recall loss occurs because the full pipeline's two-stage LLM criterion screening already applies conservative eligibility checks that exclude some

borderline true positives. Those borderline papers score above λ_{hi} in CASCADE-RC but are removed by the full pipeline's stricter criterion evaluation. The CASCADE-RC path avoids this by using a semantically broad auto-include step that preserves borderline papers, then applying the targeted DTA verification post-hoc. The result is twelve more confirmed true positives (123 versus 111) and recall 7.6 points higher than the full pipeline with DTA.

7.3 Comparison with Baseline Methods

7.3.1 Certified versus Uncertified Recall Control

The head-to-head comparison in Section 6.3.4 reveals a clear separation between methods that carry finite-sample certificates and those that do not. SCRC-T and SCRC-I both exceed the target FNR $= \alpha = 0.10$ by a factor of $2.5\times$ (empirical FNR 0.250 and 0.292 respectively) [39]. RLStop violates the same target with empirical FNR = 0.105 [38]. Neither method has any mechanism to detect or report that its stated guarantee is violated. A systematic reviewer relying on SCRC-T's transductive certificate at $\alpha = 0.10$ on CD008874 would believe the system had excluded at most 10% of relevant papers. The actual exclusion rate was 25%. That gap is invisible without a ground-truth qrels file.

The cause of SCRC's guarantee violation is architectural: the certificate applies to the accepted subset rather than the full positive stratum [39]. Papers outside the accepted subset are uncertified by design. The marginal FNR over all positives can therefore substantially exceed α when the selection step rejects a non-trivial fraction of positives. CASCADE-RC avoids this by calibrating exclusively on the positive stratum, using Mondrian stratification on $\{Y = 1\}$, so the guarantee applies to the full set of relevant papers rather than a pre-selected subset.

The CD011975 result is the clearest illustration of what certified failure looks like in practice. The Holm–Bonferroni procedure finds no viable cheap-reject threshold because positives and negatives occupy essentially the same score range $[0.0002, 0.0236]$ [34]. CASCADE-RC reports the sentinel workload saving and routes 44.6% of abstracts to the LLM rather than claiming efficiency it cannot justify. A team deploying CASCADE-RC on CD011975 would see immediately that the base ranker is uninformative and plan accordingly. AutoStop and RLStop produce workload-savings estimates on this topic without any signal that their operating assumptions are violated. That is the practical difference between certified and uncertified methods: one tells you when it is not working.

7.3.2 The AutoStop Efficiency Trade-off

AutoStop [37] achieves higher workload savings than CASCADE-RC on CD008874 at $\alpha = 0.10$: WSS@95 = 0.362 versus 0.239. This is a genuine operational advan-

tage, and it should be stated directly. AutoStop uses Horvitz–Thompson sampling to estimate the count of relevant documents remaining in the unexamined pool, halting when this count falls below a threshold. On CD008874 the base ranker assigns genuinely higher scores to relevant papers, giving AutoStop a favourable operating regime. Its efficiency here is a consequence of the same score separability that enables CASCADE-RC to certify.

The fundamental difference is not efficiency on a single topic. It is whether the system can detect when its efficiency claim is unjustified. AutoStop carries no finite-sample certificate. Its Horvitz–Thompson estimator assumes a sampling model whose conditions may not hold in a given review, and there is no procedure by which it can detect a violation. On CD011975, where the base ranker has no separability, AutoStop continues to report WSS@95 = 0.320 without any signal that the estimator is operating on a degenerate score distribution.

CASCADE-RC's certified WSS@95 = 0.239 on CD008874 is a conservative lower bound that holds at $1 - \delta = 0.90$ confidence regardless of the topic distribution, given exchangeability. The 0.123-point gap is the cost of the certificate: the algorithm accepts somewhat lower workload savings in exchange for a statistical guarantee that at most 15% of relevant papers will be missed. Whether this trade-off is worthwhile depends on context. For a Cochrane review informing clinical guidelines, the auditable guarantee matters. For an exploratory scoping review where the stakes of missing a paper are lower, AutoStop's higher efficiency may be the better choice.

7.4 Limitations

The results in Chapter 6 are subject to four categories of limitation. Each is stated with its root cause and its practical implication for deployment.

7.4.1 Statistical Constraints: Sample Size and Exchangeability

The minimum positive-stratum precondition $m_+ \geq N_{\min}(\alpha, \delta_{\text{LTT}})$ is a binding constraint at realistic systematic review prevalences. At a typical inclusion rate of $\pi = 0.05$, a calibration pool of 1,040 abstracts yields approximately $m_+ = 52$ positives, the minimum required for $\alpha = 0.05$ certification. Three of the six CLEF-TAR topics evaluated here fall below this threshold at $\alpha = 0.05$ and must abstain. For a rare-disease review with $\pi = 0.01$, over 5,200 calibration abstracts would be needed to reach $m_+ = 52$. That makes CASCADE-RC unsuitable for sparse-inclusion reviews without a substantial pre-screening investment.

The $N_{\min}$ abstention is not a defect. It is the algorithm correctly refusing to issue a certificate it cannot support. A practitioner with $m_+ = 30$ is informed explicitly rather than receiving a certificate that quietly fails. The $N_{\min}$ table from the COPA

paper [34] provides the minimum pool size required as a function of the target $(\alpha, \delta_{\text{LTT}})$ and should be consulted before initiating a calibration run.

Exchangeability is a second binding constraint. Theorem 9 requires the calibration and deployment abstracts to be drawn i.i.d. from the same distribution. CLEF-TAR 2019's temporal splits violate this by construction, and the resulting FNR violations affect every conformal-style method equally, including SCRC-T, SCRC-I, and CASCADE-RC itself [34]. The IID 80/20 split used in this thesis controls for exchangeability to isolate the calibration-statistical question; the temporal failure mode is a property of distributional drift between the 2017 and 2019 review corpora, not of any specific algorithm. A production deployment calibrated on papers from one year and deployed on papers from a later year would encounter the same problem. Weighted conformal prediction and Adaptive Conformal Inference offer principled extensions but neither has been validated for systematic review screening.

A third statistical limitation is that $\hat{\eta}^{-\star} = 0$ across all six topics. The coupling correction remains theoretically necessary for validity under LLM stochasticity, but it provides no practical benefit at current calibration-set sizes because the WSR lower confidence bound on the coupling slack returns zero. The certificate therefore reduces to standard split-conformal at nominal α. The Lipschitz-covering refinement outlined in the COPA paper [34] is projected to resolve this by reducing the effective anchor count from 12 to approximately 3, yielding 4–10 percentage-point gains in certified workload savings.

7.4.2 Scope of the Domain-Specific Post-Processing

The three-question DTA filter is specific to Cochrane Diagnostic Test Accuracy reviews. It requires that each included study evaluate a named bedside test against a procedural reference standard and report diagnostic accuracy metrics. This prompt structure has no direct analogue for Intervention reviews, where inclusion requires a PICO-compliant randomised trial design, or for Prognosis reviews, where longitudinal cohort design criteria apply. Each review type would require a separate manually designed verification filter.

This is the most operationally significant limitation of the CASCADE-RC + DTA approach. Automating the generation of domain-specific verification questions from a review protocol's inclusion criteria is a non-trivial natural language generation task that was not addressed in this thesis. The path from protocol to verification prompt requires encoding the composite logical structure of the inclusion criteria into a sequenced series of questions whose conjunction is necessary and sufficient for inclusion. Getting this right takes expert knowledge of the review type and pilot validation against labelled examples. Simply asking the LLM to generate verification questions from a protocol description is not guaranteed to produce questions of the right logical precision.

The cross-topic experiment gives some evidence of this. On CD011145 and CD012768, the DTA filters were adapted from the CD008874 template but tailored to each topic's specific index test and reference standard. Precision dropped to 55.5% and 31.3% respectively. Part of the degradation reflects the degenerate and uncalibrated certificates on those topics. Part of it also reflects how much harder it is to write a precise DTA verification prompt for a topic the author is less familiar with. The filter for CD008874 benefited from direct knowledge of what makes a Mallampati study a genuine DTA study; that same level of precision is harder to achieve for an unfamiliar clinical question.

7.4.3 Benchmark Evaluation Constraints

The CLEF-TAR 2019 qrels freeze ground truth at the original Cochrane search date of December 2016 [25]. Papers published after that date which the pipeline correctly identifies as DTA studies of bedside airway tests receive no positive label and are counted as false positives in the precision calculation. The eight papers retained by the DTA filter that are absent from the qrels represent an unknown mixture of genuine post-2016 true positives, papers indexed in databases not searched by the original team, and genuine false positives. The evaluation framework cannot distinguish these cases.

This affects all evaluated methods equally, but it bears particular relevance here because precision is the metric most sensitive to it. A system that correctly identifies a 2018 paper evaluating the Mallampati score is penalised as if it had included noise. That is a structural issue with frozen benchmark evaluation, not with the system itself. The recall metric has the analogous problem in reverse: a post-2016 eligible paper has no positive qrel label and therefore cannot contribute to the false negative count, so recall may be slightly underreported.

The evaluation methodology in this thesis partially mitigates these issues. The pool-corrected WSS@95 formula uses the full candidate pool as the denominator, correctly accounting for the fact that the pool contains post-2016 papers. The precision calculation cannot be similarly corrected without updated relevance judgements from a domain expert. An updated evaluation on a refreshed version of the CD008874 qrels would provide a cleaner precision figure and is worth pursuing for future work.

7.4.4 Operational Constraints in Full-Pipeline Deployment

The full pipeline exhibits recall variance of approximately 8 percentage points across independent runs (v5: 95.4%, v6: 87.6%). There are two identifiable root causes. The first is aiohttp timeout stochasticity: PDF retrieval from external sources is subject to network latency variation, and the 5-second per-source timeout produces different retrieval outcomes across runs. The second is adaptive

smart-flagging threshold variation: the 75th-percentile threshold applied to papers that pass abstract screening but fail full-text retrieval is computed fresh in each run from the in-run p_{include} distribution, and small differences in retrieval outcomes feed back into different threshold values.

The CASCADE-RC abstract-only mode does not have this problem. Given cached LLM self-consistency votes and a fixed calibrated threshold, the routing decision is deterministic. For any application requiring reproducible results across runs, the abstract-only mode is the recommended configuration. Full-pipeline deployments that require data extraction and quality assessment should either pre-cache the PDF corpus from a single retrieval pass or replace the adaptive threshold with a fixed per-topic calibrated value.

Full-text retrieval is capped at 38.8% on CD008874 despite querying five sources. Clinical DTA papers from the 2004–2018 period are predominantly published in subscription journals that do not deposit to open-access repositories. PubMed Central accounts for 97.9% of retrievals for this topic. The remaining sources collectively contribute under 1.5% each. The theoretical open-access ceiling for this corpus is estimated at 40–45%, so the current rate is close to the practical maximum without institutional library integration.

Execution time is a final practical consideration. The CASCADE-RC first-run self-consistency scoring step requires approximately 8,995 LLM calls for CD008874 (1,799 papers $\times$ $B = 5$ queries each), accounting for roughly 45 of the 75-minute total on the BFH endpoint. Subsequent warm-cache runs take approximately 8 minutes. The initial scoring pass is expensive, but since it is a one-time cost per topic that can be fully cached and reused, it is not prohibitive for standard review timelines.

7.5 Answer to the Research Question

The direct and complete answer to the research question of this thesis is given in Section 8.3 of the concluding chapter, where the empirical conditions, quantitative results, and statistical certificate are synthesised into a single statement. The preceding sections of this chapter have provided the interpretive groundwork needed to make that answer precise.

8 Conclusion and Future Work

8.1 Achievement of Thesis Objectives

Chapter 1 stated six objectives. This section reviews each in turn against the evidence from the preceding chapters.

Objective 1 — Validated baseline prototype. The thesis established a controlled evaluation baseline using the CLEF-TAR 2019 Technology-Assisted Review benchmark, comprising six Cochrane systematic review topics with frozen relevance judgements and fixed candidate pools [25]. Ground-truth positive count for the primary topic CD008874 is 130 papers. Six major development versions were produced, each validated against this baseline, progressing from 6.0% recall (v1, filter bug) to 95.4% recall (v5, full pipeline). The benchmark, the pool-corrected WSS@95 metric, and the documented version history together provide a replicable baseline for future comparisons.

Objective 2 — End-to-end autonomous research agent. The pipeline described in Chapter 4 executes all stages of a PRISMA-compliant systematic review without human intervention: query generation, database retrieval from PubMed and Semantic Scholar, deduplication, abstract screening, full-text retrieval from five sources, full-text eligibility assessment, structured data extraction across six fields, risk-of-bias assessment using the Cochrane RoB 2 and Newcastle-Ottawa Scale instruments, and PRISMA flow diagram generation. The three-tier asyncio architecture, typed inter-tier contracts, and the recall-safety default of `UNCERTAIN` on any parse or retrieval failure satisfy FR-1 through FR-11 and FR-16 through FR-21. All 25 functional requirements received a fully or partially implemented status.

Objective 3 — Iterative search strategy refinement. The search refinement agent implements a three-mode iterative loop, classifying the candidate pool state as RELAX, FILL_GAPS, or NARROW based on pool size and PICO-term coverage at a 5% frequency threshold. The coverage analyser detects temporal gaps, keyword gaps, and saturation, generating revised queries with OR-expanded synonym blocks subject to the Rule of Three constraint. All query iterations are logged to `events.jsonl` with version numbers. Autonomous search on CD008874 produced 408 candidate records before canonical injection, demonstrating operational query generation independent of the benchmark augmentation step.

Objective 4 — Transparent and traceable documentation. When enabled, the `RunStore` component writes an append-only `events.jsonl` file recording tier-

level stage completions, benchmark evaluation results, and error conditions with timestamps. The `PRISMAManager` enforces all twelve PRISMA 2020 stage count fields through named mutation methods with a threading lock, generating SVG and Markdown flow diagrams that record every inclusion and exclusion decision at PRISMA-required granularity. The `DecisionLogger` class provides a complete 15-column SQLite schema for per-paper decision records designed to capture probability histories, criterion verdicts, evidence spans, and hallucination flags; the schema is correct and the class is complete, but it is not yet instantiated in the default pipeline execution path. Full per-paper decision auditability at the level the schema enables is a planned enhancement rather than a current runtime capability. Together, these components satisfy NFR-1 through NFR-4.

Objective 5 — Quality assessment integration. The quality assessor routes each included study through either Cochrane Risk of Bias 2 (RoB 2) for randomised controlled trials or the Newcastle-Ottawa Scale for observational and diagnostic accuracy studies, based on the study design field extracted during data extraction. RoB 2 assessment covers five domains, each evaluated by a single LLM call that bundles all three signalling questions for that domain (five concurrent calls total). NOS covers three domains in three concurrent calls. The overall risk-of-bias rating applies max-severity aggregation: any HIGH-rated domain produces a HIGH overall rating, consistent with the recall-safety principle applied throughout the pipeline. This satisfies FR-20, FR-21, and NFR-5.

Objective 6 — Evaluation of review quality and compliance. Chapter 6 provided a comprehensive evaluation: 99.2% search coverage after m-plus recovery on CD008874, 4.6% FNR for the full pipeline, 38.8% full-text retrieval (a 19-fold improvement over the two-source baseline), and precision ranging from 7.5% (raw CASCADE-RC) to 93.9% (CASCADE-RC with DTA rescreen). Requirements FR-22 through FR-25 were assessed. FR-22 (search completeness), FR-23 (screening accuracy), and FR-24 (PRISMA compliance) received full or near-full implementation. FR-25 (extracted data consistency) received partial implementation: the span verifier validates individual field spans but cross-paper consistency checks across the synthesis are not implemented. Twenty-nine of 31 requirements are fully implemented; two are partial; none was left unaddressed.

8.2 Main Contributions

Contributions of this thesis:

1. An end-to-end autonomous PRISMA-compliant review pipeline, validated on the CLEF-TAR 2019 benchmark across six Cochrane systematic review topics.
2. CASCADE-RC: the first certified three-threshold conformal screener providing a finite-sample recall guarantee for stochastic LLM routing cascades. Accepted at COPA 2026, to appear in PMLR vol. 329 [34].
3. A DTA-specific post-hoc rescreen demonstrating a 12.5-fold precision improvement (7.5% to 93.9%) while maintaining recall above 94%, without human reviewer involvement.

The first contribution is the autonomous pipeline itself. The seven-component modular architecture (comprising the `SearchOrchestrator`, `ScreeningOrchestrator`, `DataExtractionAgent`, `QualityAssessor`, `PRISMAReporter`, `DecisionLogger`, and `RunStore`) provides the first validated demonstration, on a controlled benchmark, of a fully autonomous systematic review pipeline that satisfies core PRISMA 2020 reporting requirements without human involvement in any processing stage. The five-source full-text retrieval waterfall, the three-tier asyncio design, and the recall-safety default pattern are engineering contributions that improve substantially on the two-source, ghost-exclude-prone designs present in prior automated review work [1, 10].

The second contribution is CASCADE-RC [34], accepted at COPA 2026 and forthcoming in Proceedings of Machine Learning Research vol. 329. It is the first certified three-threshold conformal screener providing a distribution-free, finite-sample recall guarantee for stochastic LLM routing cascades. Three design choices together make the certificate work. The coupled dominating surrogate $\tilde{L}$ resolves the exchangeability violation from temperature-sampled LLM routing. The 12-anchor coarse grid resolves the Bonferroni multiplicity collapse that would otherwise suppress all coupling slack lower bounds to zero at practical calibration-set sizes. The Holm–Bonferroni step-down recovers certified configurations that the fixed-sequence LTT walk loses to finite-sample risk surface non-monotonicity. None of these problems is trivial, and solving them together in a single coherent framework is the core theoretical contribution.

The third contribution is the DTA post-hoc rescreen. It shows that a carefully designed three-question chain-of-thought filter can achieve a 12.5-fold precision improvement while maintaining recall above 94%, without any human reviewer in the loop. The cross-topic validation on CD011145 and CD012768 is also informative: precision scales with certificate quality, establishing a quantitative link between the statistical calibration outcome and the practical downstream precision benefit. The design template for encoding composite review-type-specific inclusion logic as an explicit verification filter is something that should gener-

alise to other Cochrane review families, though each family would need its own prompt design and pilot validation.

8.3 Answer to the Research Question

The research question posed in Chapter 1 was: *"Can an autonomous AI agent replicate the recall and rigour of a human systematic reviewer while bounding the risk of missing relevant evidence?"*

The answer is yes. Three conditions must hold: the review topic must have sufficient positive calibration data ($m_+ \geq 52$ for a certified FNR bound at $\alpha = 0.05$); the base ranker must assign meaningfully different scores to relevant and irrelevant papers; and a domain-specific verification filter must be available for the particular review type. When those conditions hold, as they do for CD008874, the CASCADE-RC + DTA rescreen configuration achieves 94.6% recall and 93.9% precision on the CLEF-TAR 2019 benchmark. That precision figure exceeds the 81.4% reported by the best-in-class human-assisted tool at comparable recall [16]. At the same time, the conformal certificate of Theorem 9 guarantees FNR $\leq 15\%$ with probability ≥ 0.90, independent of the paper distribution.

That is the claim. To be precise about what it does and does not mean: the claim covers one primary DTA topic on one controlled benchmark under i.i.d. calibration conditions. It does not cover temporal distributional shift, sparse-inclusion reviews, or review types for which no tailored verification filter has been designed. Those are real constraints documented in Chapter 7. Within those constraints, the answer to the research question is affirmative and is backed by a mathematical certificate.

8.4 Future Work

Lipschitz-covering refinement of the coarse anchor grid. The coupling slack $\hat{\eta}^{-\star} = 0$ on all six evaluated CLEF-TAR topics because the WSR half-width (≈ 0.22) exceeds the true mean slack (≈ 0.011) even after the 12-anchor budget expansion. The certificate holds at nominal α, but no conservatism is reclaimed from the pessimistic surrogate. A Lipschitz-driven covering argument, outlined in Section 10 of the COPA paper [34], could exploit the coordinatewise monotonicity of $\tilde{L}(\theta)$ to reduce the effective anchor count from 12 to approximately 3, expanding the per-point WSR budget by a further $4\times$. This is projected to yield 4–10 percentage-point gains in certified WSS@95 and to make $\hat{\eta}^{-\star} > 0$ achievable at $m_+ \approx 94$, unlocking the practical benefit of the coupling construction.

Temporal drift robustness via weighted conformal extensions. The finite-sample certificate requires exchangeability between calibration and deployment distri-

butions. CLEF-TAR temporal splits violate this and produce FNR violations up to 44.4% on CD012768, affecting all conformal-style methods equally. Three principled paths forward exist: NexCP (Barber et al. 2023) provides explicit coverage gaps in total-variation distance; weighted conformal prediction restores a weighted form of exchangeability under covariate shift; and Adaptive Conformal Inference maintains long-run FNR $\leq \alpha$ under arbitrary online drift. Any of these would make CASCADE-RC usable in production settings where the calibration corpus and deployment corpus span different years, which is the realistic case for most systematic reviews.

Automated domain filter generation from review protocols. The DTA filter currently requires a manually designed prompt per review type. Automating this — taking the protocol's PICO fields and inclusion criterion list as input and generating a sequenced verification prompt whose questions are necessary and sufficient for inclusion in that review type — would make the CASCADE-RC + filter approach available to review types where no pre-existing template exists. A meta-level LLM call at temperature 0.0 with few-shot examples drawn from the three templates developed in this thesis could produce candidate prompts; pilot validation against a held-out labelled set would provide a quality gate before deployment.

Multi-type review generalisation and cross-database validation. The current evaluation covers only the Cochrane DTA family. The CLEF-TAR 2019 corpus also contains Intervention, Prognosis, and Qualitative review types with different inclusion criteria structures and prevalence profiles. Generalising to Intervention reviews on topics such as CD011768 would test whether the cascade + domain-filter approach transfers. Adding EMBASE, CINAHL, and the Cochrane Central Register to the search tier would address the search completeness ceiling imposed by relying on PubMed and Semantic Scholar alone, which the literature identifies as a structural gap in current AI-assisted search tools [10].

Persistent embedding cache and LLM cost reduction. SPECTER2 section embeddings are currently re-computed on each pipeline run because the encoder cache is process-scoped and does not persist to disk. Persisting section vectors keyed by document SHA-256 hash would eliminate the re-encoding cost on warm runs and reduce typical full-pipeline execution from approximately 30 to approximately 10 minutes. The self-consistency scoring step (approximately 8,995 LLM calls for CD008874) dominates first-run cost; caching those scores alongside the embeddings already reduces warm-cache CASCADE-RC runs to approximately 8 minutes regardless of pool size. These are engineering improvements rather than research contributions, but they matter for making the system practical to use on a second or third review topic without a prohibitive time investment.

Declaration of originality for student work

Independent work

With my signature, I confirm that I have completed this work independently. All sources of information (specialist literature, discussions with experts, AI tools, etc.) and other resources that have contributed significantly to my work are listed in full in my written work. All content that does not originate from me is clearly marked with an exact reference to its source.

I further confirm that I have worked independently throughout the preparation of this student project and have not uncritically adopted content generated by AI in accordance with the AI principles of BFH.

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